CHEMISTRY GRADE 10: CHEMICAL BONDING

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (13 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.4: Chemical Bonding
Total Duration	13 lessons × 40 minutes = 520 minutes total
Sub-Strand Content	– Valence electrons in bonding — the octet (and duplet) rule; stability through electron gain, loss, or sharing – Bond types — ionic bonding (electron transfer between metals and non-metals) – Bond types — covalent bonding (electron sharing in non-metal molecules, e.g. O_2, H_2O, CO_2) – Bond types — dative (coordinate) covalent bonding (one atom donates both electrons, e.g. NH_4^+, H_3O^+) – Bond types — hydrogen bonding (special dipole–dipole interaction in H_2O, HF, NH_3) – Bond types — Van der Waals forces (London dispersion and permanent dipole–dipole forces) – Bond types — metallic bonding (sea of delocalised electrons in metal lattices, e.g. aluminium, copper) – Lewis (dot-and-cross) structures — drawing diagrams for ionic, covalent, dative covalent, hydrogen-bonded and metallic systems – Resultant structures — giant ionic lattice (e.g. NaCl), simple molecular (e.g. H_2O, I_2), giant covalent/atomic (e.g. diamond, SiO_2), giant metallic (e.g. Al) – Physical properties of structures — solubility, electrical conductivity (solid and molten/aqueous), thermal conductivity, melting point and boiling point – Relationship between bond type, resultant structure and physical properties — comparative investigation using NaCl, graphite, diamond, aluminium – Uses of substances linked to bond type and structure — graphite as lubricant/electrode, diamond in cutting tools, NaCl in food preservation, metals in construction and wiring – Project — building 3-D models of selected bonded structures (NaCl, SiO_2, graphite, diamond) using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) illustrate bond types in elements, molecules and compounds; b) investigate the relationship between bond types and physical properties of elements, molecules and compounds; c) relate bond types and resultant structures to the uses of elements, molecules and compounds; d) appreciate the uses of different substances based on their bond types and structures in day-to-day life.
Core Competencies	– Learning to Learn: The learner learns independently while drawing dot (.) and cross (x) diagrams to illustrate bonding in elements and compounds, building self-directed study habits as they construct and self-check Lewis structures for ionic, covalent, dative covalent and metallic systems. – Digital Literacy: The learner uses digital technology to effectively accomplish own task while watching animations, simulations

	and videos on chemical bonding — including simulations of electron-sharing in O_2 and the giant ionic lattice of NaCl. – Creativity and Imagination: The learner experiments with material to develop models to illustrate bonding in selected elements/compounds using locally available materials (e.g. clay, bottle tops, wire, maize cobs) to construct 3-D representations of NaCl, graphite, diamond and SiO_2. – Critical Thinking and Problem Solving: The learner interprets and infers with peers the relationship between bond types and resulting structures, explaining why diamond is hard and graphite is slippery despite both being pure carbon.
Core Values	– Love: The learner shows care by avoiding hurting others as they carry out activities to investigate the physical properties of different structures — handling glassware, conductivity apparatus and chemical samples safely and with consideration for classmates.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students generate and refine questions about why diamond and graphite — both pure carbon — have dramatically different hardness, conductivity and appearance, anchoring inquiry throughout the unit. – Developing and Using Models: Students draw Lewis dot-and-cross diagrams and build physical 3-D models (NaCl, SiO_2, graphite, diamond) to represent bonding at the particulate level and use those models to predict properties. – Planning and Carrying Out Investigations: Students design and conduct a structured experiment comparing solubility, electrical conductivity, melting point and thermal conductivity across giant ionic (NaCl), simple molecular (iodine/sugar), giant covalent (sand/SiO_2) and giant metallic (aluminium foil) samples. – Analysing and Interpreting Data: Students record and compare quantitative and qualitative data from the physical-properties investigation, identifying patterns that link bond type to observable behaviour. – Constructing Explanations: Students use electron-sharing and electron-transfer models to explain the physical properties observed, writing structured explanations that connect particulate-level bonding to macroscopic evidence. – Obtaining, Evaluating and Communicating Information: Students watch simulations/animations on chemical bonding, evaluate the accuracy of the models shown, and communicate findings through group presentations and updated class Driving Question Boards.
Pertinent & Contemporary Issues (PCIs)	– Socio-Economic and Environmental Issues — Environmental Conservation: The learner uses locally available materials (clay, bottle tops, maize cobs, wire) to make models, minimising waste and promoting resourcefulness within the Kenyan context. – Life Skills — Creative Thinking: The learner models bonding in selected elements/compounds (e.g. SiO_2, NaCl, graphite and diamond) using imaginative construction approaches that reflect real molecular geometry. – Socio-Economic and Environmental Issues — Social Cohesion: The learner discusses with peers different types of chemical bonds in diverse small groups, building respect for others' contributions and collaborative problem-solving skills.
Career Connections	– Materials Science and Engineering: Understanding bond types and resultant structures underpins the design of construction materials, ceramics, polymers and composites used in Kenyan infrastructure projects. – Mining and Gemology: Kenya's gemstone industry (rubies, garnets, tanzanite trade) requires knowledge of giant covalent and ionic structures, hardness and crystal properties rooted in chemical bonding. – Metallurgy and Manufacturing: Metallic bonding concepts explain the ductility, malleability and conductivity of metals such as aluminium and copper used extensively in Kenyan manufacturing and electrical industries. – Medicine and Pharmacy: Ionic and hydrogen bonding govern drug solubility, protein folding and drug–receptor interactions — foundational knowledge for careers in medicine, nursing and pharmaceutical science. – Agriculture and Food Science: Understanding Van der Waals forces and hydrogen bonding explains the physical behaviour of water in soils, food preservation (e.g. salt/NaCl curing of nyama choma) and crop storage. – Environmental Science: Knowledge of molecular structures and bond polarity is essential for understanding pollutant behaviour, water treatment chemistry and the chemistry of the atmosphere over the Rift Valley and Lake Victoria basin.
Focus for Lessons	This unit anchors learning in the astonishing puzzle that diamond and graphite are both 100% carbon yet have completely opposite

	physical properties — one the hardest natural substance on Earth, the other a soft, slippery, electrically conducting solid. Over 13 lessons, students build from valence electrons and the octet rule through all major bond types (ionic, covalent, dative covalent, hydrogen, Van der Waals and metallic), drawing Lewis structures and conducting hands-on investigations into physical properties. The unit culminates in a project where learners build 3-D models of selected bonded structures using locally available Kenyan materials, connecting particulate-level bonding to the real-world uses of substances in Kenyan daily life, industry and the environment.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do diamond and graphite — both made of only carbon atoms — have such completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for? KICD KEY INQUIRY QUESTIONS: 1. How do valence electrons determine the type of bond formed between atoms? 2. How does the type of chemical bond in a substance determine its physical properties and uses?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Carbon Paradox: Diamond and Graphite A teacher displays two samples side by side: a brilliant, transparent diamond chip used in a Nairobi jeweller's cutting tool, and a stick of graphite from an ordinary pencil. Both are presented with the label "100% Carbon — no other elements." Students are invited to scratch each sample against a ceramic tile: the graphite smears a grey mark instantly while the diamond leaves a white scratch on the tile itself. A conductivity tester is then touched to each: the graphite lights a bulb brightly; the diamond does not conduct at all. The puzzle: If both substances contain only carbon atoms with the same number of protons, neutrons and electrons, why is one the hardest naturally occurring solid known while the other is soft enough to write with? Why does one conduct electricity and the other is a perfect electrical insulator? Why is one transparent and the other opaque grey? The answer lies entirely in HOW the carbon atoms are bonded to one another — four bonds in a 3-D tetrahedral network (diamond) versus three bonds in flat hexagonal layers with one delocalised electron per atom free to move (graphite). This phenomenon makes chemical bonding visible, tangible and genuinely puzzling, motivating the entire unit's investigation.
Storyline Thread	Lesson 1 — PHENOMENON LAUNCH & PREDICT PHASE: Students observe the diamond vs. graphite demonstration (hardness, conductivity, appearance). They record predictions for why the properties differ and populate an initial Driving Question Board. Teacher introduces the unit question. No bonding theory yet — pure curiosity generation. Lesson 2 — VALENCE ELECTRONS & THE OCTET RULE: Students review electron configuration and identify valence electrons for elements in periods 1–3. They discuss atomic stability (gaining, losing, sharing electrons) and the octet/duplet noble-gas rule as the driving force behind all bonding. Lewis dot diagrams for individual atoms are introduced. Lesson 3 — IONIC BONDING: Students investigate electron transfer using NaCl as the primary example (linking to salt used in Kenyan cooking and food preservation). They draw dot-and-cross diagrams for Na^+Cl^- formation, discuss lattice energy conceptually, and connect ionic bonding to the giant ionic structure. Students predict properties of NaCl before the Lesson 5 investigation. Lesson 4 — COVALENT & DATIVE COVALENT BONDING: Students draw Lewis structures for simple covalent molecules (H_2, O_2, N_2, H_2O, CO_2, CH_4) and explore single, double and triple bonds. Dative covalent bonding is introduced through NH_4^+ and H_3O^+

	— relating to acid-base chemistry. Students compare dot-and-cross diagrams for covalent vs. ionic substances. Lesson 5 — HYDROGEN BONDING & VAN DER WAALS FORCES: Students investigate why water has anomalously high boiling/melting points relative to H_2S and H_2Se. Hydrogen bonding between H_2O molecules is modelled. Van der Waals (London dispersion and dipole–dipole) forces are introduced using iodine and noble gases as examples. Students link these to the Supporting Phenomenon of iodine sublimation. Lesson 6 — METALLIC BONDING: Students examine aluminium foil, copper wire and iron nails — all common in Kenyan households and construction. The sea-of-delocalised-electrons model is introduced to explain conductivity, ductility and malleability. Students draw a metallic bonding diagram and compare it to ionic and covalent diagrams. They connect this to the aluminium nyama choma foil supporting phenomenon. Lesson 7 — PHYSICAL PROPERTIES INVESTIGATION (Part 1 — Setup & Prediction): Students are presented with four unknowns representing giant ionic (NaCl), simple molecular (sugar/iodine), giant covalent (sand, SiO_2) and giant metallic (aluminium) structures. They predict which substances will dissolve in water, conduct electricity and have high or low melting points — recording predictions in a structured table linked to bond type. Lesson 8 — PHYSICAL PROPERTIES INVESTIGATION (Part 2 — Data Collection): Students carry out the hands-on investigation: solubility tests in water, conductivity testing (solid and dissolved), and comparison of melting-point data from data cards. They record observations systematically and begin identifying patterns. Lesson 9 — PHYSICAL PROPERTIES INVESTIGATION (Part 3 — Analysis & Explanation): Students analyse their data, construct explanations for each substance's behaviour using bond-type reasoning, and complete structured written explanations. The class compiles a shared comparison table (giant ionic / simple molecular / giant covalent / giant metallic) on the board. Students return to the DQB to update answers. Lesson 10 — RESULTANT STRUCTURES & USES: Students explore how bond type and structure determine real-world uses — diamond in cutting tools, graphite as lubricant and in dry-cell batteries (common in Kenyan homes), NaCl in food preservation, aluminium in Kenyan construction and cookware, SiO_2 in glass-making. Students brainstorm additional Kenyan examples and present findings in pairs. Lesson 11 — ANIMATIONS, SIMULATIONS & DIGITAL LITERACY: Students watch curated animations and simulations on chemical bonding and resultant structures. They evaluate each simulation for accuracy against their Lewis diagrams and physical-property findings, discussing what the digital models show well and what they simplify. Teacher facilitates a structured debrief connecting digital evidence back to the anchoring phenomenon. Lesson 12 — PROJECT LESSON — 3-D MODEL BUILDING: Students use locally available materials (clay, bottle tops, wire, maize cobs, string, cardboard) to build 3-D models of assigned structures (NaCl ionic lattice, graphite layered structure, diamond tetrahedral network, or SiO_2). Groups annotate their models with bond type, bond angle and key physical properties. This is the creativity and environmental conservation PCI integration lesson. Lesson 13 — MODEL PRESENTATIONS, DQB FINAL REVISION & UNIT CONSOLIDATION: Groups present their 3-D models to the class, explaining how the bonding arrangement causes the physical properties and real-world uses of their substance. The

	class conducts a final Driving Question Board revision — answering the unit question fully with evidence. Teacher leads consolidation: returning to the diamond-vs.-graphite phenomenon with complete, evidence-based student explanations. Formative assessment through exit tickets linking bond type → structure → property → use.
Supporting Phenomena	– The Salt-and-Sugar Conductivity Puzzle: When dissolved in water, common table salt (NaCl — used daily in Kenyan cooking of ugali and stew) conducts electricity and splits into ions, but dissolved sugar does not conduct at all — both are white crystalline solids, yet their bond types (ionic vs. covalent) produce opposite electrical behaviour. – Water's Mysteriously High Boiling Point: Water boils at 100 °C — far higher than expected for such a small molecule — because of hydrogen bonding between H_2O molecules. This is why Lake Victoria retains warmth overnight and why githeri takes so long to boil at high-altitude Nairobi compared to Mombasa, linking hydrogen bonding to everyday Kenyan cooking experience. – Aluminium Foil — Bendable but Not Brittle: Aluminium foil used to wrap nyama choma can be bent, rolled and shaped without shattering, unlike NaCl crystals that crack when struck. Metallic bonding's sea of delocalised electrons allows layers to slide without breaking, explaining ductility and malleability in metals used in Kenyan construction and cookware. – Iodine — Solid that Sublimes and Doesn't Conduct: Iodine crystals (used in Kenyan health programmes for salt iodisation) are a shiny solid at room temperature but sublime easily and never conduct electricity in any state — demonstrating how weak Van der Waals forces between simple molecular structures produce low melting points and the absence of free charge carriers.

LESSON 1 (80 minutes): The Carbon Paradox: Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon — the Carbon Paradox — by having students observe striking differences between diamond and graphite, both made entirely of carbon atoms, and to generate genuine curiosity about how atomic bonding determines the properties and uses of substances.
Knowledge	Students will know that diamond and graphite are both allotropes of carbon (pure carbon, same atomic structure) yet exhibit dramatically different hardness, electrical conductivity, and appearance. Students will know that these differences cannot be explained by atomic composition alone, establishing that the arrangement and type of bonds between atoms must be responsible.
Skills	Students will be able to record structured scientific observations using sensory and instrument-based evidence; construct initial predictive explanations using prior knowledge of atoms and electrons; and collaboratively generate investigable questions for posting on a Driving Question Board.
Attitudes	Students will appreciate that puzzling, counterintuitive phenomena — like two substances made of identical atoms behaving completely differently — are the starting point for genuine scientific inquiry. Students will show care (Love) when handling demonstration materials so as not to hurt themselves or others.
Key Inquiry Question	How do valence electrons determine the type of bond formed between atoms? How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	This is the opening lesson of the Chemical Bonding unit. No bonding theory is introduced here. The sole purpose is to make the Carbon Paradox — diamond vs. graphite — so vivid and puzzling that students feel a genuine need to explain it. Every subsequent lesson in the unit will return to this phenomenon as students build the conceptual tools to resolve the paradox. The DQB opened today will be updated throughout all 26 lessons.
Safety Notes	Handle the ceramic tile carefully — edges can be sharp. The conductivity tester uses a low-voltage cell; ensure no exposed wires are touched. Students must not attempt to scratch the diamond chip with their fingers. The teacher conducts the scratch test as a front demonstration only. Remind students: 'Show care — avoid actions that could hurt yourself or a neighbour' (PCIs: Love value).

B. LESSON OVERVIEW
Teacher displays a diamond chip (used in a Nairobi jeweller's cutting tool) and a graphite pencil stick, both labelled '100% Carbon — no other elements.' Students observe a scratch test on a ceramic tile and a conductivity test with a simple bulb-and-battery tester. From these two dramatic, hands-on observations, students write individual predictions explaining the property differences, then share predictions in groups and post their burning questions on an opening Driving Question Board. No chemical bonding theory is taught. The lesson closes with the teacher formally posing the unit Driving Question and framing it as a mystery the class will spend the next 26 lessons solving.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before seeing any demonstration, students are shown two sealed, labelled sample bags: one containing a small graphite stick (from a standard Kenyan HB pencil), one containing a diamond chip (replica or genuine, mounted safely). Both bags carry the same label: '100% Carbon — same number of protons, neutrons and electrons per atom.' Students silently examine the bags (2 min), then individually write responses to three prompt questions in their science journals: (1) 'What do you think each sample will feel like or look like up close?' (2) 'Predict: if I scratch both against a rough tile, what will happen to each — and why?' (3) 'Predict: if I connect both to a simple electrical circuit with a bulb, what will happen — and why?' Students then turn-and-talk with a seat partner (2 min) to compare predictions before whole-class sharing.	Display both labelled samples at the front before any talking. Say exactly: 'Do NOT tell me what you know yet — I want your best PREDICTION based only on what you can see right now. Write in your journal. You have 3 minutes of silent thinking.' [Allow 3 minutes strict silence — do not circulate and comment; presence only.] After journaling, say: 'Now compare your prediction with your neighbour. You have 2 minutes.' [Circulate, listen, note 2–3 interesting or misconception-laden predictions for cold-calling.] After pair talk, cold-call exactly 4 students (select one high-confidence prediction, one low-confidence, one misconception, one unexpected idea): 'Amina, what did you and your partner predict about the scratch test — and what was your reasoning?' [Allow 10 seconds WAIT TIME after each student response before acknowledging.] Record ALL four predictions verbatim on the board under the heading 'Our Predictions — Day 1.' Do NOT correct any prediction. Say: 'Every one of these is a legitimate scientific hypothesis. We are going to test them right now.'	Individual Predict-Before-Observe Journaling — students activate prior knowledge about atoms and properties before seeing evidence, creating a cognitive 'hook' (a knowledge gap between prediction and observation) that drives motivation to explain. Recording predictions publicly without correction models scientific culture and gives students ownership of the inquiry.	Teacher scans journals during pair-talk circulation for the following observable indicators: (a) Does the student reference atomic structure (protons/electrons) in their prediction, showing they connect atomic identity to properties? (b) Does the student distinguish between the two samples at all, or do they predict identical behaviour? (c) Are predictions written as testable 'if-then' statements or as vague guesses? Teacher notes names of students who show strong prior-knowledge activation and those who conflate composition with structure — these students will be targeted for strategic cold-calls in Phase 3.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding	The teacher performs two sequential demonstrations at the front bench, narrated live. DEMO 1 — Scratch Test: Teacher firmly presses the graphite stick across a white unglazed ceramic tile. Students observe a broad grey smear. Teacher then presses the diamond chip (held in tongs/mount) across the same tile. Students observe a white scratch gouge in the tile surface itself — the diamond damages the tile	Before Demo 1, say: 'Your only job right now is to observe — not explain. Watch every detail.' Perform the scratch test slowly and deliberately, ensuring all students can see. After the graphite smear, hold up the tile and say: 'What do you SEE? Don't explain it — just describe it.' [10 seconds WAIT TIME. Take 2 non-volunteer observations via cold-call: 'Kipchoge, describe only what you see on the tile.'] After the	Structured Observation Table with Surprise Rating — separating 'observe' from 'explain' is a core NGSS Science and Engineering Practice (SEP 3: Planning and Carrying Out Investigations). The Surprise Rating column (1–5) gives students a low-stakes emotional entry point and surfaces which observations create the strongest cognitive dissonance — these become the highest-priority questions for the DQB. The muted	Teacher circulates during the 3-minute silent table-completion and checks for these observable indicators: (a) Does the student accurately record BOTH observations (scratch AND conductivity) for BOTH samples — or do they conflate, skip, or invent? (b) Are descriptions in observation language ('the bulb lit / did not light'; 'a grey mark appeared') rather than explanation language ('because graphite has free

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	while remaining undamaged. DEMO 2 — Conductivity Test: Teacher touches the probes of a simple 6V battery-bulb-and-wire conductivity tester to the graphite stick. The bulb lights up brightly. Teacher then touches the same probes to the diamond chip. The bulb does not light at all. Students record ALL observations in a structured 'Observation Table' in their journals with four columns: Property | Diamond | Graphite | My Surprise Rating (1–5). After recording (3 min), teacher projects a 4-minute video clip showing animated close-up footage of graphite layers sliding over each other and diamond's 3-D network (no audio explanation of bonding theory — visual only, narration muted or a neutral 'structure visualisation' clip). Students add any new observations to their table.	diamond scratch, say nothing for 5 full seconds — let the visual land. Then: 'Add that to your table. Surprise rating?' Before Demo 2, say: 'Now watch the bulb — not me.' Perform conductivity test. After graphite lights the bulb, say: 'Observation only — what happened? Otieno?' [10 seconds WAIT TIME.] After diamond fails to light the bulb, say: 'Complete your table in silence — 3 minutes.' During the video clip, instruct: 'Watch for anything about how the carbon atoms are arranged — no words needed, just look. Add anything new you notice to your observation table.' After the clip, say: 'Turn and tell your partner one thing you noticed in the video that surprised you — 1 minute.'	video adds a second sensory channel (visual structure) without pre-empting student sense-making.	electrons')? Students who slip into explanation language are gently redirected: 'That's a great idea — hold that for our next phase. Right now, just describe what your eyes saw.' Teacher also listens during pair-share after the video to identify students who have spontaneously noticed the layered vs. 3-D structure difference in the animation — these students will be invited to share in Phase 3.
Explain Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their original three predictions and annotate them in a different colour pen: circling any prediction that matched observation, crossing out any that did not, and writing one sentence beginning 'I was surprised that...' for any observation that contradicted their prediction. Students then work in groups of 4 to construct a 'Group Explanation Attempt' — a written paragraph (or annotated sketch) answering: 'Using only what you already know about atoms, electrons, and structure, propose the most reasonable explanation you can for why diamond is hard and non-conducting while graphite is soft and conducting — even though both are carbon.' Groups are explicitly told: 'You are not	Say: 'Go back to your predictions. In a different-coloured pen, annotate them — circle what matched, cross out what didn't, and write your surprise sentence. 4 minutes.' [Strict timer. Circulate and observe — do not intervene.] Then: 'In your groups of 4, you have 8 minutes to write your best explanation attempt. Remember — logical and evidence-based beats correct. Go.' [During group work, circulate to 3–4 groups. Use prompts only: 'What in your observation table is your explanation built on?' and 'What would have to be true about carbon atoms for your explanation to work?' — do not give answers.] Before gallery walk, say: 'When you read another group's explanation, leave a STAR if it	Predict-Observe-Explain (POE) Cycle completed with Gallery Walk — the annotation step creates explicit metacognitive reflection on the gap between prior knowledge and new evidence (NGSS SEP 6: Constructing Explanations). The gallery walk externalises group thinking, allows students to see the diversity of explanation attempts, and surfaces the questions that will populate the DQB. Crucially, the teacher frames incomplete explanations as the scientific norm, reducing anxiety and increasing intellectual risk-taking.	Observable indicators during group explanation writing: (a) Does the group's explanation reference the observation evidence (scratch / conductivity data), or is it purely speculative? (b) Does any group spontaneously invoke electrons, atomic arrangement, or structure — even informally? (c) Are question-mark sticky notes clustered around specific conceptual gaps (e.g., 'what are electrons doing?', 'how can the same atom make different things?') — teacher photographs the sticky note patterns for use in lesson 2 to sequence the DQB. Indicators of mastery: student annotation accurately identifies which predictions were confirmed/disconfirmed; explanation attempt is internally

	expected to know the answer yet. A good scientific explanation attempt is one that is logical and based on evidence — not one that is correct.' Groups share their best explanation attempt with the class in a gallery-walk format: each group's paragraph is posted on the wall, and students do a silent 3-minute walk to read all other groups' attempts, leaving a star (agreement) or question mark (something that puzzles them) sticky note on each.	matches your thinking, and a QUESTION MARK if it raises a new question for you. Silent gallery walk — 3 minutes. No talking.' After gallery walk, cold-call 3 students to share one question mark they left and why: 'Wanjiku, what question did you leave on Group 3's poster — and what made you uncertain?' [10 seconds WAIT TIME after each.] Close by saying: 'Notice that we have great observations, honest predictions, and brave explanation attempts — but no group has a complete explanation yet. THAT is exactly where good science begins. The question we all need to answer is coming up next.'		consistent with observed evidence even if scientifically incomplete.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives three sticky notes and is asked to write one question per note — questions that today's demonstration raised but did not answer. Sentence starters are provided on the board: 'I wonder why...', 'How can it be that...', 'What would happen if...' Students write individually (3 min), then bring their sticky notes to a large class DQB chart paper posted at the front. The teacher reads each question aloud as it is posted (or invites the student to read it), and the class helps cluster questions into emerging theme columns that the teacher labels collaboratively with students: e.g., 'Questions about electrons,' 'Questions about structure/arrangement,' 'Questions about properties,' 'Questions about uses.' The teacher then formally reads the Unit Driving Question aloud — 'Why do diamond and graphite — both made of only carbon atoms — have such completely different properties,	Distribute sticky notes and say: 'Write one genuine question per sticky note — questions that today left you truly puzzled. You have 3 minutes. These do not need to be 'smart' questions — they need to be YOUR questions.' [3 minutes silence. Circulate and observe; do not prompt content.] Invite students to the DQB one row at a time. As each student posts, say: 'Read your question aloud so we all hear it.' After 8–10 questions are posted, pause and say: 'Class — I'm seeing some questions that seem to be about the same big idea. Where shall we put this one?' [10 seconds WAIT TIME. Take volunteer or cold-call: 'Chebet — which column does Otieno's question belong in, and why?'] After all questions are posted, read the Driving Question aloud twice — once by teacher, once by the whole class chorally. Say: 'This board is OURS for the whole unit. Every time we learn something new, we will come back and	Driving Question Board (DQB) — a phenomenon-anchored question-tracking tool that makes the unit's intellectual trajectory visible to students throughout all 26 lessons. Clustering questions into themes previews the conceptual structure of the unit without didactic instruction. The collaborative categorisation act (NGSS SEP 1: Asking Questions and Defining Problems) gives students agency over the inquiry arc. Returning to the DQB in every subsequent lesson creates a coherent storyline and prevents isolated 'topic hopping.'	Observable indicators: (a) Do students' DQB questions address the phenomenon specifically (diamond vs. graphite, carbon, bonding, properties) — or are they generic/off-topic? Questions that reference today's demonstration directly indicate that the phenomenon launch was effective. (b) Do any students ask questions that anticipate unit content — e.g., 'Do the electrons move between atoms?' or 'Is there a way to draw how atoms connect?' — indicating strong prior knowledge activation? (c) Are questions framed as investigable (testable/researchable) or as statements in question form? Teacher notes the 3–5 most 'generative' questions (those that will drive the most lesson sequences) and circles them on the DQB to revisit in Lesson 2's opening.

	and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for?' — and posts it at the top of the DQB. Students identify which of their questions are captured by the driving question and which might be answered in a different unit.	answer a question, or add a new one. Nothing on this board is wrong. Every question here is a reason we are studying chemical bonding.'		
Model Building Phase 📹 **VIDEO:** [Elements and atoms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students draw an 'Initial Model' of what they think is happening at the atomic level inside diamond and inside graphite — using only pencil and paper. The prompt is: 'Draw what you imagine the carbon atoms look like inside each substance. How are they connected? Are there any particles moving freely? Use circles, lines, dots — whatever symbols make sense to you. Label your drawing.' This is explicitly framed as a 'before' model — students sign and date it, and it is stored in a dedicated 'Model Envelope' that will be returned at the end of the unit for comparison with their final model. Students who finish early are invited to annotate their model with a sentence: 'The biggest thing my model cannot yet explain is...' The lesson closes with a 2-minute pair-share of models, followed by the teacher's closing statement linking the model to the unit journey ahead.	Say: 'Final task — and this one is personal. In your journal, draw your initial model of carbon atoms inside diamond and inside graphite. There is no correct answer today. I want to see YOUR thinking — however rough. You have 6 minutes.' [6-minute timer. Play soft background music if available. Circulate silently and observe without commenting on accuracy.] After 6 minutes: 'Now pair-share your model — explain your drawing to your partner in 1 minute each. What did you draw, and why?' [2 minutes pair-share.] Bring class back together and say: 'I am now going to collect your models in envelopes. You will see them again at the end of this unit — Lesson 26. I predict you will be amazed at how much your thinking changes. That change IS the learning.' Close with: 'Today you observed something genuinely strange — two substances made of exactly the same atom, behaving like they come from different planets. A Nairobi jeweller uses diamond to cut the hardest materials. A Kericho student uses graphite to write their exam answers. Same atom. Wildly different uses. The reason is chemical bonding — and that is what we will spend the next 25 lessons figuring out together.'	Initial Model Construction — a core NGSS Science and Engineering Practice (SEP 2: Developing and Using Models). The 'Model Envelope' strategy creates a tangible artefact of students' initial understanding that can be retrieved and compared at the unit's end, making conceptual growth visible and personally meaningful. Drawing forces students to commit to a spatial, structural representation — not just words — which reveals the depth (or gaps) of their atomic-level thinking. Framing it as a 'before' model removes the pressure of correctness and positions revision as the goal.	Observable indicators in initial models: (a) Does the student draw any distinction between diamond and graphite models — or draw identical structures for both? (b) Does the model show any attempt at representing connections/bonds between atoms (lines, shared regions), or only isolated circles? (c) Does the model include any sub-atomic particles (electrons, dots) — indicating prior knowledge of atomic structure from earlier learning? (d) Does the 'cannot yet explain' annotation identify a genuine conceptual gap (e.g., 'I don't know why one conducts and one doesn't't') rather than a surface observation? Teacher collects all model envelopes and reviews them before Lesson 2 to identify the 3 most common structural misconceptions to address in the bonding theory sequence.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 2:

1. PHENOMENON UPTAKE: Did the scratch test and conductivity demonstration generate genuine surprise and curiosity — or did students seem unmoved? If uptake was low, consider repeating the conductivity demo with student volunteers in Lesson 2's opening.
2. DQB QUALITY: Review the sticky-note questions. Are they anchored to the phenomenon (diamond/graphite/carbon/bonding), or are they generic? If most questions are surface-level ('Why is diamond expensive?'), plan a structured question-refinement activity at the start of Lesson 2 using the 'I wonder why... because...' frame.
3. PREDICTION MISCONCEPTIONS: What were the most common misconceptions in student predictions? Common ones include: 'Diamond is harder because it has more atoms,' 'Graphite conducts because it is black/dark,' or 'They must have different elements inside.' Document these — they are the exact alternative conceptions that the bonding theory sequence must explicitly address.
4. MODEL DIVERSITY: After reviewing the model envelopes, do students show any spontaneous structural thinking (layers, networks, arrangements), or are all models showing isolated atoms? If models show no structural thinking at all, plan to open Lesson 2 with a brief 'arrangement matters' analogy (e.g., same maize grains — loose in a bag vs. tightly packed in a cob — behave very differently in cooking) before introducing atomic structure.
5. EQUITY CHECK: Were all four cold-called students able to respond, or did wait-time need to be extended? Were students from all seating areas included in questioning? Adjust cold-call strategy in Lesson 2 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that graphite (pencil stick, labelled 100% Carbon) smeared a grey mark on a ceramic tile and lit a conductivity bulb brightly, while diamond (also labelled 100% Carbon) scratched and gouged the tile surface without being damaged, and did not conduct electricity at all. Students also viewed a silent animation showing the contrasting atomic arrangements of both allotropes.
What did I learn?	Students learned that diamond and graphite are both made entirely of carbon atoms — the same element, same atomic structure — yet have completely opposite physical properties: hardness vs. softness, electrical insulator vs. conductor, transparent vs. opaque. Students learned that differences in properties between substances made of the same atoms must be caused by differences in how those atoms are arranged and bonded — not by differences in what atoms are present. Students also learned that writing initial models and posting genuine questions on a Driving Question Board are core scientific practices.
How does this explain the phenomenon?	No bonding theory was explained in this lesson. The lesson was deliberately designed to open a knowledge gap — to make students feel the puzzle — rather than to resolve it. The full explanation (tetrahedral covalent network bonding in diamond vs. layered hexagonal covalent bonding with delocalised electrons in graphite) will be built across the unit as students learn about valence electrons, bond types, Lewis structures, and the relationship between structure and physical properties.

LESSON 2 (80 minutes): Valence Electrons & the Octet Rule: The Electron Bookkeeping Behind the Carbon Paradox	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To establish valence electrons and the octet/duplet rule as the foundational driving force behind all chemical bonding, and to represent individual atoms using Lewis dot diagrams.
Knowledge	Learners identify valence electrons for elements in Periods 1–3 using electron configuration; state the octet rule (and duplet rule for Period 1); explain why atoms gain, lose, or share electrons to achieve noble-gas stability; and draw Lewis dot diagrams for individual atoms (H, He, Li, C, N, O, F, Ne, Na, Mg, Al, Si, P, S, Cl, Ar).
Skills	Learners write electron configurations, extract valence electron counts, construct Lewis dot diagrams accurately, and use dot diagrams to predict how many bonds an atom is likely to form — connecting this directly to carbon's four valence electrons in both diamond and graphite.
Attitudes	Learners appreciate that a single rule — atoms 'want' a full outer shell — underlies the enormous diversity of substances in the world, including the carbon paradox of diamond and graphite. Learners show care (Love) by working collaboratively without disrupting peers during group activities.
Key Inquiry Question	How do valence electrons determine the type of bond formed between atoms?
Purpose in Storyline	Lesson 1 presented the puzzle: diamond and graphite are both pure carbon, yet behave oppositely. Before we can explain HOW carbon bonds differently in each, learners must master the WHY — atoms bond because their outer electron shells are incomplete. Carbon's four valence electrons (half-filled outer shell) make it uniquely versatile: it can form four covalent bonds in any geometry it 'needs.' This lesson gives learners the electron-counting tools (configurations, valence electrons, Lewis dot diagrams, octet rule) that every subsequent lesson in the unit will rely on. By the end, learners should be able to look at the carbon Lewis dot diagram and articulate exactly why carbon can bond four times — the key to unlocking the diamond vs. graphite paradox.
Safety Notes	No hazardous chemicals are used in this lesson. If physical atomic model kits are used, small beads or clay pieces should be kept off the floor to prevent slipping. Learners with visual impairments should be seated near the demonstration area. Remind learners of the value of Love — handle shared materials carefully and return them intact.

B. LESSON OVERVIEW

Learners open by revisiting the carbon paradox phenomenon (diamond vs. graphite) and are challenged with a bridging question: 'Carbon has 6 electrons — but which electrons actually do the bonding?' This motivates a structured review of electron configuration for Periods 1–3, leading to the concept of valence electrons. Through a guided group activity using Lewis dot diagrams drawn on mini-whiteboards, learners discover the octet/duplet pattern across Period 3. They then apply the octet rule to predict how many bonds each atom needs to 'complete' its outer shell. The lesson closes with learners annotating a shared Driving Question Board entry and revising their initial mental model of the carbon atom to show its four unpaired valence electrons — a direct bridge to Lesson 3's exploration of covalent bonding in graphite and diamond.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Noble gas configuration] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Valence Electrons (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the two carbon samples from Lesson 1 (diamond chip and graphite pencil) briefly re-displayed at the front. The teacher poses a written prompt on the board: 'Carbon's atomic number is 6. Write its electron configuration, then circle the electrons you think are responsible for bonding. Predict: how many bonds can one carbon atom form?' Learners work silently and individually for 4 minutes in their exercise books, drawing a sketch of a carbon atom and annotating which electrons they believe are 'bonding electrons.' They then share their prediction with a shoulder partner (Think-Pair-Share, 2 minutes). Pairs are cold-called to share predictions with the class; the teacher records 3–4 different predictions on the board without evaluating them, labelling them 'Prediction A, B, C…' for later revisiting. Learners also complete a quick Entry Ticket: a 2-column table with headers 'What I remember about electron configuration' and 'What I am unsure about.'	Re-display the diamond and graphite samples from Lesson 1. Say verbatim: 'Last lesson we saw that diamond and graphite are both 100% carbon. Today we go one level deeper — we are going to look inside the carbon atom itself and figure out which electrons are actually doing the bonding work.' Write the Predict prompt on the board and read it aloud. Set a visible 4-minute timer. Circulate silently — do NOT correct or hint. After 4 minutes, say: 'Turn to your shoulder partner. You have 90 seconds — share your prediction and listen to theirs.' After 90 seconds, cold-call exactly 3 pairs using the random name stick. For each response, say: 'Thank you — I am writing that on the board as Prediction [A/B/C]. We will come back to these.' Do NOT affirm or correct any prediction. Collect Entry Tickets as a quick scan (read 5–6 at random) to gauge prior-knowledge gaps before proceeding.	Think-Pair-Share with Prediction Board: By externalising predictions publicly without evaluation, the teacher creates cognitive dissonance — learners who gave different answers are motivated to find out whose prediction was correct. Recording predictions on the board also creates accountability; learners will revisit and self-correct them later in the lesson, reinforcing metacognitive awareness of their own learning.	Observable indicator: Each learner produces a written prediction in their book (electron configuration sketch + bond-count prediction) within 4 minutes. The teacher scans Entry Tickets for common misconceptions (e.g., 'all 6 electrons bond,' 'only 2 electrons bond'). If more than 50% of Entry Tickets show confusion about electron shells vs. subshells, the teacher inserts a 3-minute bridging micro-lecture on shell notation (2,8,8 rule) before Phase 2.
Observe Phase 📹 **VIDEO:** [Noble gas configuration] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Valence Electrons (Flexbook)]	The teacher distributes a structured worksheet: 'Mapping Outer Electrons Across Periods 1–3.' The worksheet lists 16 elements (H, He, Li, Be, B, C, N, O, F, Ne, Na, Mg, Al, Si, P, S, Cl, Ar) in a table with columns: Atomic Number | Full Electron Configuration | Number of Occupied Shells | Electrons in Outermost Shell (= Valence Electrons) | Noble Gas? (Y/N). Learners complete the table in groups of four (8 minutes), using	Distribute the worksheet. Say: 'In your groups of four, complete the full table. Use your Periodic Table and your knowledge of electron configuration from Grade 9. You have 8 minutes.' Set timer. Circulate, pausing at each group for 30–45 seconds. Listen for the misconception that transition metals' d-electrons are valence electrons — if heard, say: 'For now, we are only working with main-group elements in Periods 1–3. Note that for these elements,	Data Table Construction with Colour-Coding: Building the table collaboratively makes the periodic pattern of valence electrons an observed, data-driven discovery rather than a told fact. Colour-coding (red circles) creates a visual anchor. The 'reset' question bridges the table to the periodic law, connecting valence electrons to the structure of the Periodic Table — which learners will need again in Sub-Strand 1.5 (Periodicity).	Observable indicator: Completed worksheet tables are collected at the end of Phase 2. The teacher checks specifically that carbon is correctly assigned 4 valence electrons (configuration 2,4) and that learners are NOT counting inner-shell electrons. A quick show-of-fingers check ('Hold up the number of valence electrons in a chlorine atom') allows instant whole-class diagnosis before moving to Phase 3.

Source: CK-12 🔍 Search ARES for similar readings	their Grade 9 notes and the Periodic Table provided. Groups then share their completed rows for Period 2 only, with one group scribe writing agreed answers on the class board. The teacher uses a colour-coded approach: valence electrons are circled in RED on the board configuration. The class reads aloud the valence electron counts across Period 2 (1,2,3,4,5,6,7,8) and the teacher asks: 'What do you notice about where this sequence resets?' (Answer: at Na, Period 3 starts again at 1.) Learners annotate their worksheets with the pattern. The teacher then introduces the term 'valence electrons' formally and writes the definition on the board: 'Valence electrons are the electrons in the outermost energy level of an atom — they are the electrons involved in chemical bonding.'	valence electrons are always in the s and p subshells of the outermost shell.' After 8 minutes, cold-call one group to give Period 2 answers row by row. After each row, ask the class: 'Does your group agree? Thumbs up, thumbs down, or sideways.' Correct errors immediately on the board. Circle all valence electrons in red marker. Then ask — wait 10 seconds — cold-call 2 students: 'Akinyi, what pattern do you see in the valence electron count as we move across Period 2?' and 'Kamau, why do you think the count resets when we get to sodium?' Write the formal definition of valence electrons on the board and instruct learners to copy it with a star (*) for importance.		
Explain Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Valence Electrons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the concept of atomic stability through a relatable Kenyan analogy: 'Think of a matatu with 8 seats. Atoms are most stable — most comfortable — when their outer shell has 8 electrons, like a full matatu. Atoms with fewer than 8 will do whatever it takes — gain, lose, or share passengers — to fill those seats.' (For Period 1: the 'matatu' only has 2 seats — the duplet rule.) Learners discuss in pairs for 2 minutes: 'Using the matatu analogy, how many electrons does carbon need to gain, lose, or share to fill its outer shell? What about sodium? What about chlorine?' After the pair discussion, the teacher demonstrates drawing Lewis dot diagrams on the board for H, He,	Introduce the matatu analogy with enthusiasm: 'Imagine the outer electron shell as a matatu with 8 seats. Nairobi matatus are not happy until they are full — and neither are atoms.' Pause for laughter/engagement. Write on board: 'OCTET RULE: Atoms react to achieve 8 valence electrons (or 2 for Period 1 — the duplet rule), matching the nearest noble gas.' Direct pair discussion for 2 minutes on the matatu-analogy question. Then say: 'Watch me draw a Lewis dot diagram for carbon. Every step matters — copy this exactly.' Draw on board slowly: symbol → 4 sides → place one dot per side first → then add remaining dots. Draw H, He, C, Na explicitly. Say: 'Now on your mini-whiteboards, draw the Lewis dot	Analogy Anchoring (Matatu Analogy) + Iterative Diagram Practice: The matatu analogy ties an abstract quantum-mechanical concept to a culturally immediate Kenyan experience, lowering the cognitive load for new vocabulary. Iterative whiteboard practice (draw → show → correct → repeat) provides immediate formative feedback cycles, ensuring procedural accuracy before conceptual application. The explicit connection of carbon's 4 unpaired dots back to the anchoring phenomenon maintains narrative coherence across the lesson.	Observable indicator: Mini-whiteboard show — the teacher can see in real time whether Lewis dot diagrams are correctly drawn (one dot per side before pairing; correct count of valence electrons). Specific error to watch for: placing all dots on one side, or drawing 6 dots for carbon instead of 4. The 2-sentence written explanation ('Carbon has 4 valence electrons, so it can form 4 bonds') is collected as an exit-level indicator of conceptual understanding, distinguishing procedural drawing skill from conceptual grasp.

	C, and Na step by step, narrating each step aloud. Learners copy each diagram and then independently draw Lewis dot diagrams for N, O, F, Ne, Mg, Al, Si, P, S, Cl, Ar on mini-whiteboards (or in books). Pairs peer-check each other's diagrams against the rule: 'Dots must be placed one at a time on each of the four sides before pairing.' The teacher then returns to the board and, using the carbon Lewis dot diagram (showing 4 unpaired dots), asks: 'Carbon has 4 unpaired valence electrons. What does that tell us about how many bonds it can form?' After 10 seconds wait time, cold-call 3 students. Learners write a 2-sentence explanation in their books connecting carbon's 4 valence electrons to its bonding capacity. The teacher closes by writing on the board: 'The octet rule is the ENGINE that drives all chemical bonding — every bond type we will study (ionic, covalent, metallic) is simply a different STRATEGY atoms use to achieve a full outer shell.'	diagram for nitrogen. You have 90 seconds. Go.' Circulate and observe. After 90 seconds, say: 'Hold up your boards.' Scan the room. Address the most common error seen. Continue for O, F, Ne, then Period 3 elements. After all diagrams, point to the carbon diagram and ask: 'Carbon has four dots — all unpaired. Wanjiku, what does that tell us?' Wait exactly 10 seconds. Cold-call Wanjiku, then 2 more students. Affirm and build: 'Exactly — carbon is primed to form exactly four bonds. Remember that number — it is the key to the carbon paradox.' Write the closing statement on the board and ask learners to copy it and underline 'ENGINE.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Noble gas configuration] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Valence Electrons (Flexbook)]	Learners return to the class Driving Question Board established in Lesson 1. Each learner receives a sticky note in a designated colour (e.g., yellow = 'new understanding,' blue = 'new question'). On their yellow note, learners write one specific thing they now understand that they did not understand at the start of the lesson — it must reference either valence electrons, the octet rule, or Lewis dot diagrams, AND must connect to the diamond/graphite paradox. On their blue note, they write one new question the lesson	Distribute sticky notes (two per learner). Say: 'The Driving Question Board is our living record of what we are figuring out together. It is not decoration — every note must be specific and must connect to diamond or graphite or carbon bonding.' Give 4 minutes for silent writing. Circulate — if a learner writes something vague like 'I learned about valence electrons,' prompt: 'Can you connect that to carbon specifically? How many valence electrons does carbon have and what does that mean for bonding?'	Two-Colour DQB Protocol: Separating 'new understanding' (yellow) from 'new questions' (blue) makes the iterative, incomplete nature of scientific knowledge explicit — understanding and questioning are both valued outcomes. Reading blue-note questions aloud validates curiosity and creates a forward narrative pull into subsequent lessons, maintaining engagement across the 13-lesson sequence.	Observable indicator: Yellow notes must contain at least one specific scientific term (valence electrons, octet rule, Lewis dot diagram, 4 bonds) AND a reference to carbon, diamond, or graphite to be counted as satisfactory. The teacher reviews all posted yellow notes after class — any that are too vague are flagged for follow-up with those learners in Lesson 3's opening.

Source: CK-12 🔍 Search ARES for similar readings	has raised. Example yellow note: 'Carbon has 4 valence electrons and 4 unpaired dots — that is why it can form 4 bonds in diamond AND in graphite.' Example blue note: 'If carbon can form 4 bonds, why does graphite only form 3 bonds per carbon atom? Where does the 4th electron go?' Learners post their notes on the DQB under the headings 'What We Now Understand' and 'What We Still Wonder.' The teacher reads 3–4 blue notes aloud and says: 'These are excellent questions — they are exactly what Lessons 3 and 4 will answer.' Learners record in their books: 'DQB Update — Lesson 2: [their yellow note text].'	After posting, read 3 blue-note questions aloud, select ones that preview covalent bonding (Lesson 3) and metallic/delocalized electrons (graphite). Say verbatim: 'This question — [read it] — is going to be the first thing we investigate in Lesson 3. So whoever wrote this: well done, you are already thinking like a chemist.' Photograph the DQB for the ARES platform record.		
Model Building Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Valence Electrons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the initial annotated diagram of the carbon atom they drew in Lesson 1 (stored in their Science Investigation Journals or exercise books). Using a different colour pen, they now revise their model to: (1) correctly label the two inner electrons (shell 1) and four outer electrons (shell 2) using the correct electron configuration notation (2,4); (2) draw the four valence electrons as Lewis dots around the carbon symbol; (3) write a caption: 'Carbon has 4 unpaired valence electrons — it needs 4 more electrons to complete its octet. It will SHARE electrons to bond.' Learners annotate with an arrow: 'These 4 electrons are responsible for ALL the differences between diamond and graphite.' In pairs, learners compare their revised models and check each other for accuracy (correct dot placement, correct electron count). The teacher	Say: 'Take out the carbon atom diagram you drew in Lesson 1. You are going to upgrade it using everything you learned today. Use a different colour so we can see your thinking grow.' Give 5 minutes for individual revision. Circulate and look for: correct 2,4 configuration, 4 Lewis dots, caption referencing sharing. After 5 minutes, select two books to display on the visualiser (ask permission). Say: 'Here are two models. I am not telling you which is more accurate yet. In pairs, discuss: What is correct in Model A? What would you change in Model B?' Wait 90 seconds. Cold-call 2 pairs. Then reveal which model is more accurate and why. Close by saying: 'Every lesson in this unit, you will come back to this model and add to it. By Lesson 13, your model will explain everything about diamond, graphite, salt, metals — and much more.' Direct learners to write in their journals:	Cumulative Model Revision: Returning to the same carbon atom diagram each lesson and revising it in a new colour creates a visible, layered record of conceptual growth — a hallmark of NGSS Science and Engineering Practice 2 (Developing and Using Models). The anonymous two-model critique builds evaluative reasoning and normalises the idea that models are always improvable, not definitively correct, reflecting authentic scientific practice.	Observable indicator: The revised carbon diagram must show: (a) 2,4 electron configuration labelled, (b) exactly 4 Lewis dots around the carbon symbol, (c) a written caption connecting the 4 valence electrons to the carbon paradox. The journal sentence stem 'I changed/added _____ because _____' is a metacognitive writing indicator. The teacher collects 6 journals at random at the end of class to check depth of caption reasoning — this informs the opening review in Lesson 3.

	selects two contrasting models (one strong, one with a common error) to display anonymously on the visualiser and leads a 3-minute whole-class critique using the prompt: 'What is correct in this model? What would you improve?'	'Model Revision — Lesson 2: I changed/added _____ because _____.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your ARES teacher journal:

1. PREDICTION ACCURACY: How many of the board predictions (A, B, C…) from Phase 1 were revised by learners by the end of the lesson? Did learners explicitly self-correct, or did they need prompting? If most predictions were not revisited, restructure Phase 4 to include a formal 'Prediction Revisit' step.
2. ANALOGY EFFECTIVENESS: Did the matatu analogy land well — did it reduce confusion or introduce new misconceptions (e.g., 'atoms move around like passengers')? If the matatu analogy caused confusion, consider replacing it with a githeri analogy: 'Atoms are like a githeri pot — they are most stable when the pot is full (8 beans in the outer ring).' Adapt for next lesson.
3. LEWIS DOT DIAGRAM FLUENCY: During the mini-whiteboard show in Phase 3, what was the most common error? (Likely: incorrect dot placement order, or drawing the wrong number of dots for carbon/nitrogen.) Plan a 5-minute warm-up at the start of Lesson 3 to address the top two errors seen.
4. DQB QUALITY: Were the blue-note questions specific enough to preview covalent bonding (Lesson 3)? If most blue notes were vague ('I wonder how bonding works'), the DQB protocol needs more scaffolding — consider providing a sentence frame: 'I now understand that carbon has ___ valence electrons, which means ___. I still wonder ___.'
5. MODEL REVISION DEPTH: Did learners' Phase 5 captions go beyond 'carbon has 4 valence electrons' to explain WHY this means it forms 4 bonds? If captions were superficial, open Lesson 3 by reading two exemplary captions aloud (anonymised) and asking the class what makes them strong.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners completed an electron configuration table for 16 elements across Periods 1–3, colour-identified valence electrons in red, and noticed the periodic reset pattern from Period 2 to Period 3.
What did I learn?	Valence electrons are the outermost-shell electrons that participate in bonding; atoms gain, lose, or share electrons to achieve an octet (or duplet for Period 1) matching the nearest noble gas; carbon has exactly 4 valence electrons shown as 4 unpaired Lewis dots, priming it to form 4 bonds.
How does this explain the phenomenon?	Carbon's 4 unpaired valence electrons mean it can form 4 bonds — the octet rule is the engine driving this. In diamond and graphite, carbon uses all 4 valence electrons in different bonding geometries, which is why two substances made of identical atoms behave so differently.

LESSON 3 (80 minutes): Ionic Bonding: Electron Transfer and the Giant Ionic Structure	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how electron transfer between atoms produces ionic bonds, and to connect the giant ionic lattice structure of sodium chloride to its observable physical properties.
Knowledge	Learners will know that: (1) ionic bonds form by complete transfer of one or more valence electrons from a metal atom to a non-metal atom; (2) the resulting oppositely charged ions (cations and anions) are held together by strong electrostatic attraction in a giant ionic lattice; (3) lattice energy is the energy released when gaseous ions pack into a lattice, and conceptually explains why ionic compounds are stable solids; (4) the giant ionic structure of NaCl accounts for its high melting point, brittleness, solubility in water, and ability to conduct electricity only when molten or dissolved.
Skills	Learners will be able to: (1) draw accurate dot-and-cross diagrams showing electron transfer in NaCl formation; (2) represent Na^+ and Cl^- ions correctly, including charges and electron configurations; (3) predict physical properties of NaCl from its bonding and structure; (4) connect ionic bonding back to the anchoring phenomenon by explaining why NaCl is neither as hard as diamond nor as soft and conductive as graphite.
Attitudes	Learners appreciate (Value: Love — showing care in group work) how the arrangement of electrons in everyday substances like table salt — used daily in Kenyan cooking of ugali, nyama choma, and sukuma wiki — is directly responsible for observable properties. Learners also appreciate (PCI: Social cohesion) how discussing ideas with peers deepens understanding of invisible sub-atomic events.
Key Inquiry Question	How do valence electrons determine the type of bond formed between atoms? (KICD Key Inquiry Question 1)
Purpose in Storyline	In Lesson 1 students were puzzled by the Carbon Paradox and built the Driving Question Board. In Lesson 2 they reviewed valence electrons and the octet rule. Now in Lesson 3 — the first bond-type lesson — students investigate ionic bonding using NaCl as their primary case study. This is the first of three bond-type lessons (ionic → covalent → metallic) that will collectively explain WHY diamond and graphite are so different. By the end of this lesson students will have a working model of electron transfer and a prediction set for Lesson 5's ionic properties investigation, advancing the class's collective answer to the driving question.
Safety Notes	No hazardous chemicals are used in this lesson. If using a conductivity tester (connecting lesson to phenomenon), ensure wires are insulated and the power source does not exceed 6 V. Handle ceramic tiles carefully — edges can be sharp. Learners with cuts on hands should not handle salt solutions. Remind learners of the Value of Love: handle all apparatus and group members' ideas with care.

B. LESSON OVERVIEW

In this 80-minute lesson, learners first revisit the anchoring phenomenon (diamond vs. graphite) and are challenged to think about a third carbon-free substance — common table salt (NaCl) — and how its bonding might differ from both carbon allotropes. Through a structured predict–observe–explain sequence, learners investigate electron transfer by modelling NaCl formation with electron-transfer cards, draw dot-and-cross diagrams, watch a short simulation of lattice formation, and predict the macroscopic properties that emerge from the giant ionic structure. The lesson closes with learners updating the Driving Question Board and revising their cumulative bonding model. All activities are anchored to the driving question: the class is building a toolkit of bond-type knowledge that will ultimately explain the Carbon Paradox.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three labelled samples on the demonstration desk: the diamond chip, the graphite pencil stick (both from the anchoring phenomenon), and a small transparent crystal of table salt (NaCl) — labelled 'iodised salt, used in ugali preparation.' The teacher poses the Predict Prompt on the board: 'NaCl contains sodium (a metal) and chlorine (a reactive gas) — yet together they form a harmless white solid we eat every day. Before we look at any diagrams, predict: (a) What do you think happens to electrons when sodium and chlorine bond? (b) Will NaCl be hard or soft? Conduct electricity or not? Why?' Learners write individual predictions in their ARES prediction journals for 4 minutes, then share with a shoulder partner for 2 minutes. Three pairs are cold-called to share predictions with the class. Predictions are recorded on a sticky-note column on the Driving Question Board labelled 'Our predictions about ionic bonding — Lesson 3.'	Display the three samples prominently. Say verbatim: 'Look at this salt crystal. Yesterday a farmer in Kericho probably used salt just like this to preserve nyama choma. But what is actually happening between sodium and chlorine atoms inside this crystal? Write your prediction now — no wrong answers at this stage.' WAIT 4 minutes (individual writing). Then: 'Share with your partner — 2 minutes.' WAIT 2 minutes. Cold-call exactly 3 pairs using a random name stick. Record each prediction verbatim on the board without correcting. Ask: 'Do we all agree? Thumbs up, sideways, or down.' Note the spread of agreement for formative data. Before moving on, say: 'We are going to find out whether your predictions are right — keep them visible because we will come back to them at the end of the lesson.'	Prediction Journaling with Public Commitment — by writing and then publicly sharing predictions, learners activate prior knowledge about metals, non-metals, and electron configuration (from Lesson 2) and create a cognitive 'hook' that makes subsequent evidence more memorable. Recording predictions on the DQB creates accountability and drives the need to explain.	Observable indicator: Teacher scans written predictions during cold-call sharing. Look for: (1) whether learners reference valence electrons or electron transfer (shows connection to Lesson 2); (2) whether any learners spontaneously mention charges or ions (indicates prior knowledge to leverage); (3) the proportion of thumbs-up vs. thumbs-down after peer share (gauges class confidence). Record two or three specific misconceptions heard (e.g. 'they share electrons') to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12	Activity A — Electron Transfer Card Model (10 min): Each group of four receives an envelope containing: one 'Na atom' card (showing 2,8,1 electron shells), one 'Cl atom' card (showing 2,8,7 electron shells), and 11 small circular 'electron' tokens. Groups physically move the single outer electron token from the Na card to the Cl card, observe that Na now has a full outer shell (2,8) and Cl now has a full outer shell (2,8,8), then label each card with its resulting ion charge (Na^+ and Cl^-).	Distribute envelopes before the lesson starts to save time. Say: 'Your job right now is NOT to explain — just to observe and record what you see happening to the electrons.' During Activity A, circulate to all six groups within 5 minutes. At each group ask: 'Which atom gave up the electron? Which atom received it? How do you know?' WAIT 10 seconds after each question before accepting an answer. For Activity B, pause the simulation at each marked timestamp (use a sticky note on	Concrete-Representational-Abstract (CRA) Progression — the physical electron token cards (concrete) bridge to the simulation (representational/visual) before learners encounter formal dot-and-cross diagrams (abstract). This progression is especially important for the sub-atomic scale, which learners cannot directly observe. The simulation also makes the lattice's 3-D geometry visible, pre-loading understanding needed for the Model Building phase.	Observable indicator: During Activity A, check that all groups correctly place the electron token on the Cl card (not the Na card) and write 'Na^+' and 'Cl^-' (with correct charges). Common error to watch for: groups writing 'Na^-' or failing to write charges at all — intervene immediately with the prompt 'If sodium loses a negative electron, what sign does sodium's overall charge become?' During Activity B, collect one observation journal per group after the simulation; scan for the three

🔍 Search ARES for similar readings	Groups record: 'What changed? What stayed the same?' in their ARES observation journals. Activity B — Simulation (10 min): The teacher plays a 3-minute PhET-style ionic bonding simulation (or ARES offline animation) showing Na and Cl atoms approaching, electron transfer occurring, and the resulting ions packing into a 3-D lattice. Learners watch once, then watch a second time pausing at three key moments: (i) just before transfer, (ii) just after transfer (ions formed), (iii) lattice assembled. At each pause learners write one observation sentence. Groups then compare observation sentences with an adjacent group.	the screen edge as a cue). At pause (ii) ask the class: 'What is the charge on sodium now? Chlorine now? How do you know?' Cold-call 2 learners who have not yet spoken today. At pause (iii) say: 'Look at the pattern of ions in the lattice. How is this arrangement similar to — or different from — the 3-D tetrahedral network we described for diamond in Lesson 1?' WAIT 15 seconds. Accept 2 responses.		required observation sentences — flag any group that wrote fewer than three.
Explain Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Part A — Dot-and-Cross Diagrams (12 min): Learners individually draw a complete dot-and-cross diagram for NaCl formation in their ARES journals, following a worked example on the board that the teacher builds step-by-step. The diagram must show: (i) Na atom with 1 dot (valence electron), (ii) Cl atom with 7 crosses (valence electrons), (iii) the transfer arrow, (iv) Na^+ ion (empty outer shell, square brackets, charge), (v) Cl^- ion (full outer shell of 8, square brackets, charge), (vi) the electrostatic attraction arrow labelled 'ionic bond.' Learners then attempt a second diagram independently: MgO (magnesium oxide — used as a refractory lining in Kenyan steel plants in Mombasa). Part B — Conceptual Lattice Energy (5 min): Teacher uses the analogy of the Nairobi City Market — 'Imagine thousands of traders (ions) packed into a tight, organised market structure. It	For Part A: Build the dot-and-cross diagram on the board in silence for the first 60 seconds, then narrate each step aloud. Say: 'I am using dots for sodium's electrons and crosses for chlorine's electrons so we can track which electrons belong to which atom — this is the convention used in all Kenyan Chemistry examinations.' After completing the NaCl example, say: 'Now draw MgO independently. Magnesium is in Group 2 — how many valence electrons does it have? Chlorine— sorry — oxygen is in Group 6 — how many does it have? Begin.' WAIT 4 minutes. Cold-call 1 learner to draw their MgO diagram on the board. Class gives a 'glow and grow' (one praise, one suggested improvement). For Part B: Tell the Nairobi City Market analogy, then ask: 'Using this analogy, why does NaCl have a high melting point?' WAIT 10 seconds. Cold-call 2 learners. For Part C: Circulate as	Worked Example → Guided Practice → Independent Transfer (I Do – We Do – You Do) — the teacher models NaCl, the class co-evaluates MgO, building learner confidence for independent diagram work. The City Market analogy (culturally grounded in Nairobi) makes the abstract concept of lattice energy tangible. The Property Prediction Table operationalises the NGSS practice of 'Constructing Explanations' by requiring learners to link sub-atomic structure to macroscopic properties using causal language ('because the ions…').	Observable indicator: Collect 5 randomly selected MgO dot-and-cross diagrams (use a random name stick). Mark against a four-point checklist: (1) correct valence electrons shown for Mg and O, (2) transfer of 2 electrons shown, (3) correct charges Mg^{2+} and O^{2-} in brackets, (4) ionic bond labelled. Score distribution informs whether reteaching is needed before Lesson 4. In the Property Prediction Table, the minimum acceptable response is a prediction with a reason referencing 'ions' or 'charges' — blanks or reasons referencing 'molecules' indicate a misconception to address.

	takes a LOT of energy (heat) to separate them all. That energy needed to pull the lattice apart is related to what chemists call lattice energy.' Learners write a two-sentence summary of lattice energy in their own words. Part C — Property Prediction Table (3 min): Learners complete a pre-printed table predicting: melting point (high/low?), solubility in water (yes/no?), electrical conductivity as solid / as solution / when molten (yes/no?), and hardness/brittleness. They write a reason for each prediction using the phrase 'because the ions…'	learners complete the prediction table. Stamp or tick completed tables — this table feeds directly into Lesson 5's investigation.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes: (1) GREEN — 'One thing I now understand about bonding that I did not understand at the start of Lesson 3'; (2) ORANGE — 'One new question this lesson raised for me.' Learners place sticky notes on the class DQB in the designated Lesson 3 row. The teacher reads 3–4 orange questions aloud and the class votes (show of hands) on which new question is MOST important for the sequence to answer. The winning question is circled and labelled 'Priority Question — to be answered by Lesson ___' (teacher fills in the lesson number based on the storyline). The teacher then explicitly connects today's learning back to the anchoring phenomenon by asking: 'We now know NaCl forms a giant ionic lattice. Go back to our original puzzle — diamond also forms a giant structure, but a covalent one. What is ONE difference you expect between ionic and covalent giant structures?' Learners discuss in pairs for 90 seconds, then one	Distribute two sticky notes per learner before this phase. Say: 'Green note: one genuine understanding — not 'I learned about bonding' but something specific, like 'I now understand why sodium becomes positive.' Orange note: one genuine question — something you are still confused about or curious about.' WAIT 3 minutes for writing. Direct learners to the DQB. Read 4 orange questions aloud using a neutral tone — do not answer them yet. Say: 'Which of these questions do you most want science to answer? Raise your hand for the question that puzzles you most.' Tally hands visibly. Circle the winner. Then ask the bridging question about diamond vs. ionic structures. WAIT 90 seconds for pair talk. Cold-call 1 pair. Affirm the response and say: 'Hold that thought — in Lesson 4 we investigate covalent bonding, and in Lesson 7 we will look directly at diamond's giant covalent structure. Your question is exactly the right one to be asking.'	Driving Question Board as Epistemic Tracker — the two-colour sticky note system (understanding vs. question) makes metacognition visible for both learners and the teacher. The class vote on the priority orange question gives learners genuine agency in the inquiry sequence and models how scientists prioritise research questions. The explicit bridging prompt back to diamond/graphite prevents lesson-level knowledge from becoming isolated and reinforces the unit's coherent storyline.	Observable indicator: Scan green sticky notes for specificity — vague notes ('I learned ionic bonding') indicate surface processing; specific notes ('I understand that Na loses 1 electron to become Na^+ because it then has a full outer shell') indicate deeper processing. Orange sticky notes reveal productive confusions (e.g. 'Why doesn't NaCl conduct as a solid?') versus conceptual gaps (e.g. 'What is an electron?') — the latter requires individual follow-up. Photograph the DQB at the end of each lesson for the teacher's ongoing assessment record.

	pair shares.			
Model Building Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Bonding Model' booklet (started in Lesson 1). Today they add an Ionic Bonding page. Using locally available materials (dried maize kernels for Na^+ ions, dried beans for Cl^- ions — referencing githeri as a Kenyan food context for mixed components), learners construct a small 2-D cross-section of a NaCl lattice on a square of card, alternating maize and bean 'ions' in a checkerboard pattern. They then annotate the model with: (1) ion charges, (2) direction of electrostatic attraction arrows between adjacent ions, (3) a caption reading 'Giant ionic structure — strong electrostatic forces in all directions → high melting point, brittle, conducts when dissolved or molten.' Finally, each learner adds a 'Phenomenon Connection' sentence at the bottom of their model page: 'NaCl has a giant ionic structure, which means it is [similar to / different from] diamond because _____, and [similar to / different from] graphite because _____.' Learners keep this model booklet — it will be updated every bonding lesson through Lesson 13.	Distribute materials (maize, beans, card, glue sticks) efficiently — have one set per table ready before the phase begins. Say: 'You are building a model — not a picture of a model. A model must show the pattern and be something you can point to and explain. Every ion must be labelled.' Circulate and ask each group: 'Point to one ionic bond in your model. What holds these two ions together? What would you need to do to separate them?' WAIT 10 seconds per group for a response before giving any hint. With 4 minutes remaining say: 'Stop building, start annotating and writing your Phenomenon Connection sentence.' With 1 minute remaining say: 'Hold up your model page. Check: do you have ion charges? Attraction arrows? A caption? A Phenomenon Connection sentence? If any of these are missing, add them now.' Collect Phenomenon Connection sentences (written on a half-sheet) as an exit ticket.	3-D Physical Modelling with Locally Available Materials (PCI: Environmental Conservation / Creativity and Imagination) — using dried maize and beans as ion proxies makes the abstract lattice tangible without requiring expensive kits, aligns with KICD's PCI on using locally available materials, and creates a memorable sensory-motor memory of the alternating ion arrangement. The Phenomenon Connection sentence is the critical sensemaking move: it forces learners to integrate today's ionic bonding knowledge into the unit's overarching comparison, directly advancing the driving question.	Exit ticket — the Phenomenon Connection sentence is the primary formative assessment artifact for this lesson. Minimum acceptable response: learner correctly identifies at least ONE similarity and ONE difference between NaCl and diamond AND ONE similarity and ONE difference between NaCl and graphite, using vocabulary from the lesson (giant structure, ionic, covalent, electrostatic, delocalised electrons, etc.). Responses are sorted into three piles overnight: (A) ready to proceed to Lesson 4, (B) partial understanding — needs a targeted warm-up in Lesson 4, (C) significant gap — teacher schedules a brief 1-on-1 or small-group reteach before Lesson 5's investigation.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your ARES teacher journal:

1. PREDICT PHASE: What were the most common predictions learners made about electron transfer? Did any learners spontaneously use the word 'transfer' versus 'sharing'? What does this tell you about the effectiveness of Lesson 2's valence electron review?

2. OBSERVE PHASE: Did the physical electron-token card activity successfully distinguish ionic bonding (transfer) from covalent bonding (sharing) at this early stage, or did it create confusion? If groups struggled, consider adding a contrast card activity at the start of Lesson 4.

3. EXPLAIN PHASE: How many of the 5 sampled MgO dot-and-cross diagrams met all four checklist criteria? If fewer than 3 out of 5 were correct, plan a 5-minute targeted reteach at the opening of Lesson 4 before introducing covalent bonding.

4. DQB PHASE: What was the winning priority orange question? Does it align with the planned Lesson 4 or 5 content, or does it reveal an unexpected gap that needs addressing?

5. MODEL BUILDING: Were learners able to articulate the Phenomenon Connection sentence without scaffolding, or did they need the sentence frame? If more than 50% of exit tickets were categorised as B or C, increase the explicit phenomenon connection prompts in Lesson 4.

6. EQUITY CHECK: Were the same learners responding during cold-call moments throughout the lesson? Review your name-stick records and ensure different learners are targeted in Lesson 4. Specifically note any learners who have not yet been cold-called across Lessons 1–3 and prioritise them in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed (1) electron transfer modelled physically using Na and Cl atom cards with moveable electron tokens, (2) the resulting formation of Na^+ and Cl^- ions with complete outer electron shells, (3) a simulation of the giant ionic lattice showing alternating positive and negative ions in a 3-D arrangement, and (4) a physical model of the NaCl lattice cross-section built from dried maize (Na^+) and beans (Cl^-).
What did I learn?	Students learned that: (1) ionic bonds form when a metal atom transfers one or more valence electrons to a non-metal atom, satisfying the octet rule for both; (2) the transferred electrons produce oppositely charged ions (cations and anions) held by strong electrostatic attraction; (3) the giant ionic lattice of NaCl — with strong forces in all directions — explains its high melting point, brittleness, and conductivity only when dissolved or molten; (4) dot-and-cross diagrams are a standard tool to represent electron transfer in ionic bond formation.
How does this explain the phenomenon?	Students explained the Carbon Paradox connection by noting that NaCl's giant ionic structure (electron transfer, charged ions, electrostatic lattice) is fundamentally different from both diamond's giant covalent network (electron sharing, no ions, directional covalent bonds) and graphite's layered structure (delocalised electrons, weak inter-layer forces) — reinforcing that the TYPE of bonding, not merely the presence of a giant structure, determines properties and uses.

LESSON 4 (80 minutes): Covalent & Dative Covalent Bonding: Drawing Lewis Structures and Exploring Electron Sharing	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to draw Lewis dot-and-cross diagrams for covalent and dative covalent molecules, distinguish between single, double and triple bonds, and connect electron-sharing arrangements to the properties of real substances including graphite and diamond.
Knowledge	Learners will know: (1) covalent bonds form when atoms share pairs of electrons to achieve a noble gas configuration; (2) single, double and triple covalent bonds differ in the number of shared pairs; (3) dative (coordinate) covalent bonds occur when one atom donates both electrons of the shared pair, as in NH_4^+ and H_3O^+; (4) Lewis (dot-and-cross) diagrams represent bonding and lone pairs in covalent molecules; (5) graphite's three covalent bonds per carbon atom plus one delocalised electron explains both its layered structure and electrical conductivity.
Skills	Learners will be able to: (1) draw accurate dot-and-cross diagrams for H_2, O_2, N_2, H_2O, CO_2 and CH_4; (2) identify and label lone pairs and bonding pairs in Lewis structures; (3) construct dot-and-cross diagrams for NH_4^+ and H_3O^+ showing dative covalent bonds; (4) compare covalent and ionic dot-and-cross diagrams; (5) use Lewis structures to predict bonding arrangements in graphite and relate these to the anchoring phenomenon.
Attitudes	Learners will appreciate: (1) that the precise arrangement of shared electrons — not the identity of atoms — determines the properties of a substance, reinforcing wonder about the Carbon Paradox; (2) the value of peer collaboration in constructing and critiquing diagrams; (3) how chemical bonding knowledge underpins everyday Kenyan technologies such as pencil manufacture, jewellery cutting tools and acid-base chemistry in food preservation.
Key Inquiry Question	How do valence electrons determine the type of bond formed between atoms, and how does sharing electrons in different ways produce substances with completely different properties?
Purpose in Storyline	In Lessons 1–3, learners established that ionic bonding involves electron transfer and produces giant lattice structures with characteristic physical properties. In Lesson 4, learners pivot to covalent bonding — the bonding mode that governs both graphite and diamond. By drawing Lewis structures and discovering that carbon can form four covalent bonds, learners begin to see WHY graphite (three bonds + one delocalised electron per carbon) differs from diamond (four bonds in a 3-D network). Dative covalent bonding extends the concept and links to acid-base chemistry encountered later in the course. This lesson is the conceptual engine of the Carbon Paradox: students now have the tool — Lewis structures — to model the very phenomenon that opened the unit.
Safety Notes	No hazardous chemicals are used in this lesson. If pencil graphite samples or ceramic tiles are handled, remind learners not to rub eyes after touching graphite powder. Coloured markers used for diagram work should be non-toxic. Learners with dust sensitivities should avoid handling loose graphite; a sealed transparent bag can be used instead.

B. LESSON OVERVIEW

Learners begin by revisiting the Carbon Paradox phenomenon: the teacher re-displays the graphite pencil and diamond chip and asks learners to recall what they know about electron arrangements in carbon. The lesson then moves through a structured sequence: (1) a Predict phase where learners attempt to draw how carbon atoms might share electrons; (2) an Observe phase using a guided Lewis-structure construction activity with molecular models and a short ARES simulation showing electron clouds overlapping in H_2, O_2 and N_2; (3) an Explain phase where groups draw and compare dot-and-cross diagrams for H_2O, CO_2, CH_4, NH_4^+ and H_3O^+, distinguishing ordinary from dative covalent bonds; (4) a Driving Question Board update where learners post new evidence cards explaining how carbon's four covalent bonds connect to diamond versus graphite; and (5) a Model-Building phase where groups revise their cumulative class model by adding a covalent bonding panel showing graphite's

hexagonal layer and diamond's tetrahedral network using dot-and-cross notation. Throughout, the teacher uses cold-calling, targeted wait time, and gallery-walk formative assessment to monitor understanding.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the graphite pencil and diamond chip from the anchoring phenomenon and reminded of the puzzle: both are 100% carbon, yet behave oppositely. The teacher writes on the board: 'Carbon has 6 electrons. Its electron configuration is 2,4 — so it has 4 valence electrons.' Learners individually complete a PREDICT CARD: (A) 'Sketch how you think two carbon atoms might share electrons to bond together — use dots and crosses.' (B) 'How many bonds do you predict one carbon atom can form with other atoms? Explain your reasoning.' (C) 'Do you think carbon will form the same type of bonds in graphite as in diamond? Why or why not?' Learners write silently for 4 minutes, then share predictions with a shoulder partner for 2 minutes. Three volunteers share whole-class — teacher annotates predictions on the board WITHOUT correcting them yet, labelling each 'Prediction A', 'Prediction B', 'Prediction C' to revisit later.	Re-display the two carbon samples prominently. Say: 'Before we learn anything new today, I want your best scientific guess — not the right answer, your reasoning.' Distribute Predict Cards. After 4 minutes of silent work, say: 'Turn to your shoulder partner. You have 90 seconds each — tell them what you drew and why.' Cold-call 3 learners using sticks: 'Amina, what did you sketch for carbon sharing electrons?' After each response, WAIT 12 seconds before responding. Do NOT affirm or correct — say: 'Interesting — let's hold that idea and test it.' Annotate predictions on the board using a distinct colour (e.g., orange) labelled 'Our Predictions — Lesson 4 Start'. Leave visible for the entire lesson.	PREDICT-CARD STRATEGY: By externalising prior conceptions in writing and sketches before instruction, learners activate schema about electron configuration (Lessons 1–2) and ionic bonding (Lesson 3). The act of committing a prediction to paper creates cognitive dissonance when evidence contradicts it — deepening encoding of correct models. Annotating predictions on the board makes thinking public and creates a reference point for revision throughout the lesson.	Circulate during the 4-minute silent predict phase. Look for: (1) learners who correctly identify 4 as the number of bonds carbon can form — these learners can be used as peer explainers later; (2) learners who draw electron transfer (ionic-style) rather than sharing — flag these for targeted questioning during Phase 3; (3) learners who leave the sketch blank — sit beside them and prompt: 'What do you remember about carbon's electron configuration from Lesson 2?' Collect 5 Predict Cards to scan quickly before Phase 3 begins.
Observe Phase 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive Simulations (English)	ACTIVITY A — ARES SIMULATION (8 minutes): Learners open the ARES offline simulation 'Electron Sharing in Covalent Bonds.' They observe animated electron-cloud overlap for H_2 (single bond), O_2 (double bond) and N_2 (triple bond). The	Introduce ARES simulation: 'Open the simulation now. You have 8 minutes — fill in every row of your observation table. I will cold-call four people when we regroup.' Circulate: if learners skip lone pairs, point and ask: 'What happens to the electrons that	TWO-MODE OBSERVATION STRATEGY (simulation + hands-on construction): The simulation provides dynamic, visual evidence of electron clouds merging — making the abstract quantum concept concrete. The card-based construction kit adds a kinesthetic	During Activity B, use a 'board pass' check: after pairs complete CH_4, ask every pair to hold up their diagram. Scan for: (1) correct placement of 4 bonding pairs around carbon with no lone pairs on C — CORRECT; (2) lone pairs incorrectly placed on carbon —

🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	simulation highlights: number of shared pairs, lone pairs remaining on each atom, and whether each atom achieves a noble gas configuration. Learners complete an OBSERVATION TABLE in their notebooks with columns: Molecule | Total valence electrons | Shared pairs (bonds) | Lone pairs per atom | Noble gas config. achieved? ACTIVITY B — GUIDED DOT-AND-CROSS CONSTRUCTION (12 minutes): In pairs, learners receive a printed 'Lewis Structure Construction Kit' — cards showing element symbols with their valence electron counts (H=1, C=4, N=5, O=6). Pairs physically arrange the cards and draw dot-and-cross diagrams step by step for: H_2, O_2, N_2, H_2O, CO_2, CH_4. Teacher models H_2 live on the board using the 4-step method: (1) count total valence electrons, (2) draw skeletal structure, (3) fill octets starting with outer atoms, (4) form multiple bonds if needed. Pairs then work independently. A reference poster with the 4-step method remains visible throughout.	aren't shared? Where do they go on the diagram?' After simulation, transition to Activity B. Model H_2 on the board, narrating aloud: 'Step 1 — hydrogen has 1 valence electron. I use a dot for one atom and a cross for the other so we can TRACK which electrons come from which atom. This is why we call it dot-AND-cross, not just dot.' Emphasise: 'The cross electrons come from atom 2 — this is crucial for dative bonds in Phase 3.' Cold-call 4 learners after Activity B to draw one molecule each on the board: 'Kamau — draw O_2 for us. Everyone else — watch and give a thumbs-up when he places the lone pairs correctly.' WAIT 12 seconds after each drawing before class responds.	layer: physically placing cards forces learners to count electrons deliberately. Using different symbols (dots vs. crosses) for electrons from different atoms is a disciplinary practice of chemistry — it plants the seed for recognising dative bonds (Phase 3) where BOTH electrons originate from one atom. The observation table structures noticing into recordable evidence that feeds directly into Phase 3 explanation.	prompt: 'Count carbon's electrons in the diagram — has it got 8 around it including shared pairs?'; (3) hydrogen given more than 2 electrons total — prompt: 'Hydrogen only needs 2 to be like helium — what noble gas rule applies here?' Cold-call the 4 board-drawers and ask the class to identify any errors using the 4-step checklist before you correct.
Explain Phase 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES	PART A — DATIVE COVALENT BONDS (12 minutes): Teacher introduces NH_4^+ using a Kenyan context: 'Ammonium compounds are used as fertilisers on tea farms in Kericho — the NH_4^+ ion is how nitrogen reaches the soil. Let us figure out how it forms.' Learners first draw NH_3 using their 4-step method, identifying the lone pair on nitrogen. Teacher then poses the question: 'When a H^+ ion (no electrons at all) approaches the nitrogen lone pair in NH_3, what happens?' Learners discuss in groups of 4 for 3 minutes, then the	PART A: Draw NH_3 on the board first and ask: 'Nitrogen has 5 valence electrons — it forms 3 bonds with hydrogen. Count carefully — how many electrons are left on nitrogen as a lone pair?' WAIT 10 seconds. Cold-call: 'Wanjiku — tell me what you count.' Then introduce H^+: 'This proton has ZERO electrons. Can it share electrons if it has none to offer? Discuss with your group.' After 3 minutes, cold-call one group: 'Otieno's group — what did you decide?' Draw the dative bond arrow on the board and say: 'We	THREE-LAYER EXPLANATION STRATEGY: Part A uses a Kenyan agricultural context (Kericho tea fertiliser) to anchor dative bonding in a real application, then uses the 'gift analogy' (donation) to make the asymmetric electron contribution memorable. Part B is the critical sensemaking moment of the entire lesson — learners must USE Lewis structures to EXPLAIN the phenomenon that opened the unit. By writing a 3-sentence explanation using specific vocabulary, learners engage in the	Collect all 3-sentence Carbon Paradox explanations from Part B (one per group). While learners begin the DQB phase, quickly read for: (1) correct use of 'delocalised' to explain graphite's conductivity — TARGET UNDERSTANDING; (2) explanations that say graphite 'has free electrons' without explaining WHY — prompt in DQB phase: 'What causes those electrons to be free — what is carbon NOT doing with that 4th electron in graphite?'; (3) explanations that confuse ionic and covalent — use T-chart from

for similar readings	teacher demonstrates the dative bond: nitrogen DONATES both electrons of its lone pair to form the N→H bond in NH_4^+. Learners annotate their diagrams with an arrow showing the direction of donation. Learners then independently draw H_3O^+ using the same logic (water's lone pair donating to H^+). PART B — CONNECTING TO THE CARBON PARADOX (8 minutes): Learners receive a 'Bonding Comparison Card' showing: (i) a partial Lewis structure of graphite's hexagonal layer with 3 bonds per carbon atom and a question mark where the 4th electron goes, and (ii) a partial Lewis structure of diamond's tetrahedral network with 4 bonds per carbon atom. Groups answer: 'Why can graphite conduct electricity but diamond cannot? Use the words: delocalised, bonding pair, lone pair, octet.' Groups write a 3-sentence explanation. PART C — DOT-AND-CROSS COMPARISON (5 minutes): Learners compare their covalent diagrams from Phase 2 with the ionic dot-and-cross diagram of NaCl from Lesson 3, identifying: (1) what is transferred vs. shared, (2) where the square brackets appear and why, (3) which diagram shows discrete molecules versus a continuous lattice.	call this DATIVE because one atom DONATES — like giving a gift — both electrons. The bond is identical in strength to a regular covalent bond once formed — the arrow just records its origin.' PART B: Distribute Bonding Comparison Cards. Say: 'You have 8 minutes. Your explanation must include at least three vocabulary words from the board.' Post vocabulary: delocalised, bonding pair, lone pair, octet, tetrahedral, hexagonal layer. Circulate and challenge strong groups: 'If graphite has 3 bonds per carbon, where is the 4th valence electron — and why is THAT the key to the whole Carbon Paradox?' PART C: Display NaCl diagram from Lesson 3 beside CH_4. Ask whole class: 'Give me one difference — hands down, I will cold-call.' Cold-call 5 learners in 90 seconds. Record responses as a T-chart: IONIC | COVALENT.	NGSS Science and Engineering Practice of 'Constructing Explanations.' Part C uses a structured compare-and-contrast (T-chart) to consolidate the distinction between bond types accumulated across Lessons 3 and 4 — building a relational knowledge structure rather than isolated facts.	Part C as a re-teaching reference. Use a 3-point rubric: 3 = correct mechanism + vocabulary; 2 = correct idea, imprecise vocabulary; 1 = misconception present.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive	Each group receives two sticky notes: one GREEN (new understanding) and one YELLOW (new question). Groups discuss for 3 minutes: 'What do we now understand about the Carbon Paradox that we did not understand at the start of Lesson 1? What new questions has	Distribute sticky notes. Say: 'Green cards: write something you can NOW explain about the Carbon Paradox using Lewis structures. Yellow cards: write the deepest question you still cannot answer.' Give 3 minutes for group discussion. As groups write, circulate and prompt: 'Can you	DRIVING QUESTION BOARD — EVIDENCE TRACKING: The DQB is a publicly visible record of the class's growing causal model of the phenomenon. By distinguishing 'evidence gathered' (green) from 'questions remaining' (yellow), learners experience science as a cumulative, iterative	Read all yellow question cards after class. Categorise into: (1) questions addressed in upcoming lessons (e.g., 'Why is diamond 3-D but graphite is flat?' → Lesson 5 giant covalent structures); (2) questions that reveal misconceptions (e.g., 'Is dative bonding weaker than normal

Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	today's lesson raised?' GREEN cards are posted under the DQB column: 'EVIDENCE WE HAVE GATHERED.' YELLOW cards are posted under: 'QUESTIONS WE STILL HAVE.' The class then does a 2-minute silent gallery walk to read all new cards. The teacher reads three green cards aloud and asks: 'Does our class now have enough evidence to explain WHY graphite conducts and diamond does not?' Learners vote with thumbs — up (yes), sideways (partially), down (no) — and the teacher records the class verdict on the DQB.	explain what the delocalised electron in graphite has to do with conductivity? If yes — that goes on your green card.' After posting, read three green cards aloud verbatim, then ask: 'Class, do we now have a FULL explanation — thumbs up if yes.' After the vote, say: 'Most of us are at sideways — that is exactly where we should be at Lesson 4 of 13. The upcoming lessons on giant covalent structures and metallic bonding will fill in the remaining gaps.' Update the DQB master record.	process — not a single-lesson revelation. The class thumb-vote makes metacognitive monitoring social and low-stakes. Connecting today's Lewis structures explicitly to the DQB forces learners to articulate HOW today's evidence advances (but does not yet fully answer) the driving question — a key feature of NGSS-aligned storyline teaching.	covalent bonding?' → address at start of Lesson 5 as a warm-up correction); (3) questions that show sophisticated thinking (e.g., 'Could you have a molecule with BOTH ionic and covalent bonds?' → affirm and preview Lesson 6 on bond polarity). Use yellow card categories to plan the Lesson 5 opening hook.
Model Building Phase 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups return to their cumulative class bonding model (a large A2 poster begun in Lesson 1 and updated each lesson). Today's task: add a COVALENT BONDING PANEL to the model. Using coloured markers, groups draw: (1) a dot-and-cross diagram of CH_4 with all bonding pairs and lone pairs correctly labelled; (2) a simplified Lewis structure of graphite's hexagonal ring showing 3 bonds per carbon and an arrow labelled 'delocalised electron — free to move → conducts electricity'; (3) a simplified Lewis structure of diamond's tetrahedral unit showing 4 bonds per carbon and a label 'no free electrons → insulator.' Groups also add a 'DATIVE BOND CORNER' showing NH_4^+ with the donation arrow. At the bottom of the panel, groups write one sentence: 'In covalent bonding, atoms [share/transfer] electrons to achieve [noble gas/octet] configuration, and the number and arrangement of bonds determines the substance's properties.'	Distribute A2 posters and coloured markers. Say: 'Today you are adding the covalent bonding panel. It must include three things — I am writing them on the board.' Write: (1) CH_4 dot-and-cross, (2) graphite Lewis structure with delocalised electron labelled, (3) diamond tetrahedral unit with insulator label. Give 7 minutes. At 5 minutes, circulate and prompt groups who have not yet drawn graphite: 'The graphite structure is the KEY to the whole Carbon Paradox — make sure it is on your panel before time is up.' At 7 minutes, say: 'Swap posters with the group next to you. You have 60 seconds — find one thing they drew that you agree with and one thing you would change.' Cold-call two groups: 'Nairobi group — what would you change on Mombasa group's diagram?'	CUMULATIVE MODEL REVISION STRATEGY: The A2 poster model grows lesson by lesson, making the class's developing understanding of the Carbon Paradox literally visible on the wall. Adding the covalent panel alongside the ionic panel (Lesson 3) enables learners to see the contrast spatially — a graphic organiser at the scale of the whole sub-strand. Drawing graphite's Lewis structure is not a routine exercise: it is a direct attempt to MODEL THE PHENOMENON. The one-sentence summary forces linguistic consolidation of the lesson's core idea. The peer swap introduces a low-stakes critique practice aligned with NGSS Science and Engineering Practice 7 (Engaging in Argument from Evidence).	During the poster swap, note which groups correctly place the delocalised electron label on graphite and which omit it. After class, photograph all group panels. Use the images to: (1) select 2–3 exemplary panels to display and discuss at the start of Lesson 5; (2) identify the most common error (likely: drawing 4 bonds on every carbon in graphite, as in diamond) — plan a targeted 3-minute correction at the Lesson 5 opening using the board. A correctly completed covalent panel must show: CH_4 with 4 bonding pairs, no lone pairs on C; graphite ring with 3 bonds per C and a mobile electron arrow; diamond tetrahedron with 4 bonds and 'insulator' label; NH_4^+ dative arrow.

	Groups compare their panels with the neighbouring group and identify one similarity and one difference in their diagrams.			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to independently draw dot-and-cross diagrams for CH_4 and CO_2 without scaffolding by the end of Phase 2 — or do they still need the 4-step card? If most still needed the card, build a 5-minute retrieval practice warm-up into Lesson 5. (2) Did learners correctly explain WHY graphite's delocalised electron causes conductivity, or did they simply state it as a fact? If the latter, the Phase 3 Part B 3-sentence task needs stronger vocabulary scaffolding — consider providing a sentence frame: 'Graphite conducts electricity because __________, which means that __________.' (3) Were dative covalent bonds understood as a type of covalent bond (not a different, weaker bond)? Check yellow DQB cards for this misconception. (4) Was the Kericho fertiliser context (NH_4^+) effective in grounding dative bonding in a Kenyan application — or did it distract from the chemistry? Adjust the contextual hook in Lesson 6 accordingly. (5) Did the cumulative model posters correctly distinguish graphite (3 bonds + delocalised e^-) from diamond (4 bonds) — the central insight needed for Lesson 5 on giant covalent structures? If not, begin Lesson 5 with a targeted correction gallery walk before introducing new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed ARES simulation animations showing electron-cloud overlap forming single (H_2), double (O_2) and triple (N_2) covalent bonds, and completed an observation table recording shared pairs, lone pairs and noble gas configuration for each molecule. Learners also physically constructed dot-and-cross diagrams for H_2O, CO_2 and CH_4 using the Lewis Structure Construction Kit cards, and examined a Bonding Comparison Card showing partial Lewis structures of graphite's hexagonal layer and diamond's tetrahedral network.
What did I learn?	Learners learned that covalent bonds form when atoms SHARE pairs of electrons (rather than transfer them as in ionic bonding), that carbon's 4 valence electrons allow it to form 4 covalent bonds, that dative (coordinate) covalent bonds occur when one atom donates BOTH electrons of a shared pair (as in NH_4^+ and H_3O^+), and that in graphite each carbon forms only 3 covalent bonds leaving one electron per atom delocalised and free to move — explaining graphite's electrical conductivity — while in diamond all 4 bonds are used in a 3-D tetrahedral network leaving no free electrons, explaining why diamond is an insulator.
How does this explain the phenomenon?	Learners explained (in their group 3-sentence summaries and cumulative model panels) why graphite conducts electricity while diamond does not, using the concepts of bonding pairs, lone pairs, delocalised electrons and the octet rule — directly addressing the Carbon Paradox anchoring phenomenon and advancing the driving question: the way carbon atoms bond to one another, not the identity of the atoms themselves, determines all the observable properties of the substance.

LESSON 5 (80 minutes): Hydrogen Bonding & Van der Waals Forces: Why Water Defies Expectation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and explain why intermolecular forces — hydrogen bonding and Van der Waals forces — determine the physical properties of substances such as water, iodine, and noble gases.
Knowledge	Learners will be able to: (1) define hydrogen bonding and state the conditions required for it to form; (2) explain why water has anomalously high boiling and melting points compared to H_2S and H_2Se; (3) describe London dispersion forces and dipole–dipole interactions as types of Van der Waals forces; (4) relate intermolecular force strength to physical properties (boiling point, melting point, volatility).
Skills	Learners will be able to: (1) draw diagrams showing hydrogen bond formation between water molecules; (2) analyse boiling point data for Group VI hydrides and identify the anomaly for water; (3) construct and revise particle-level models to explain iodine sublimation and the low boiling points of noble gases; (4) interpret graphical data to compare intermolecular force strengths across substances.
Attitudes	Learners appreciate that invisible forces between molecules — not just atoms — shape the everyday properties of substances like water from Lake Victoria or iodine antiseptic used in Kenyan clinics, thereby developing curiosity about the molecular world. Learners show care (Love) by handling iodine safely and working collaboratively without endangering peers.
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	Lessons 1–4 established ionic, covalent, dative covalent, and metallic bonding — all bonds WITHIN molecules or lattices. Lesson 5 zooms out to ask: what holds SEPARATE molecules together? By investigating water's surprisingly high boiling point and iodine's easy sublimation, students discover that intermolecular forces (hydrogen bonding and Van der Waals forces) are the 'hidden layer' of bonding that controls melting points, boiling points, and states of matter. This directly advances the driving question: in graphite, delocalised electrons between layers create only weak Van der Waals forces, which is exactly WHY layers slide — connecting back to the Carbon Paradox anchoring phenomenon.
Safety Notes	Iodine is toxic and a skin/eye irritant: learners must not handle iodine crystals directly — teacher demonstrates only. Ensure adequate ventilation when heating iodine. If iodine vapour is produced, do not inhale. Learners with asthma should stand back. Water baths should use water below 60 °C to avoid burns. Report any spillage to the teacher immediately.

B. LESSON OVERVIEW

In this 80-minute lesson, learners investigate the concept of intermolecular forces by examining two puzzling phenomena: (1) Why does water — a small molecule — boil at 100 °C while the larger H_2S boils at −60 °C? and (2) Why does iodine (I_2) transition directly from solid to purple vapour when gently warmed, without becoming a liquid first? Through data analysis of Group VI hydride boiling points, guided diagramming of hydrogen bonds between water molecules, and a teacher-led demonstration of iodine sublimation, learners construct an understanding of hydrogen bonding and Van der Waals (London dispersion and dipole–dipole) forces. The lesson anchors back to graphite: the weak Van der Waals forces between carbon layers in graphite are the molecular-level reason layers slide, giving graphite its softness — the very property students observed in Lesson 1 when graphite smeared on the ceramic tile.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Van der Waals Forces - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrogen and Alkali Metals (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a data table on the board listing the boiling points of Group VI hydrides: H_2O (100 °C), H_2S (−60 °C), H_2Se (−41 °C), H_2Te (−2 °C). Without any instruction, they are asked to (a) draw a trend line they EXPECT from $H_2Te \rightarrow H_2S \rightarrow H_2O$ based on decreasing molecular mass, and (b) predict where water should fall on that trend. Learners write individual predictions in their science notebooks using the sentence frame: 'I predict water will boil at approximately ___ °C because ___.' They then share predictions with a shoulder partner before 4 cold-calls to the class.	Display the boiling point table using the ARES platform or a hand-drawn chart. Say: 'Look at this data carefully — do NOT try to explain it yet. Just look at the numbers for H_2Te, H_2Se, and H_2S. What pattern do you notice as molecules get smaller?' Allow 10 seconds WAIT TIME. Cold-call 3 learners: 'Achieng, what trend do you see? — Kamau, do you agree or disagree? — Fatuma, what does your predicted trend tell you water's boiling point should be?' Then reveal the actual boiling point of water (100 °C) by uncovering it on the chart. Say: 'Water does NOT follow the trend. It boils more than 160 °C HIGHER than we would predict. Today we are going to find out why.' Record the class's range of predictions on the board for revisiting at the end of the lesson.	Anomalous Data Prediction — Learners first construct an expected pattern from known data, then confront a real anomaly. This cognitive conflict (water boils far above prediction) creates a genuine need-to-know that motivates the lesson's explanation. It mirrors how Kenyan scientists like Prof. Thomas Odhiambo approached unexpected results — as puzzles worth solving.	Teacher circulates and reads 5–6 notebook predictions. Observable indicator: learner's predicted trend line extrapolates BELOW 0 °C for water (approximately −80 °C), confirming they understand the pattern before the anomaly is revealed. If most learners predict water near 100 °C without reasoning, probe: 'What was your basis for that prediction?' — distinguishing prior knowledge from genuine trend analysis.
Observe Phase 📹 **VIDEO:** [Van der Waals Forces - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrogen and Alkali Metals (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — Iodine Sublimation Demonstration (8 min): Teacher places a few iodine crystals in a sealed flask and gently warms it over a hot water bath. Learners observe purple vapour forming directly from the solid with minimal liquid phase. They record observations using a 3-column table: 'What I saw | What I expected | What puzzles me.' ACTIVITY B — Boiling Point Data Analysis (12 min): Learners work in groups of 4 to analyse an extended data set on ARES (or printed worksheet) comparing boiling points across: noble gases (He, Ne, Ar, Kr), hydrogen halides (HF, HCl, HBr, HI), and Group VI hydrides. Each group plots a graph of boiling point vs. molecular mass	For the iodine demonstration: 'Everyone observe the solid iodine carefully before I heat it. Mwangi, describe the colour and state for the class.' Heat slowly. Say: 'Watch — I am not boiling it. The temperature of this water bath is only 50 °C. Record EXACTLY what you see, not what you think is happening.' Allow 10 seconds WAIT TIME after the vapour appears before speaking. For data analysis: 'Your group has 12 minutes. Task 1: plot noble gas boiling points — assign one person as grapher, one as data caller, one as checker, one as reporter. Task 2: find the pattern. Task 3: compare iodine to xenon — they have similar mass but very different boiling points. Your group	Contrasting Cases Analysis — Learners compare substances of similar molecular mass but different boiling points (e.g., Xe vs I_2, or H_2O vs H_2S). By holding molecular mass roughly constant and changing molecular polarity/H-bonding ability, learners isolate the variable (intermolecular force type) that explains the difference. This is the same reasoning strategy used in controlled experiments.	Collect and scan 2–3 group graphs. Observable indicators: (1) noble gas graph shows a clear upward trend — larger atoms have higher boiling points; (2) learners correctly identify that I_2 has a much higher boiling point than a noble gas of similar mass; (3) observation tables distinguish 'purple vapour appears directly from solid' from vague statements. Red flag: if learners state iodine 'melted' — intervene and replay the observation.

	for noble gases and identifies the trend. They then locate iodine's boiling point (184 °C) and compare it to noble gases of similar mass (Xe, 108 a.m.u. boils at −108 °C vs I_2, 254 a.m.u. boils at 184 °C). Groups annotate graphs with the question: 'What is causing the difference?'	reporter will share one observation.' Circulate to 4 groups; prompt with: 'What happens to boiling point as noble gas atoms get bigger and heavier?' Cold-call 2 group reporters after activity.		
Explain Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Hydrogen Bonding (13 min): Teacher introduces hydrogen bonding using the water anomaly as the hook. Learners draw the Lewis structure of H_2O (revisiting Lesson 3 skill), identify the two lone pairs on oxygen and the partial charges (δ+ on H, δ− on O). They then draw a diagram showing THREE water molecules with dashed lines representing hydrogen bonds between δ+ H of one molecule and the lone pair of O on an adjacent molecule. Using a printed or ARES-displayed template, learners label: donor atom, acceptor atom, and the H-bond itself. They apply the rule: 'H-bonding occurs when H is covalently bonded to N, O, or F.' They compare water to H_2S: 'S is not electronegative enough — no hydrogen bond.' PART B — Van der Waals Forces (12 min): Teacher explains London dispersion forces using noble gases: temporary dipoles form when electron clouds momentarily shift. Learners sketch 'before' (symmetrical electron cloud) and 'after' (temporary dipole) diagrams for Ar atoms. Teacher explains dipole–dipole forces using HCl as example. Learners then apply both concepts: 'Why does I_2 have such high boiling point despite being non-polar? (Large electron cloud → strong London dispersion	Part A: 'Before I tell you anything, look at your Lewis diagram of water from Lesson 3. How many lone pairs does oxygen have?' WAIT 10 seconds. Cold-call: 'Otieno — two lone pairs, correct. Now — those lone pairs have a slight negative charge. What do you think they might be attracted to on a NEIGHBOURING water molecule?' WAIT 12 seconds. Cold-call 2 learners. After drawing H-bonds on the board: 'This dashed line is NOT a covalent bond. It is a weak electrostatic attraction. But water has up to FOUR of these per molecule — two from lone pairs, two from H atoms. That is why water's boiling point is so much higher than expected.' For Part B, use the analogy: 'Think of electron clouds in noble gas atoms like a herd of cattle in Rift Valley — usually spread out evenly, but sometimes all bunch up to one side by chance. When that happens, the atom has a temporary dipole. The nearby atom responds. That is London dispersion force.' Write on board: 'Larger atom = more electrons = easier to distort cloud = stronger London dispersion force.' Then ask: 'Apply this — which should have a higher boiling point: helium or krypton?' Cold-call 3 learners. Pivot to graphite: 'Between the flat carbon layers in	Analogical Reasoning + Mechanistic Explanation — The 'bunching cattle' analogy makes the abstract concept of temporary dipoles concrete and locally resonant. Mechanistic explanation (step-by-step: electron shift → temporary dipole → induced dipole in neighbour → attraction) builds causal understanding rather than rote definition. Returning to graphite at the end of Part B directly advances the driving question and makes the phenomenon relevant.	Learners individually complete an exit mini-task (2 minutes): 'Draw two water molecules and show ONE hydrogen bond between them. Label δ+, δ−, and the H-bond with a dashed line.' Observable indicators: (1) dashed line connects H of one molecule to O (not H-to-H or O-to-O); (2) δ+ placed on H atoms, δ− on O; (3) at least one lone pair shown on the acceptor O. Teacher scans 6 drawings during the Van der Waals explanation. Common error to watch for: learners drawing H-bonds as solid lines (confusing with covalent bonds) — correct immediately.

	forces.) Why does graphite's layers slide? (Only Van der Waals forces between layers — much weaker than the covalent bonds within layers.)'	graphite — what type of bond holds the layers together?' WAIT 15 seconds. Record answers on board.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the PINK note they write one NEW question this lesson has raised for them about bonding and properties. On the GREEN note they write one connection they can now make to the anchoring phenomenon (diamond vs. graphite). Groups of 4 discuss their sticky notes for 3 minutes, then one representative from each group places the notes on the class Driving Question Board under two columns: 'New Questions' and 'Connections to Carbon.' The class reads 3–4 posted connections aloud. Learners then update their personal 'Bonding Concept Map' (started in Lesson 1) by adding 'Intermolecular Forces' as a new node, with branches for 'Hydrogen Bonding' and 'Van der Waals Forces,' linked to examples (H_2O, I_2, noble gases, graphite layers).	Say: 'You now know about five types of bonding or force: ionic, covalent, dative covalent, metallic, and now intermolecular forces. Go back to the big question on your board — WHY does graphite conduct and slide while diamond does not? On your GREEN sticky note, write specifically what intermolecular forces have to do with graphite's properties. You have 2 minutes — write something, even if you are not certain.' After posting, read 2–3 green notes aloud. Affirm correct connections: 'Exactly — between graphite layers, only weak Van der Waals forces hold them, which is why they slide. There are no such weak links in diamond's 3-D network.' Point to any misconceptions on pink notes and say: 'These are great questions — we will return to them in Lessons 6 and 7 when we study giant structures.'	Driving Question Board (DQB) Curation — The two-colour sticky note protocol forces learners to distinguish between new uncertainties (pink) and consolidated understanding (green). Updating the personal concept map builds a cumulative, visible knowledge structure that learners own and revise throughout the unit, supporting metacognitive awareness of their own learning progression.	Teacher photographs the updated DQB for review after class. Observable indicators: (1) green notes reference graphite layers, Van der Waals forces, or 'weak forces between layers' — showing transfer to the anchoring phenomenon; (2) concept maps now include 'Intermolecular Forces' node linked to at least one example; (3) pink notes reveal genuine curiosity (e.g., 'Does H-bonding exist in proteins?') rather than confusion about core content taught today. Flag any green notes that confuse intermolecular forces with intramolecular bonds — address at lesson start of Lesson 6.
Model Building Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Hydrogen and Alkali Metals (Flexbook) Source: CK-12	Learners revise their cumulative particle model of carbon structures (begun in Lesson 1, extended in Lessons 2–4). Today they focus on the graphite portion of their model. Using pencil and ruler on a fresh model sheet (or the ARES digital canvas), learners draw: (1) two flat hexagonal layers of carbon atoms connected by covalent bonds within each layer; (2) a 'gap' between the layers; (3) dashed lines in the gap labelled 'weak Van der Waals forces.' They then annotate: 'These weak forces	Distribute or display the model sheet. Say: 'Your model of carbon has grown over five lessons. Today we are adding the final piece that explains graphite's softness and conductivity. Draw two layers of hexagons first — the covalent bonds you know from Lesson 2. Then add the gap. Now — what goes in that gap? Not a covalent bond. Not an ionic bond. Use today's lesson.' WAIT 15 seconds. Circulate to 5–6 learners. Prompt those who draw solid lines between layers: 'Is that a strong or	Cumulative Model Revision — Each lesson adds one layer of explanatory power to the same carbon model, making the growth of understanding visible and tangible. Peer review with explicit criteria (intramolecular vs. intermolecular distinction; line type convention) develops scientific modelling literacy and reinforces the precise vocabulary needed for KCSE responses.	Collect 4–5 model sheets at end of lesson. Observable indicators: (1) covalent bonds within graphite layers shown as solid lines with correct hexagonal arrangement; (2) inter-layer forces shown as dashed lines labelled 'weak Van der Waals forces'; (3) sentence completion shows understanding — e.g., 'Diamond has no Van der Waals forces between layers because all carbon atoms are covalently bonded in a continuous 3-D tetrahedral network with no separate layers.' Red flag: learner

🔍 Search ARES for similar readings	allow layers to slide — that is why graphite writes on paper and conducts electricity (delocalised electrons) while being mechanically soft.' Learners compare their graphite model to their diamond model from Lesson 2 and write one sentence: 'Diamond has no Van der Waals forces between layers because ___.' They share models with a partner for peer review using two criteria: (a) Is the intramolecular bonding (covalent) correctly shown? (b) Are the Van der Waals forces between layers correctly labelled as weak/dashed?	weak force? Show me that with your line type.' After 8 minutes of drawing, say: 'Swap with your partner. Using our two criteria on the board, give your partner one strength and one improvement. You have 2 minutes.' Bring class back together: cold-call 2 pairs to share one improvement they gave. Close with: 'Next lesson, we will look at giant ionic and giant covalent structures — and you will see exactly why diamond is so different from NaCl even though both are giant structures. Your model today is your foundation.'		draws dashed lines WITHIN the graphite layer (confusing inter- and intramolecular) — this must be corrected before Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the boiling point anomaly of water generate genuine cognitive conflict, or did learners already know the answer? If many knew, consider whether the predict phase needs a less familiar anomaly next time. (2) Were learners able to correctly distinguish intermolecular forces (between molecules) from intramolecular bonds (within molecules) — the most common source of confusion in this topic? Check the model sheets collected. (3) Did the graphite connection in Phase 5 feel natural or forced? If learners struggled to link Van der Waals forces to graphite's softness, spend 5 minutes at the start of Lesson 6 revisiting this before introducing giant ionic structures. (4) Was the iodine demonstration sufficiently ventilated and safe? Note any safety issues for future repetition. (5) Which DQB pink-note questions were most intellectually rich? Consider incorporating 1–2 of these as the hook for Lesson 6. Record the names of learners whose model sheets showed the inter/intramolecular confusion for targeted follow-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Iodine crystals turned directly into purple vapour when gently warmed (sublimation with minimal liquid phase). The boiling point data showed water boiling at 100 °C — far above the expected trend extrapolated from H_2S (−60 °C) and H_2Se (−41 °C). Noble gases showed increasing boiling points with increasing atomic mass.
What did I learn?	Water's anomalously high boiling point is caused by hydrogen bonding — a strong intermolecular attraction between the δ+ hydrogen of one water molecule and a lone pair on the δ− oxygen of a neighbouring molecule. Hydrogen bonding requires H to be bonded to N, O, or F. Van der Waals forces include London dispersion forces (temporary dipoles in all molecules/atoms — stronger in larger, more polarisable electron clouds) and dipole–dipole forces (between permanently polar molecules). Iodine's relatively high boiling point despite being non-polar is explained by strong London dispersion forces from its large, polarisable electron cloud. Graphite's layers are held together by weak Van der Waals forces — this explains why layers slide, making graphite soft and able to write on paper.
How does this explain the phenomenon?	The Carbon Paradox is now deeper: diamond has no intermolecular forces because it is a continuous covalent network — every carbon is bonded to four others in 3-D, so there are no 'separate molecules' and no layers. Graphite, however, HAS layers, and between those layers only weak Van der Waals forces act. This is precisely why graphite is soft (layers slide past each other

	easily) while diamond is the hardest natural substance (no weak links anywhere in the structure). Intermolecular forces are the 'hidden layer' of chemistry that controls boiling points, melting points, and physical states of matter.

LESSON 6 (40 minutes): Metallic Bonding: The Sea of Electrons — From Copper Wire to Aluminium Foil	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will construct the sea-of-delocalised-electrons model of metallic bonding to explain why metals conduct electricity, bend without breaking, and feel different from ionic crystals or covalent solids.
Knowledge	Students will be able to: (1) define metallic bonding as the electrostatic attraction between a lattice of positive metal ions and a sea of delocalised electrons; (2) explain how delocalised electrons account for electrical and thermal conductivity; (3) explain how the non-directional nature of metallic bonding accounts for ductility and malleability; (4) compare metallic bonding diagrams to ionic and covalent diagrams drawn in previous lessons.
Skills	Observing and recording physical properties of metals; constructing a 2-D diagrammatic model of metallic bonding; comparing and contrasting three bond types using evidence; communicating scientific reasoning in writing and orally.
Attitudes	Curiosity about the invisible electron structure that gives familiar household objects their useful properties; appreciation of how chemistry underpins everyday Kenyan life and construction materials; openness to revising prior models when new evidence is introduced.
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses? — addressed specifically by showing that the presence of delocalised electrons in metals uniquely produces high conductivity and plastic deformability, properties absent in ionic and simple covalent substances.
Purpose in Storyline	Lesson 6 introduces the third major bond type — metallic bonding — completing the bonding-type trilogy (ionic, covalent, metallic) before students investigate all four structure types in Lessons 7-9. It also provides the electron-sea explanation for graphite conductivity that students have been carrying as an open question since Lesson 1, partially resolving the anchoring phenomenon and motivating the full structure comparison ahead.
Safety Notes	No hazardous chemicals are used in this lesson. Students handle aluminium foil, copper wire and iron nails — remind students not to use sharp wire ends to poke others. If a conductivity tester with a battery is used for demonstration, ensure the circuit is set up by the teacher and that battery contacts are not short-circuited for extended periods.

B. LESSON OVERVIEW
Students begin by handling three familiar Kenyan metal objects — aluminium nyama choma foil, copper electrical wire, and an iron roofing nail — and recording sensory observations and predictions about what makes metals different from the ionic and covalent substances studied so far. The teacher then introduces the sea-of-delocalised-electrons model through a guided explanation with analogy, and students draw a labelled 2-D bonding diagram. A structured comparison table consolidates all three bond types. The lesson closes with a DQB update linking metallic bonding back to graphite conductivity and the anchoring carbon paradox.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are each given a small piece of aluminium foil (the	Distribute materials before students arrive or as they settle to	Elicit-before-explain: collecting written predictions before	Teacher reads 6-8 predictions while circulating. Target

📹 **VIDEO:** The periodic table - transition metals Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Metallic Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	supporting phenomenon object), a short length of copper wire, and an iron nail — all labelled with their chemical symbol. Working in pairs, students spend 4 minutes completing a three-column observation chart in their notebooks with the headings: WHAT I SEE / WHAT I FEEL / WHAT IT CAN DO. They record observations such as shininess, flexibility, ability to be bent without snapping, and coolness to the touch. Students then respond individually to the written prompt: 'In NaCl (ionic) the electrons were transferred. In H2O (covalent) they were shared between two specific atoms. What do you predict happens to the electrons in a metal like aluminium — where every atom is the same element? Write one sentence.' Predictions are not discussed yet — students hold them for comparison after the explanation.	save time. As students write, circulate and read predictions silently — do not correct or affirm yet. After 4 minutes say: 'Keep your prediction written exactly as it is. We will come back to it in about 15 minutes to see how close your thinking was to the actual model chemists use. This is your personal evidence of your thinking today.' Prompt pairs who finish early: 'Can you bend the iron nail? Can you bend the copper wire? Both are metals — what is the same about what happens?'	instruction ensures students have a committed prior model to revise, maximising conceptual change when the sea-of-electrons model is introduced. The three objects are chosen to show that metallic properties are consistent across different metals, priming students to look for a common structural explanation.	misconceptions to listen for: (a) students who say electrons stay on their own atom in a metal; (b) students who confuse metallic bonding with ionic bonding because metals form positive ions. Note these for targeted questioning in the Explain Phase. No marks are awarded — this is diagnostic only.
Observe Phase 📹 **VIDEO:** The periodic table - transition metals Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Metallic Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher conducts a 6-minute teacher-led demonstration at the front, treating it as a structured observation task. Students record each observation in a second table with columns: WHAT HAPPENED / WHAT QUESTION DOES THIS RAISE. Demonstration sequence: (1) Teacher bends the copper wire into a curve — it holds its shape. Teacher asks: 'Did it crack like a chalk stick or a salt crystal would?' (2) Teacher rolls aluminium foil flat, then scrunches it — it deforms without breaking. (3) Teacher uses a simple conductivity tester (9V battery, LED bulb, two crocodile-clip leads) to test aluminium foil — the bulb lights. Then tests an ionic crystal (NaCl solid) — bulb does	Before the conductivity test say: 'Remember in Lesson 1 we tested graphite and diamond for conductivity. Graphite conducted — diamond did not. Now watch what the metals do.' [Pause — allow students to make the connection silently for 5 seconds before proceeding.] After the solid NaCl test say: 'Why does solid NaCl not conduct even though it has ions? Think back to Lesson 3.' [WAIT 8 seconds — cold call one student.] After the full demonstration say: 'In your table, write one question that these observations raise that our current models — ionic bonding and covalent bonding — cannot answer.' [WAIT 10 seconds of silent writing time.]	Phenomenon-linked observation: every demonstration is explicitly connected to substances students have already modelled (NaCl from Lesson 3, graphite from Lesson 1) so students see metallic bonding as filling a gap in their existing framework rather than as an isolated topic. The data strip on melting points begins building the quantitative evidence base for the Lessons 7-9 investigation.	Collect one question per pair on a sticky note or verbal share-out. Target questions to probe: 'Why does bending not break the bonds?' and 'Where are the electrons if they can move?' These questions are posted on the DQB to be answered in the Explain Phase. If no student raises the bending question, the teacher asks it directly: 'Our ionic lattice shattered when we bent it — why does the copper wire not shatter?'

	not light. Then tests a piece of candle wax (simple molecular covalent) — bulb does not light. (4) Teacher holds a piece of copper wire briefly and says: 'Metal feels cold to the touch even at room temperature — that is thermal conductivity moving heat away from your hand rapidly.' (5) Teacher projects or draws on the board a simple data strip: Melting point of Al = 660 degrees C, Cu = 1085 degrees C, Fe = 1538 degrees C — all much higher than iodine (184 degrees C) or water (0 degrees C) but lower than diamond (above 3500 degrees C). Students record these numbers and note the pattern.			
Explain Phase 📹 **VIDEO:** The periodic table - transition metals Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Metallic Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a 12-minute guided explanation using a combination of analogy, board diagram and student note-taking. Students construct a labelled 2-D metallic bonding diagram alongside the teacher explanation. Step 1 — Analogy introduction: students are told to imagine the Nairobi City Market fish section where fish (the positive metal ions) are packed tightly in a regular arrangement in ice (but imagine the ice as a moving river of water flowing between and around all the fish — those water molecules represent the delocalised electrons). The fish stay in position; the water moves freely everywhere. Step 2 — Model definition: Teacher writes on the board and students copy: 'In metallic bonding, metal atoms release their valence electrons into a shared pool. What remains are positive metal ions arranged in a regular lattice. The released electrons are delocalised —	During diagram drawing, circulate and check that students are labelling both the positive ions AND the delocalised electrons — common error is drawing only the ions without showing the electron sea. If students draw metallic bonding as dots-and-crosses between two atoms (covalent-style), gently redirect: 'In covalent bonding the electrons were shared between which atoms specifically? In metallic bonding, which specific atoms do these electrons belong to?' Use the contrast to sharpen understanding. After Question B, explicitly contrast with ionic bonding: 'In Lesson 3 we said ionic crystals shatter when struck because ion layers shift and like charges repel. Today you have explained why metals do NOT shatter. That is a real scientific explanation — you used a model to predict behaviour and it matched observation.' This affirms student reasoning and models scientific practice. When students	Structured teacher-guided explanation with embedded student talk moves (pair discussion, wait time, cold-calling) prevents passive note-taking. The fish-market analogy is locally grounded for Kenyan students. The three-question sequence scaffolds from recall (electrons move) to structural reasoning (non-directional bonding) to comparative quantitative reasoning (melting point trends), building progressively higher-order thinking. Returning to written predictions enacts metacognitive reflection.	After Question A, B and C, teacher records which pairs could answer independently and which required scaffolding — this informs grouping for Lesson 7. The prediction-revision writing is collected at the end of the lesson (or photographed) as written evidence of conceptual change. Specific check: every student diagram must show (a) labelled positive ions, (b) labelled delocalised electrons shown as a cloud or distributed dots, (c) a title identifying the substance. Diagrams without all three components are flagged for peer correction before the end of lesson.

	meaning they do not belong to any single ion but move freely throughout the entire structure. The electrostatic attraction between the lattice of positive ions and the sea of delocalised electrons IS the metallic bond.' Step 3 — Board diagram: Teacher draws a 4x4 grid of circles labelled Cu^+ (or Al^{3+} or Fe^{2+} as appropriate), then draws a cloud of small dots around and between all the circles labelled 'delocalised electrons (sea of electrons)'. Students copy and label their own diagram. Step 4 — Explaining properties using the model: Teacher asks three successive questions with wait time, students discuss in pairs then report back: Question A — 'Using the sea-of-electrons model, explain why metals conduct electricity.' [WAIT 15 seconds pair talk.] Expected answer: the delocalised electrons are free to move and carry charge when a voltage is applied. Question B — 'When we bend copper wire, the layers of positive ions slide past each other but the wire does not break. Why not?' [WAIT 15 seconds.] Expected answer: the electron sea surrounds the ions from all directions — there is no directional bond to break, so ions can slide and the attraction is maintained in the new position. Contrast with ionic: sliding in an ionic lattice brings like charges together causing repulsion and cracking. Question C — 'Why do metals generally have high melting points compared to simple molecular substances like iodine?' [WAIT 10 seconds.] Expected answer: strong electrostatic attraction between many delocalised	return to their predictions, say: 'Changing your mind because of evidence is exactly what scientists do. If your prediction was close, explain why. If it was different, explain what was missing from your earlier thinking. Both are valuable.'		

	electrons and many positive ions requires much energy to overcome. Step 5 — Return to predictions: Teacher says: 'Now look at the prediction you wrote at the start of the lesson. In one sentence below your prediction, write how your thinking has changed or been confirmed.' Students write silently for 2 minutes.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** The periodic table - transition metals Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Metallic Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students spend 5 minutes on a targeted DQB update. Each pair receives one sticky note (or writes in their DQB section of the notebook if physical sticky notes are unavailable). The pair must complete this frame: 'We now know that [substance / phenomenon] can be explained by metallic bonding because [mechanism]. This connects to the carbon paradox because [link].' Students are specifically prompted to address the graphite conductivity anomaly that has been on the DQB since Lesson 1 — graphite conducts because each carbon in graphite contributes one delocalised electron per atom into the plane of the layer, giving graphite a partial metallic-bonding character even though it is a covalent solid overall. The teacher provides this connection explicitly if no student pair reaches it independently, framing it as a preview of the deeper graphite discussion in Lesson 9.	Before the DQB update, point to the Lesson 1 observation still posted on the board: 'On Day 1 we saw that graphite conducts electricity. We said that was strange for a non-metal. You now have a concept — delocalised electrons — that begins to explain it. Does graphite sound more like a metal or a covalent solid in terms of its electrons?' [WAIT 10 seconds — take 2-3 responses.] Then say: 'This is one of the most fascinating puzzles in the whole unit — graphite sits between two categories. We will return to this fully in Lesson 9. For now, write your current best explanation on the DQB.' Post new sticky notes next to the Lesson 1 graphite-conductivity question to show the DQB is evolving with evidence.	The DQB update makes the storyline progression visible — students physically see their earlier questions being answered by new learning. Linking graphite back to the anchoring phenomenon reinforces the unit narrative and signals that the driving question is now partially answerable, motivating continued engagement through the investigation lessons ahead.	Read all DQB sticky notes before Lesson 7. Target: at least 80 percent of pairs should correctly identify delocalised electrons as the cause of conductivity in both metals and graphite. Pairs who write vague responses ('electrons move around') without connecting to the lattice of positive ions are identified for a brief clarifying conversation at the start of Lesson 7.
Model Building Phase 📹 **VIDEO:** The periodic table - transition metals	In the final 8 minutes, students complete a structured three-column comparison table in their notebooks titled 'Three Bond Types — Evidence Summary'. Columns: BOND TYPE / HOW	As students fill in the table, circulate and look specifically at the ionic row — a common regression error is students writing 'electrons shared' for ionic bonding (confusing it with covalent). If	The comparison table functions as a portable conceptual tool that consolidates three lessons of bonding theory into a single retrievable schema. Self-correction with a coloured pen externalises	Collect or photograph 5 randomly selected tables before students leave. Check: (1) ionic row correctly states electron transfer and explains brittleness and conductivity in solution; (2)

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Metallic Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ELECTRONS BEHAVE / PROPERTIES EXPLAINED. Rows: Ionic bonding (from Lesson 3), Covalent bonding (from Lesson 4), Metallic bonding (today). Students fill in all three rows from memory and lesson notes before checking against a completed model displayed on the board at the 6-minute mark. Students self-correct using a coloured pen, writing a brief note explaining any row they had wrong or incomplete. The table will be used as a reference tool in the Lessons 7-9 investigation.	observed, ask: 'In NaCl, did sodium share its electron with chlorine, or did it give the electron away completely?' Use Socratic questioning rather than correction to reinforce retrieval. At the 6-minute mark, reveal the completed table and say: 'Use a different coloured pen to add or correct. The colour difference is important — it shows you and me exactly what your memory had right and what you needed to revise. That is useful data about your own learning.' Before dismissal say: 'In the next three lessons you will test four unknown substances and use exactly this table to explain every observation you make. Keep it safe.'	metacognitive awareness and creates a visible record of learning gaps. Framing the table as a tool for the upcoming investigation gives it authentic purpose beyond rote copying.	covalent row correctly states electron sharing and explains low melting points for simple molecular; (3) metallic row correctly states delocalised electrons and explains conductivity and ductility. Any systematic error across multiple tables is addressed in the first 2 minutes of Lesson 7 as a whole-class correction.

D. TEACHER REFLECTION

After the lesson, consider: Did students spontaneously connect the sea-of-electrons model to graphite conductivity, or did this require teacher prompting? If prompting was needed, plan a brief 3-minute retrieval task at the start of Lesson 7 asking students to explain graphite conductivity in writing before the investigation begins. Were the pair-discussion wait times sufficient — did most pairs produce an answer before cold-calling, or did silence persist beyond 15 seconds? If silence persisted, consider providing a sentence starter in Lesson 7 pair discussions. Did the fish-market analogy land well for students — were there alternative local analogies suggested by students (e.g. matatu passengers tightly packed with space for a hawker to move freely between them)? Record any student-generated analogies and use them in future iterations. Check that the aluminium nyama choma foil supporting phenomenon has been explicitly named and explained — if time ran short, open Lesson 7 with a 2-minute revisit of why aluminium foil can be shaped, conducts heat to cook meat, yet does not burn or dissolve in cooking fat.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Aluminium foil, copper wire and an iron nail could all be bent without breaking; all three metals lit the conductivity bulb when tested; solid NaCl and candle wax did not conduct electricity; metals feel cool to the touch and have melting points hundreds of degrees higher than simple molecular substances like iodine or water.
What did I learn?	Metallic bonding is the electrostatic attraction between a regular lattice of positive metal ions and a surrounding sea of delocalised electrons that are free to move throughout the entire structure and belong to no single ion.
How does this explain the phenomenon?	Electrical conductivity in metals is explained by the free movement of delocalised electrons carrying charge; ductility and malleability are explained by the non-directional nature of the electron sea — ion layers slide without bond-breaking; relatively high melting points reflect the strong electrostatic attraction across the entire lattice; graphite conducts because one electron per carbon atom is delocalised within each layer, giving it partial metallic character despite being a covalent solid.

LESSON 7 (80 minutes): Physical Properties Investigation — Part 1: Setup & Prediction	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To predict and begin investigating how bond type determines the physical properties (solubility, electrical conductivity, and melting point) of four unknown substances representing giant ionic, simple molecular, giant covalent, and giant metallic structures.
Knowledge	Learners will know that different structural types — giant ionic (NaCl), simple molecular (sugar/iodine), giant covalent (SiO_2/sand), and giant metallic (aluminium) — arise from different bonding arrangements, and that these arrangements predict observable physical properties including solubility in water, electrical conductivity, and relative melting point.
Skills	Learners will be able to: (i) link bond type and structure to predicted physical behaviour; (ii) design and set up a fair comparative investigation using four unknowns; (iii) record structured predictions in an evidence table with bond-type justifications; (iv) handle laboratory apparatus and chemical samples safely.
Attitudes	Learners appreciate that careful, evidence-based prediction — rather than guessing — reflects the disciplined thinking of practising scientists such as Kenyan chemist Prof. Julius Mwangi, who models how systematic reasoning precedes laboratory work. Learners show care (Love) for peers by handling chemicals and electrical equipment responsibly so no one is harmed.
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	In Lessons 1–6 learners established what types of bonds exist and why atoms form them. Now, in Lesson 7, the investigation pivots to consequence: if we know the bond type, can we PREDICT what a substance will do? By treating four real substances as 'unknowns', learners experience the predictive power of bonding theory — the same power that explains the Carbon Paradox: diamond (giant covalent, 3-D network) versus graphite (giant covalent, layered, delocalised electrons), both 100% carbon yet utterly different. Today's predictions set the stage for experimental confirmation in Lesson 8.
Safety Notes	● Iodine crystals are toxic and a skin/eye irritant — learners must wear safety goggles and nitrile gloves when handling iodine; teacher demonstrates decanting. ● Electrical conductivity testers use low-voltage batteries (≤6 V) — inspect all wires for exposed copper before use. ● Aluminium pieces may have sharp edges — handle with tongs. ● Sand (SiO_2) dust can irritate airways — avoid blowing on samples; wet sand before disposal. ● In the spirit of the CBC Value of Love, learners must alert the teacher immediately if any chemical contacts skin or eyes; an eyewash station or clean running water must be identified at lesson start.

B. LESSON OVERVIEW

Learners are presented with four labelled 'Unknown' samples (A = table salt/NaCl, B = sugar, C = fine sand/SiO_2, D = aluminium foil strip) without being told their identities. Using their accumulated knowledge of giant ionic, simple molecular, giant covalent, and giant metallic bonding from Lessons 1–6, they predict — with written justifications — how each unknown will behave across three property tests: (1) solubility in water, (2) electrical conductivity in solution or solid form, and (3) relative melting point (high/low). Predictions are entered into a structured evidence table that explicitly links predicted behaviour to bond type and structural reasoning. The lesson closes with groups sharing their most confident and least confident predictions, populating the Driving Question Board with new 'testable claim' cards, and building an initial column in the cumulative bonding model. No testing is carried out today — that is Lesson 8 — so the lesson's energy is entirely in rigorous, evidence-based reasoning that revisits the Carbon Paradox: learners must also predict where diamond (giant covalent) and graphite (giant covalent with delocalised electrons) would sit in the same table.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners enter to find four small trays on each group table, each holding a mystery sample labelled Unknown A, B, C, D (no chemical names given). Each learner silently examines the samples for 3 minutes using only eyes and (gloved) fingertips — no water, no electricity yet. They write three 'I notice / I wonder' statements per sample in their science journals. After silent observation, groups of 4 discuss: 'Based only on appearance and what we know about bond types, which structural category — giant ionic, simple molecular, giant covalent, or giant metallic — do you think each unknown belongs to?' Each learner independently completes Column 1 ('Predicted Structure Type') of the Evidence Prediction Table before any group discussion is finalised, protecting individual thinking. Groups then reach a consensus prediction and record it on a mini-whiteboard for a gallery comparison.	1. SETUP (before learners arrive): Place pre-measured samples in four clearly labelled trays per group: A = ~5 g NaCl (white crystals), B = ~5 g white sugar (white crystals), C = ~5 g fine dry sand (grey-white), D = one 4 cm × 1 cm strip of aluminium kitchen foil, folded into a pellet. Ensure goggles and gloves are on every desk. 2. HOOK (2 min): Hold up the diamond chip and the graphite pencil from the anchoring phenomenon. Say verbatim: 'Seven lessons ago we asked — how can two substances made of ONLY carbon atoms behave so differently? Today we flip that question around. If I tell you the bond type, can YOU predict the behaviour — before you even test it? That is what real chemists do.' 3. SILENT OBSERVATION (3 min): 'Put on your goggles and gloves. You have three minutes of silence to observe your four unknowns. Do NOT add water or use the tester yet. Write at least three I-notice/I-wonder statements per sample.' Start a visible 3-minute timer. 4. INDIVIDUAL PREDICTION (5 min): 'Now, WITHOUT talking to your group, fill in Column 1 of your Evidence Prediction Table. Which structural type — giant ionic, simple molecular, giant covalent, or giant metallic — do you predict each unknown is? Write a one-sentence justification using the words bond type and structure.' WAIT 5 minutes in full silence;	Strategy: INDIVIDUAL-THEN-GROUP PREDICTION PROTOCOL. Learners first commit individually (preventing social loafing and anchoring bias), then negotiate group consensus. This mirrors the peer-review process in science and makes each learner personally accountable for their reasoning. Connecting to the phenomenon: learners must also predict where diamond and graphite would appear in the same structural-type column — reinforcing that the Carbon Paradox is not an exception but the clearest example of the rule.	Observable indicator: Each learner's Evidence Prediction Table Column 1 is completed with a bond-type justification before group discussion. Teacher scans for: (i) learners who use particle-level language ('ions', 'molecules', 'delocalised electrons') vs. those who guess by colour/appearance only — the latter need re-engagement with Lesson 4/5 notes; (ii) groups that predict both NaCl and sugar as 'giant ionic' — a key misconception to address in Phase 3.

		circulate and read over shoulders to identify misconceptions. 5. COLD-CALL REVEAL (4 min): After individual time, cold-call FOUR different students (one per unknown) using lolly-stick random selection: 'Amina — what is your prediction for Unknown A and WHY?' Apply 12-second wait time before accepting an answer. After each response ask a second student: 'Kiplangat — do you agree or disagree, and what is your evidence from Lessons 1–6?' Do not confirm or correct yet. 6. GROUP CONSENSUS (3 min): 'Now discuss with your group and write your consensus structural prediction on the mini-whiteboard.' Circulate; look for groups that confuse NaCl and sugar (both white crystals) — note these groups for targeted questioning in Phase 3.		
Observe Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Rather than running the experiments today (Lesson 8), learners engage in a structured EVIDENCE FROM PRIOR KNOWLEDGE observation phase. Each group receives a set of four 'Data Cards' — laminated A5 cards showing real published data: melting points in °C, whether each substance type dissolves in water (with a photograph of the dissolution or non-dissolution), and whether a conductivity bulb lights for the solid and for the dissolved/molten form. The data is presented WITHOUT the substance names — only structural labels ('Substance Type W, X, Y, Z'). Learners must match W/X/Y/Z to the four structural categories AND to their four unknown samples A/B/C/D. This is	1. DATA CARD DISTRIBUTION (2 min): Hand out Data Card sets. Say: 'These cards show real laboratory data for four structural types. Your job — as a group — is to match Type W, X, Y, Z to the four structural categories and then to your unknowns A, B, C, D. You have 8 minutes. Annotate every card with at least one sticky note explaining the particle-level reason for that data point.' 2. CIRCULATE with targeted questions (8 min): Visit each group twice. Use these specific prompts — do NOT give answers: • 'Why does Substance Type W have a melting point above 1000 °C? What is happening at the particle level when it melts?' • 'This card shows the conductivity bulb does NOT light	Strategy: DATA-DRIVEN PREDICTION REVISION (Predict–Observe–Explain cycle, Phase 1 of POE). By presenting real data as a matching puzzle — rather than a confirmation exercise — learners engage in the same reasoning process as analytical chemists who identify unknown substances from property data. The animation anchors particle-level thinking, preventing learners from treating 'dissolves' or 'conducts' as magic rather than as emergent consequences of bond type. Connecting to phenomenon: the teacher's pause on graphite during the animation explicitly reconnects to the Carbon Paradox — graphite's delocalised electrons explain conductivity in the same framework learners are using for	Observable indicator: Learners' sticky-note annotations on Data Cards use particle-level vocabulary (ions, molecules, lattice, delocalised electrons, intermolecular forces). Teacher checks that 'My thinking changed because…' boxes contain a bond-type reason, not just 'the data showed me.' Groups that have not matched the conductivity pattern to ion mobility need immediate re-teaching using the question: 'What needs to be FREE and MOVING for electricity to flow?'

	a two-step matching task. Groups annotate the data cards with sticky notes explaining HOW each data point connects to bond type. They then watch a 4-minute ARES-platform animation showing particle-level visualisations of: (i) NaCl dissolving into Na^{+} and Cl^{-} ions (Mombasa salt flats context), (ii) sugar molecules dispersing intact, (iii) sand remaining intact in water, and (iv) aluminium atoms remaining in metallic lattice. After the animation learners update their predictions if needed, recording any changes in a 'My thinking changed because…' box.	for the solid but DOES light when dissolved. Which bond type does that pattern match — and why?' • 'Kericho tea farmers dissolve fertiliser salts in water every season. Which of your unknowns behaves the same way — and why?' 3. ANIMATION LAUNCH (1 min instruction + 4 min viewing): 'Now we will watch a short animation showing what is happening at the particle level for each structural type. As you watch, tick or cross each of your predictions — and write in the change box if your thinking shifts.' Play the ARES platform animation. Pause at the NaCl dissolution scene and ask aloud (no answer expected yet): 'Notice how the ions separate. Now imagine the graphite layers in your pencil — do the carbon atoms separate like this when graphite dissolves in water? Why or why not? Hold that thought.' 4. SILENT UPDATE (3 min): 'Individually update Column 2 of your table — Revised Structural Prediction — and record any changed predictions with a reason.' WAIT in full silence. 5. COLD-CALL SHARE (3 min): Cold-call TWO students who you observed changing a prediction: 'Wanjiru — you changed your prediction for Unknown B. Walk us through your reasoning.' Apply 10-second wait time. Prompt with: 'Use the words intermolecular forces in your explanation.'	aluminium.	
Explain Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan	Each group constructs a completed EVIDENCE PREDICTION TABLE with all three property columns filled in for all four unknowns PLUS diamond and	1. TABLE COMPLETION PROMPT (2 min): 'You now have enough evidence to complete all three property columns for all six substances — A, B, C, D, plus	Strategy: CLAIM–EVIDENCE–REASONING (CER) PEER REVIEW. CER forces learners to distinguish between descriptive observation language and	Observable indicator: Each cell in the Evidence Prediction Table contains a particle-level justification with at least two of these terms: bond, structure,

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	graphite (added as 'bonus unknowns E and F'). For each cell, learners write a particle-level justification (2–3 sentences) linking bond type → structure → property. Groups then peer-review one another's tables using a structured CLAIM–EVIDENCE–REASONING (CER) protocol: they circle any cell where the justification is an observation ('it is hard') rather than a particle-level explanation ('strong covalent bonds in a 3-D tetrahedral network require a large amount of energy to break, so the melting point is very high'). Groups then compose a 'Bond-Type Property Rule' — a generalised if-then statement for each structural type — and post it on the classroom wall as a reference for Lesson 8.	diamond (E) and graphite (F). Complete your tables individually first — 5 minutes of silence.' Set timer. 2. SILENT INDIVIDUAL WRITING (5 min): Circulate and specifically look for whether learners correctly handle graphite's anomaly — it is giant covalent YET conducts electricity. Identify 2–3 students who get this right to share later. 3. GROUP TABLE CONSOLIDATION (4 min): 'Compare your tables within your group. Where you disagree, the person who uses particle-level language wins the argument — not the person who sounds most confident.' 4. PEER REVIEW — CER PROTOCOL (5 min): Groups exchange tables with the neighbouring group. Say: 'Read their justifications. Circle any cell where the explanation is an OBSERVATION, not a particle-level EXPLANATION. Write one specific improvement suggestion using the phrase — try adding the words…' 5. GRAPHITE DISCUSSION — TARGETED COLD-CALL (4 min): 'Graphite is giant covalent — same structural category as diamond and sand. Yet ONE of our three properties is completely different for graphite versus diamond and sand. Tendo — which property, and why?' Apply 15-second wait time. If Tendo is unsure, say: 'Think about what we said about graphite's structure in Lesson 1 — the layers and the delocalised electrons.' Cold-call a second student to extend: 'Fatuma — so where does graphite break the simple rule, and why is that NOT a	mechanistic explanation language — a core Science and Engineering Practice (Constructing Explanations). The deliberate inclusion of graphite as 'Unknown F' creates productive cognitive dissonance: it is giant covalent (like sand and diamond) but conducts electricity. Resolving this anomaly requires learners to recall the delocalised electron layer structure from Lesson 1, directly reconnecting to the anchoring phenomenon. The Bond-Type Property Rules posters transform individual understanding into a shared classroom knowledge artefact that will be tested in Lesson 8.	energy, ion/molecule/lattice/delocalised electron, force. Teacher collects one table per group at the end of Phase 3 for written feedback before Lesson 8. Key misconception to flag: learners who predict graphite will NOT conduct (treating all giant covalent structures as identical) — these learners need a one-to-one prompt: 'Go back to your Lesson 1 notes. What is special about graphite's structure that diamond does NOT have?'

		contradiction of bonding theory, but actually PROOF of it?' 6. BOND-TYPE PROPERTY RULES POSTER (4 min): 'Each group writes four if-then Bond-Type Property Rules on your poster strip — one per structural type. For example: IF a substance is giant ionic, THEN we predict it will… because… Post these on the wall — they will be your hypothesis board for Lesson 8.' Circulate to ensure each rule contains the words bond type, particle level, and one Kenyan context example (e.g., 'salt from Magadi Soda — giant ionic').		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky note cards: one ORANGE ('Testable Claim I am now confident about') and one BLUE ('Question I still have'). They write one entry on each card linked to today's prediction activity and post them on the class DQB under the relevant bond-type column headers. The teacher then facilitates a 5-minute whole-class DQB gallery walk where learners read the posted cards silently and place a small dot sticker on any question that matches their own uncertainty. The three most-dotted questions are read aloud and designated as 'Priority Questions for Lesson 8 Testing'. Learners copy these three priority questions into their journals as their personal investigation focus for the upcoming experiment.	1. CARD DISTRIBUTION & WRITING (3 min): 'On your ORANGE card write one testable claim you are now confident enough to bet on — use the format: I predict that [substance/structural type] will [property result] because [bond-type reason]. On your BLUE card write one genuine question you still have — something today's work did NOT fully answer for you.' Allow 3 minutes of silent writing. 2. DQB POSTING (2 min): Direct learners to post cards under the correct column header on the DQB (Giant Ionic / Simple Molecular / Giant Covalent / Giant Metallic / Carbon Paradox). 'If your question or claim spans more than one column, put it in the Carbon Paradox column — that is where complexity lives.' 3. SILENT GALLERY WALK & DOT VOTING (3 min): 'Walk silently past the DQB. Place a dot sticker on any BLUE question that is also YOUR question.' 4. PRIORITY QUESTION	Strategy: DOT-VOTING PRIORITY QUESTIONS on the DQB. This strategy makes metacognition social and visible — learners see that their uncertainties are shared, which normalises not-yet-knowing and motivates Lesson 8 testing. By anchoring one priority question explicitly to the Carbon Paradox, the teacher ensures the entire investigation sequence remains tethered to the original phenomenon rather than drifting into abstract property-listing.	Observable indicator: ORANGE cards contain a testable, bond-type-justified claim (not just 'I think Unknown A dissolves'). BLUE cards contain genuine mechanistic questions (not procedural questions like 'how do we test melting point?'). Teacher photographs the DQB at the end of class and uses it to plan differentiated support for Lesson 8 — specifically identifying learners whose BLUE cards reveal fundamental misconceptions about ion formation or electron mobility.

		ANNOUNCEMENT (2 min): Count dots aloud with the class. Read the three highest-dotted questions. Say: 'These are your class's top three open questions. Write them in your journal NOW. In Lesson 8 — the actual experiment — your job is to collect evidence that begins to answer them.' Ensure at least one priority question relates back to the phenomenon: 'If a question about diamond or graphite has not appeared, I will add one myself — here it is: Why does graphite conduct electricity but diamond does not, if both are giant covalent structures? Post this in the Carbon Paradox column.'		
Model Building Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative Bonding Concept Model — a large poster started in Lesson 1 showing the Carbon Paradox at the centre. Today they add a new outer ring: a PROPERTY PREDICTION RING. For each of the four structural types (giant ionic, simple molecular, giant covalent, giant metallic), learners draw or write the three predicted properties in colour-coded boxes (blue = solubility, red = conductivity, green = melting point) and connect each property box to the bond-type node with an annotated arrow explaining the particle-level mechanism. Diamond and graphite remain at the centre as the reference anchors. Learners also flag any property prediction that they are NOT yet certain about with a yellow question-mark flag — these become their personal experiment targets for Lesson 8. Groups share their model additions with one neighbouring group using a 90-second 'model talk': 'Here is what we added, here	1. MODEL RETRIEVAL (1 min): 'Get out your group's Bonding Concept Model poster from the folder. Today you are adding the Property Prediction Ring — the outer ring that connects bond type to physical behaviour.' 2. DRAWING INSTRUCTIONS (2 min): 'Draw four new nodes — one per structural type — in the outer ring. For each node, attach three colour-coded property boxes: blue for solubility, red for conductivity, green for melting point. Inside each box write your prediction AND the particle-level reason in one sentence. Use your Bond-Type Property Rules poster from Phase 3 as a reference.' 3. MODELLING TIME (8 min): Circulate. Prompt with: 'Your arrow from Giant Metallic to Conductivity says conducts — but WHY does it conduct? Draw the delocalised electrons on the arrow.' And: 'Your Giant Covalent node — you have diamond and graphite both here. Are their property boxes	Strategy: CUMULATIVE MODEL REVISION with COLOUR-CODED PROPERTY RING. Adding a new outer ring to the existing model makes conceptual growth visible and tangible — learners can literally see their understanding expanding from 'bond types exist' (inner ring, Lessons 1–4) to 'bond types predict properties' (outer ring, Lesson 7). The yellow flag protocol transforms uncertainty from a source of anxiety into a scientific resource — flagged areas become investigation priorities. Reconnecting to the Carbon Paradox at the lesson close reinforces that every structural type on the model is a variation of the same underlying principle that separates diamond from graphite.	Observable indicator: Property Prediction Ring is present on all group models with colour-coded boxes and annotated arrows. Teacher checks that graphite's conductivity box is flagged as DIFFERENT from diamond's within the giant covalent node — groups that treat them identically are directed to Lesson 1 notes before Lesson 8. Yellow flags provide a group-by-group map of unresolved understanding that guides the teacher's differentiation plan for the Lesson 8 experiment.

	is why, and here is what we are still unsure about.'	IDENTICAL? If not, which box is different and why?' Apply 12-second wait time before offering hints. 4. YELLOW FLAG TASK (1 min): 'Place a yellow sticky flag on any property box where you are predicting but NOT yet certain. These are your personal investigation targets for Lesson 8.' 5. MODEL TALK (3 min): '90 seconds — explain your model additions to the neighbouring group. Tell them what you added, why, and what yellow flags you have.' After 90 seconds, switch direction. 6. LESSON CLOSE (1 min): Display the diamond and graphite samples from Lesson 1 again. Say verbatim: 'Diamond. Graphite. Both carbon. But look at your model — your new Property Prediction Ring tells you EXACTLY why they are different. In Lesson 8, you will test four unknowns and find out whether your predictions are right. Come ready to be surprised — and to explain the surprise using your model.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. PREDICTION QUALITY: Did learners justify predictions using particle-level language (bond type, structure, ions, delocalised electrons) or did they rely on surface features (colour, texture) alone? What does this tell you about consolidation of Lessons 1–6 content?

2. THE GRAPHITE ANOMALY: How many groups correctly identified that graphite would conduct electricity despite being giant covalent? If fewer than half did so, plan a 5-minute mini-lesson at the start of Lesson 8 revisiting graphite's layered structure and delocalised electron sheet before any testing begins.

3. DQB QUALITY: Were learners' BLUE question cards mechanistic ('Why do ionic compounds only conduct when dissolved?') or procedural ('How do we measure melting point?')? Procedural questions signal that learners are thinking about the activity rather than the concept — redirect by asking 'Why does THAT test tell us about bond type?'

4. CARBON PARADOX CONNECTION: Did the lesson successfully re-anchor to diamond and graphite, or did the four unknowns become an isolated activity? If learners did not spontaneously mention diamond/graphite during Phases 3–5, build in a more explicit callback prompt for Lesson 8.

5. DIFFERENTIATION NOTE: Learners who struggled to distinguish giant ionic from simple molecular (both dissolve in water, but for different particle-level reasons) will need a targeted prompt at the start of Lesson 8: 'When NaCl dissolves, what particles are in the solution? When sugar dissolves, what particles are in the solution? How are those different?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined four unknown solid samples (labelled A–D) using sensory observation and matched them against a set of published property Data Cards showing melting point ranges, solubility, and conductivity data for four structural types (labelled W–X–Y–Z), without being told the substance names. They also watched an ARES-platform particle-level animation showing NaCl (Mombasa salt flats context), sugar, sand, and aluminium dissolving or not dissolving in water.
What did I learn?	Students learned that each structural type — giant ionic, simple molecular, giant covalent, and giant metallic — has a characteristic pattern of physical properties (solubility, conductivity, melting point) that arises directly from its bond type and particle arrangement. They learned that graphite is an anomaly within giant covalent structures because its delocalised electrons allow electrical conductivity, unlike diamond or SiO_2. They practised using Claim–Evidence–Reasoning to distinguish particle-level explanation from surface-level observation.
How does this explain the phenomenon?	Students explained their predictions by linking bond type to particle behaviour: ionic compounds dissolve because water molecules pull apart oppositely charged ions; simple molecular compounds have low melting points because only weak intermolecular (Van der Waals) forces must be overcome; giant covalent structures have very high melting points because strong covalent bonds throughout the lattice require enormous energy to break; metallic structures conduct electricity because delocalised electrons move freely through the lattice. Graphite's conductivity was explained as a consequence of its one-delocalised-electron-per-carbon-atom layer structure — the same structural feature that makes it fundamentally different from diamond despite both being 100% carbon.

LESSON 8 (80 minutes): Physical Properties Investigation — Part 2: Data Collection (Solubility, Conductivity & Melting Point)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To carry out a systematic hands-on investigation of the physical properties — solubility, electrical conductivity (solid and dissolved), and melting-point comparison — of substances with different bond types, and to begin identifying patterns that link bond type to physical behaviour.
Knowledge	Learners will know that: (i) giant ionic compounds (e.g. NaCl) are soluble in water and conduct electricity only when dissolved or molten; (ii) simple molecular compounds (e.g. sugar/glucose) may dissolve but do not conduct electricity in solution; (iii) giant covalent/atomic structures (e.g. graphite, diamond, SiO_2) are generally insoluble and have very high melting points; (iv) giant metallic structures (e.g. aluminium) conduct electricity in the solid state and have high melting points.
Skills	Learners will be able to: (i) set up and use a simple conductivity tester safely; (ii) perform solubility tests and record qualitative observations systematically; (iii) read and interpret melting-point data cards; (iv) organise evidence into a structured comparison table; (v) identify preliminary patterns from experimental data.
Attitudes	Learners will: (i) demonstrate Love by working carefully and avoiding harm to peers during electrical testing; (ii) show social cohesion by contributing equitably within investigation groups; (iii) appreciate the real-world relevance of bond-type properties to everyday Kenyan materials (table salt from Magadi, pencils, aluminium sufurias, sand/SiO_2).
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	In Lesson 7, students designed the investigation protocol and predicted outcomes. In this lesson (Lesson 8 of 13) they execute that protocol: testing NaCl, sugar, graphite, SiO_2 sand, and aluminium foil for solubility and conductivity, then comparing melting-point data from data cards. The evidence they gather today becomes the raw material for the pattern-finding and explanation work in Lesson 9, and moves the class closer to answering why diamond and graphite — both pure carbon — behave so differently. Every data point feeds directly back to the anchoring phenomenon.
Safety Notes	SAFETY NOTES (read aloud to class before starting): (1) The conductivity tester uses a 9 V battery — voltages are safe, but learners must NOT touch bare wire ends simultaneously; (2) Do not taste any substance, including salt or sugar solutions, as laboratory chemicals may be contaminated; (3) Handle SiO_2 sand carefully — avoid inhaling fine dust; (4) Wipe up any spilled water immediately to avoid slipping; (5) Aluminium foil edges can be sharp — handle with care. VALUE LINK — Love: 'We show care for one another by following every safety step; rushing or skipping a step could hurt a classmate.'

B. LESSON OVERVIEW

Lesson 8 is the core hands-on evidence-gathering session of the physical properties investigation sequence. Working in groups of four, learners rotate through three stations — (A) Solubility Testing, (B) Conductivity Testing (solid then dissolved), and (C) Melting-Point Data Cards — collecting data on five substances: sodium chloride (NaCl), sugar (sucrose), graphite powder (pencil shavings), SiO_2 (fine sand from a Kenyan river bed), and aluminium foil. Students record all results in a pre-designed class comparison table, begin spotting patterns, and update their Driving Question Board. The lesson is framed throughout by the Carbon Paradox phenomenon: every new data point about conductivity or melting point is compared back to the known behaviour of diamond and graphite to build explanatory momentum toward Lesson 9's pattern-analysis discussion.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-page 'Pre-Investigation Prediction Card' listing the five test substances (NaCl, sugar, graphite, SiO_2 sand, aluminium foil) and three properties (solubility in water, conductivity as solid, conductivity when dissolved/molten). Learners individually fill in their predicted outcome for each cell using a simple code: ✓ (yes / high), ✗ (no / low), ? (unsure). They annotate ONE prediction with a sentence beginning: 'I predict this because the bond type is ____, which means ____.' Pairs then compare prediction cards and note any disagreements — these become 'live questions' to test during the investigation. The teacher asks two cold-call pairs to share a disagreement with the class, anchoring the session to the DQB question: 'Does bond type control whether a substance conducts electricity?'	BEFORE DISTRIBUTING CARDS (2 min): Display the anchoring phenomenon image (diamond chip + graphite pencil, side by side). Say verbatim: 'We already know graphite conducts electricity and diamond does not — both are pure carbon. Today we are going to collect evidence from four other substances to find out whether that difference is unique to carbon or whether it is a pattern that applies to ALL bond types. Your predictions RIGHT NOW are your scientific hypothesis — own them, because the data will either confirm or challenge them.' DISTRIBUTE prediction cards. Allow 4 minutes of SILENT individual writing. WAIT TIME: After 4 minutes, say: 'Pause your writing. Look at your neighbour's card — you have 60 seconds to find ONE prediction you disagree on.' After 60 seconds, COLD-CALL 2 pairs (minimum): 'Group 2, Amina and Brian — what did you disagree on? What were your two different reasons?' Scribe the disagreement on the board under 'Live Questions to Test.' Repeat for one more pair. Say: 'Perfect — science works exactly like this. We disagree, then we collect evidence. Let's go.'	STRATEGY — Prediction Card with Justification Annotation: By requiring learners to write a causal sentence ('I predict this BECAUSE the bond type is…'), the teacher surfaces existing mental models before instruction. Disagreements between pairs create genuine cognitive tension that the investigation is designed to resolve, making data collection purposeful rather than procedural.	OBSERVABLE INDICATOR: Every learner's prediction card is completed before rotation begins. Teacher scans 6–8 cards during the 4-minute window, noting whether justification sentences reference bond type (target) or only physical intuition ('it looks like it will dissolve'). Cards requiring only physical intuition are flagged for targeted questioning during Phase 2 station rotations. Criterion: at least 70% of learners reference bond type in their justification annotation.
Observe Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Groups of four rotate through three investigation stations (approximately 18 minutes per station, with 2-minute rotation transitions). Each group carries one shared Data Recording Sheet and divides roles: Investigator (handles apparatus), Recorder	STATION SETUP (before lesson): Pre-label five beakers at Station A with substance names. Pre-build two conductivity testers for Station B (test both before class). Print and laminate data cards for Station C. Prepare graphite powder by shaving 3–4 pencils over paper.	STRATEGY — Structured Station Rotation with Role Accountability: Assigning fixed roles (Investigator, Recorder, Safety Monitor, Timekeeper/Reporter) ensures every learner is accountable for a specific aspect of evidence collection, not just passive	OBSERVABLE INDICATORS: (1) At Station B — every group correctly records that NaCl solid does NOT conduct but NaCl solution DOES conduct. If any group records the opposite, teacher intervenes immediately with: 'Show me exactly where both

for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(writes data), Safety Monitor (reads safety checklist before each sub-test), and Timekeeper/Reporter (will report findings during Phase 3). STATION A — Solubility Testing: Groups add one spatula measure of each of the five substances to separate labeled beakers of 50 mL distilled water, stir for 30 seconds, and observe. They record: (i) whether a clear solution forms, (ii) whether any residue remains, (iii) the appearance of the mixture. They also test solubility in propanone (a non-polar solvent) for sugar and NaCl to notice polarity effects (extension observation). STATION B — Conductivity Testing: Using a pre-built 9 V battery conductivity tester (battery + LED bulb + two crocodile-clip probes), groups first test each substance in solid/powder form by touching both probes to a small sample on a ceramic tile. They record bulb brightness: bright / dim / off. They then test the solutions or mixtures from Station A by dipping the probes in. They record bulb brightness for dissolved/suspended samples. CRITICAL CONNECTION: Groups are instructed to test graphite powder (pencil shavings collected before class) and compare to the diamond data card provided. The data card states: 'Diamond — conductivity as solid: NONE; conductivity dissolved in water: NOT SOLUBLE, NOT APPLICABLE.' Groups circle this on their sheet and write: 'Carbon — same atoms, completely different conductivity result. WHY?' STATION C — Melting Point Data Cards: Groups receive a set of five	DURING ROTATIONS — circulate every 4–5 minutes, spending maximum 90 seconds at each group. TARGETED QUESTIONS to ask at Station A: 'What does this tell you about what type of solvent this substance needs? Think back to what we said about ionic compounds and polar water molecules.' At Station B (cold-call one learner per group by name): 'Chiamaka, the bulb is bright in the NaCl solution but was off when you tested the solid. What has CHANGED between those two tests? Take 10 seconds.' WAIT 10 seconds before accepting response. At Station C: 'Look at your bar chart. Where does diamond sit? Where does graphite sit? Why do two substances made of identical atoms have nearly the same melting point but completely different conductivity?' — leave this as an open question; do NOT answer it yet. Say: 'Write that as a question on your recording sheet — we will answer it in Lesson 9.' FLAG any group at Station B where the conductivity tester LED does not light for NaCl solution — troubleshoot immediately: 'Check that your probes are both submerged and not touching each other.' SAFETY MONITOR PROMPT: At the start of each station rotation, ask the Safety Monitor in the incoming group: 'Read me the one safety rule for this station.' Do not allow work to begin until this is done.	observers. The deliberate re-injection of diamond and graphite data cards at Station C and Station B ensures that new evidence is always interpreted relative to the anchoring phenomenon, preventing isolated fact-collection and building toward the causal explanation in Phase 3 and Lesson 9.	probes are touching.' (2) At Station C — every group places diamond and graphite at the top of the melting-point ranking. (3) Recording sheets show the written question 'Carbon — same atoms, completely different conductivity — WHY?' before leaving Station B. Teacher collects one recording sheet per group at the end of rotations to check completeness before Phase 3.

	laminated data cards (one per substance) showing melting point in °C, a structural diagram, and the bond type label. They rank the five substances from lowest to highest melting point, plot a simple bar chart on their recording sheet, and answer: 'Which two substances have the highest melting points? What bond type do they share?' They are also given a sixth card: Diamond (mp: >3,500°C) to add to their chart. A seventh card shows: Graphite (mp: >3,600°C). Students note: both carbon allotropes have extremely high melting points — yet graphite conducts and diamond does not. This re-injects the paradox into the data-collection phase.			
Explain Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups reconvene. Each group's Reporter shares their most surprising finding using the sentence frame written on the board: 'We were most surprised that [substance] [property observation] because we predicted [prediction] — this might mean that [tentative explanation linking bond type].' The class then co-constructs a master data table on the board, with each group contributing one row. Once the table is complete, teacher poses a structured thinking task: learners individually write a 3-sentence 'Pattern Statement' using the sentence starters: (1) 'Substances that conduct electricity in solution but not as solids seem to have ____ bonds because…'; (2) 'Substances with very high melting points seem to have ____ bonds because…'; (3) 'The carbon paradox — diamond vs graphite — shows that even within the same bond type (giant covalent), a	REPORTER SHARES (8 min): Cold-call Reporters from 3 groups (not volunteers): 'Group 5 Reporter — Fatuma, your most surprising finding, using the sentence frame on the board.' After each share, ask the class: 'Does anyone's data CONTRADICT what Fatuma just said? Raise your hand.' Address contradictions by returning to the raw data: 'Let's look at both recording sheets side by side — what could explain the difference?' MASTER TABLE CONSTRUCTION (5 min): Teacher scribes on board as groups call out values. If a group's value conflicts with others: 'Two groups got different results for graphite conductivity in solid form. What could cause that difference in a real experiment? Think about the contact area of the probes — 10 seconds.' PATTERN STATEMENT TASK (8 min): Distribute the written sentence starters (pre-printed at bottom of	STRATEGY — Data-to-Pattern Sentence Starters: Providing scaffolded sentence starters with explicit gaps for 'bond type' vocabulary forces learners to move from descriptive observation ('NaCl dissolved') to mechanistic pattern language ('ionic compounds dissolve because…'). The third starter directly re-engages the anchoring phenomenon, preventing the investigation from becoming a disconnected lab exercise and maintaining the storyline thread toward the ultimate explanation.	OBSERVABLE INDICATORS: (1) Each Pattern Statement sentence contains a named bond type (ionic, covalent, metallic, giant covalent) — not just a substance name. (2) The third Pattern Statement attempts to distinguish diamond from graphite using structural language (even partial: 'different arrangement,' 'one has free electrons'). (3) Thumbs-up consensus on Starters 1 and 2 reaches at least 75% of class — if not, teacher re-teaches the specific row of the master table before proceeding. Teacher photographs the board master table for display in the classroom and for inclusion in the ARES platform lesson record.

	difference in ____ changes the property of ____ because…' Students share Pattern Statements in pairs, then one pattern statement per starter is shared with the class and refined collectively. The lesson closes with learners returning to their Prediction Cards and marking each prediction ✓ (confirmed) or ✗ (disconfirmed) with a one-word note: 'bond type = ____.'	recording sheet). Allow 4 minutes individual writing — SILENT. Then 2 minutes pair comparison. Cold-call 3 individual learners (one per starter) to share. After each, ask the class: 'Thumbs up if your pattern statement agrees, thumbs sideways if partly, thumbs down if your statement says something different.' Adjust the class pattern statement on the board based on show of thumbs. For Starter 3 (the carbon paradox statement), say: 'This one is the hardest — and we may not be able to complete it fully today. That is FINE. Write your best attempt, even if you have a question mark. That question mark is what will drive our learning in Lesson 9.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes of different colours: YELLOW = 'Something I now have evidence for' and BLUE = 'A new question the data raised.' Learners spend 3 minutes writing one note of each colour and then post them on the class DQB under the relevant driving question column. The teacher reads 3–4 sticky notes aloud, clustering them visibly on the board under emerging themes: 'Conductivity Pattern,' 'Melting Point Pattern,' 'The Carbon Paradox (still open).' The teacher explicitly points to the Carbon Paradox cluster: 'Notice that we now have strong evidence that bond type controls conductivity and melting point — but the puzzle of WHY graphite conducts and diamond does not, even though both are giant covalent structures, is still sitting here unanswered. That is exactly what Lesson 9 will unlock.'	DISTRIBUTE sticky notes before Phase 3 closes so learners are composing notes mentally during the Pattern Statement sharing. POSTING (3 min): Allow learners to walk to the DQB and post simultaneously — managed noise is acceptable. READ ALOUD 4 notes — choose 2 yellow and 2 blue deliberately: pick one yellow that correctly names a bond-type pattern, one yellow that is only descriptive (to model upgrading it: 'This says NaCl dissolves — can we add WHY? What bond type does NaCl have?'), one blue that points to the carbon paradox, and one blue that anticipates a future lesson (e.g., 'Why does aluminium conduct as a solid but NaCl doesn't?'). Say: 'Every blue note on this board is a promise that this unit will answer. Let me show you where each one gets answered.' Point to the lesson sequence poster on the wall: 'The carbon paradox blue notes — Lesson 9.	STRATEGY — Two-Colour DQB Protocol (Evidence vs. New Questions): Using distinct colours for 'evidence confirmed' versus 'new questions raised' makes the epistemological status of each idea visible to every learner. Clustering notes by theme models how scientists organise evidence into patterns. Explicitly pointing to the Carbon Paradox cluster as 'still open' maintains productive uncertainty and sustains motivation through the remaining lessons.	OBSERVABLE INDICATORS: (1) Every learner posts at least one yellow and one blue note — teacher scans the board to confirm; any learner without a blue note is prompted: 'The data raised a question for you — what is one thing you still cannot explain?' (2) Yellow notes reference observable evidence from today's stations (not prior knowledge from outside the lesson). (3) At least 3 blue notes cluster around the carbon paradox / graphite-vs-diamond conductivity question, confirming that the anchoring phenomenon remains productively unresolved.

		The aluminium question — that is Lesson 10, metallic bonding. You are building the answer, piece by piece.'		
Model Building Phase 📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Bond-Type Property Model' — a table-and-diagram sheet that has been building since Lesson 5. Today they add a new column: 'Physical Property Evidence (Lesson 8).' For each of the four bond types (ionic, simple molecular, giant covalent, metallic), learners add: (i) solubility result from today's data, (ii) conductivity result (solid and dissolved), (iii) melting-point ranking position. They then revise their earlier structural diagrams if today's evidence contradicts a prior entry — crossing out old entries with a single line (not erasing) and writing the new value in a different colour, with the note 'Revised based on: Station [A/B/C] evidence, Lesson 8.' Finally, in the 'Giant Covalent' row, learners draw a two-cell sub-table distinguishing diamond (no conductivity, no free electrons) from graphite (conducts, has delocalised electrons) — even if the explanation is incomplete. This incomplete distinction is labelled: 'QUESTION STILL OPEN — to be completed in Lesson 9.'	DISTRIBUTE cumulative model sheets (or direct learners to open their ARES platform model page). Say: 'Your model is a living scientific document — it should show your thinking CHANGING as evidence comes in. A model that never changes is not science, it is just copying. Use your second colour to show what today changed.' CIRCULATE during the 7-minute model update: TARGETED QUESTIONS: (1) 'Show me one thing you changed from your Lesson 7 model. Why did the evidence change it?' (2) 'In your Giant Covalent row — can you tell diamond and graphite apart yet? What is the ONE physical property that separates them?' COLD-CALL one learner to share their two-cell diamond/graphite sub-table with the class: 'Otieno, hold up your model — what did you write in the graphite conductivity cell? What did you write in the diamond cell? And what did you put in the explanation column?' After sharing, say to class: 'This is the exact question that drove scientists for decades. We now have the evidence. Lesson 9 gives us the explanation. Make sure that cell in your model has a question mark today — it is the most important question mark in the whole unit.'	STRATEGY — Cumulative Revision Model with Visible Change Tracking: Requiring learners to cross out (not erase) prior entries and add revised values in a contrasting colour makes the process of scientific model revision tangible and normalises changing one's mind based on evidence. The deliberately incomplete diamond/graphite sub-table creates a 'knowledge gap artefact' on the learner's own page — a physical reminder of the unresolved phenomenon that sustains curiosity into Lesson 9.	OBSERVABLE INDICATORS: (1) Every learner's model sheet shows at least one visible revision (cross-out + new value) in a second colour — a model with no revisions suggests the learner is not integrating today's evidence. (2) The Giant Covalent row contains a two-cell diamond/graphite distinction for conductivity. (3) The explanation cell for graphite conductivity contains either a correct partial explanation ('free electrons,' 'delocalised,' 'layers') OR an explicit question mark — both are acceptable at this stage. Teacher collects 6 model sheets (one per table group) as exit artefacts to review before Lesson 9 and identify misconceptions to address in the opening of that lesson.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) DATA QUALITY — Did all groups obtain clear results at Station B (conductivity)? If the conductivity tester failed for any group, what is the contingency for Lesson 9 — can those groups use the class master table as their evidence base? (2) PATTERN FORMATION — Were learners able to move from descriptive observations to bond-type pattern language in their Pattern Statements, or did most statements remain at the level of 'NaCl dissolved'? If so, Lesson 9 should open with explicit modelling of how to upgrade a descriptive statement to a mechanistic one. (3) CARBON PARADOX ENGAGEMENT — Did the re-injection of diamond and graphite data cards at Stations B and C generate visible curiosity (written questions, animated discussion)? If learners seemed indifferent to the paradox, consider opening Lesson 9 by re-doing the original demonstration (graphite conducts, diamond does not) live, before launching the explanation. (4) SAFETY AND LOVE VALUE — Was the Safety Monitor role taken seriously, or was it treated as a formality? Consider awarding a specific recognition (verbal praise, DQB acknowledgement) to the best Safety Monitor performance to reinforce the value of Love in practice. (5) EQUITY IN PARTICIPATION — Were the same learners dominating the Reporter and Recorder roles? Review group composition and reassign roles for Lesson 9 to ensure every learner leads at least one phase across the investigation sequence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners tested five substances (NaCl, sugar, graphite, SiO_2 sand, aluminium foil) at three investigation stations: (A) solubility in water, (B) electrical conductivity as solid and when dissolved, and (C) melting-point ranking from data cards. They recorded that NaCl dissolved fully and conducted in solution but not as a solid; sugar dissolved but did not conduct; graphite did not dissolve but conducted as a solid; SiO_2 sand did not dissolve and did not conduct; aluminium foil did not dissolve but conducted as a solid. Melting-point data cards showed giant covalent and metallic structures at the top of the ranking. Diamond and graphite data cards were added to the melting-point chart — both near the top — yet graphite conducted and diamond did not.
What did I learn?	Learners identified that bond type appears to determine physical properties: ionic compounds (NaCl) conduct only when dissolved because ions become free to move; simple molecular compounds (sugar) do not conduct even in solution because they do not form ions; giant metallic structures (aluminium) conduct as solids because of free electrons; giant covalent structures (SiO_2, diamond) do not conduct. They recognised that graphite is an anomaly within giant covalent structures — it conducts — and that this anomaly traces directly back to the anchoring phenomenon. Pattern Statements were recorded and partially completed, with the graphite/diamond distinction flagged as an open question for Lesson 9.
How does this explain the phenomenon?	Learners updated their cumulative Bond-Type Property Model with evidence from all three stations, adding a column for physical property data and revising earlier entries where new evidence contradicted prior predictions. They posted yellow (evidence confirmed) and blue (new questions) notes on the Driving Question Board, clustering open questions around the carbon paradox. The model's Giant Covalent row now contains a two-cell diamond/graphite sub-table distinguishing their conductivity results, with the explanation cell left intentionally incomplete and marked as the key unresolved question to be addressed in Lesson 9.

LESSON 9 (40 minutes): Physical Properties Investigation Part 3: Analysis, Explanation and DQB Update	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students analyse the data collected in Lesson 8 to construct evidence-based explanations linking bond type to physical properties, compile a class comparison table, and update the Driving Question Board with new evidence that advances understanding of the diamond-graphite paradox.
Knowledge	Students will be able to: (1) explain why ionic compounds conduct electricity only when dissolved or molten in terms of free-moving ions; (2) explain why simple molecular compounds do not conduct in any state in terms of the absence of ions or free electrons; (3) explain why giant metallic structures conduct in the solid state in terms of delocalised electrons; (4) explain why giant covalent structures generally do not conduct and have very high melting points; (5) identify graphite as an anomalous giant covalent structure and explain its conductivity using the concept of one delocalised electron per carbon atom in the layered structure.
Skills	Data analysis, constructing evidence-based scientific explanations, identifying patterns and anomalies, collaborative sense-making, scientific writing, updating and revising a Driving Question Board.
Attitudes	Intellectual honesty in revising predictions in light of evidence; persistence in constructing complete explanations; appreciation that anomalies such as graphite are productive puzzles rather than errors.
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	This lesson is the analytical pivot of the investigation sequence. Learners move from raw data (Lesson 8) to explanatory understanding. The graphite anomaly — flagged but unresolved at the end of Lesson 8 — is resolved here by connecting back to the anchoring phenomenon: graphite conducts because one electron per carbon atom is delocalised between layers, the same structural feature that makes it soft. This resolution gives students their most powerful partial answer yet to the driving question and directly advances the DQB.
Safety Notes	No new practical hazards in this lesson. If students handle residual samples from Lesson 8, they should wash hands after contact with iodine-stained glassware. All electrical apparatus should remain switched off unless supervised.

B. LESSON OVERVIEW
Lesson 9 opens with a 5-minute structured review of Pattern Statements from Lesson 8, during which students silently re-read their data tables. Students then work in groups to write full explanatory sentences for each of the four structure types, using a sentence-frame scaffold. A whole-class compilation builds a shared comparison table on the board. The graphite anomaly is resolved through targeted teacher questioning that reconnects to the original diamond-graphite demonstration. Students update their DQB sticky notes and write a structured exit explanation. The lesson runs across Predict, Observe, Explain, DQB Update and Model Building phases mapped to the 40-minute period.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students (5 minutes) silently re-read their completed data tables	Teacher projects or re-draws the blank data table on the board with	Activation of prior data through silent re-reading followed by pair	Teacher listens to the two named puzzles per student and uses

📹 **VIDEO:** [Creating a Poof and Not a Giant Kaboom] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	and Pattern Statement starters from Lesson 8. Each student writes two sentences in their notebooks: (1) the one result from Lesson 8 that surprised them most, and (2) the one result they feel most confident explaining and why. They then share briefly with a shoulder partner before whole-class sharing. This activates prior data and surfaces residual confusion — particularly around the graphite anomaly flagged at the end of Lesson 8.	the four unknowns labelled: NaCl (giant ionic), sugar (simple molecular), SiO2 sand (giant covalent) and aluminium (giant metallic). Teacher says: 'Before we construct our explanations, take two minutes in silence — look back at your data table from yesterday and identify the one result you cannot yet explain. Write it down.' WAIT TIME: 2 full minutes of silence. Teacher then asks: 'Who has a result they cannot explain? Do not give the answer yet — just name the puzzle.' Cold-calls two or three students. Teacher records the puzzles on the board, ensuring the graphite conductivity anomaly is captured. If no student names graphite, teacher prompts: 'Remember the conductivity tester we touched to the graphite pencil mark on the ceramic tile in Lesson 1 — what did it do? How does that connect to what you measured yesterday?'	sharing. Surfacing of anomalies before explanation phase so that the graphite puzzle is positioned as a productive question rather than a mistake.	these to gauge which aspects of the data remain uninterpreted. Students who cannot name any surprise may need additional scaffolding during the Explain phase. Students who name the graphite anomaly demonstrate readiness to engage with the unit-level phenomenon.
Observe Phase 📹 **VIDEO:** [Creating a Poof and Not a Giant Kaboom] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students (7 minutes) work in their investigation groups from Lesson 8 to review their collective data tables, reconcile any discrepancies between group members' records, and identify the three clearest patterns across all four substances. Each group also identifies the one result that does not fit the pattern they have identified — this is intended to surface graphite if it was tested, or the contrast between SiO2 and graphite if the teacher added a graphite conductivity result as a teacher demonstration data point. Groups record their three patterns and their anomaly on a half-sheet of paper.	Teacher circulates and asks each group: 'What pattern links the substances that conduct in the solid state? What do those substances have in common at the particle level?' WAIT TIME: allow groups to discuss for 30 seconds before teacher offers a hint if needed. Teacher also asks: 'Which of your four substances has the highest melting point based on the data cards? What does that tell you about the strength of the forces holding it together?' For groups that have already identified graphite as anomalous, teacher says: 'You flagged graphite. Hold that thought — we will build an explanation for it together in the Explain phase. For now, record	Group data reconciliation followed by explicit pattern identification and anomaly naming. Separating pattern recognition (Observe) from causal explanation (Explain) is deliberate — it models scientific reasoning and prevents premature closure.	Teacher collects the half-sheets at the transition to the Explain phase. A quick scan reveals whether groups have correctly identified: (a) metallic and ionic-in-solution as the conducting categories; (b) SiO2 and diamond as high-melting non-conductors; (c) sugar as low-melting non-conductor. Groups that conflate ionic conduction with metallic conduction will need targeted questioning during Explain.

		exactly what the data shows: does graphite conduct? Is it soft or hard?' Teacher does not yet explain the graphite anomaly — the explanation is reserved for the Explain phase to preserve the explanatory payoff.		
Explain Phase 📹 **VIDEO:** Creating a Poof and Not a Giant Kaboom Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students (18 minutes) work through a structured written explanation task using a sentence-frame scaffold provided on the board or a printed card. The scaffold reads: 'Substance [name] belongs to the [structure type] category. Its particles are held together by [bond type]. This means that [conductivity explanation]. Its melting point is [high or low] because [energy explanation]. This substance [does or does not] dissolve in water because [solubility explanation].' Each student completes the scaffold independently for all four substances (NaCl, sugar, SiO2, aluminium) in their notebooks. After 10 minutes of individual writing, groups share and compare sentences, then the teacher facilitates a whole-class compilation onto the board comparison table. The graphite anomaly is addressed last: teacher reveals that graphite is a giant covalent structure but conducts electricity, and asks students to recall from Lesson 1 what the conductivity tester showed on the graphite pencil mark. Teacher then explains that in graphite each carbon forms only three bonds, leaving one electron per atom free to move between layers — the same delocalised electron that makes graphite a conductor also means the layers are held together only by weak van der Waals	Teacher introduces the sentence-frame scaffold and says: 'I want complete scientific sentences — not just yes or no. Every explanation must mention particles, forces or electrons.' WAIT TIME: 10 minutes of supervised independent writing with teacher circulating and reading over shoulders. Teacher identifies two or three exemplary sentences and two or three incomplete ones for use in the whole-class share. During whole-class compilation, teacher uses targeted questioning: 'Ahmed, you wrote that NaCl conducts when dissolved — can you explain what happens to the ions when NaCl dissolves in water? What are ions doing that electrons cannot do in the solid?' WAIT TIME: 15 seconds. 'Wanjiku, you wrote that sugar does not conduct even in solution — what does that tell us about whether sugar forms ions?' For the graphite resolution, teacher says: 'Here is the final piece of today is puzzle. Both diamond and graphite are giant covalent structures — all carbon — but one conducts and one does not. Look at your Lesson 1 notes. What did the tester show on graphite?' After student responses, teacher draws a simple two-panel diagram: diamond with four bonds per carbon, no free electrons; graphite with three bonds per carbon, one free electron per	Scaffolded scientific writing using sentence frames to ensure explanations are mechanistic, not merely descriptive. Whole-class compilation of the comparison table makes individual thinking public and correctable. Explicit phenomenon connection at the graphite resolution anchors the explanation to the anchoring phenomenon and advances the driving question.	Teacher reads the four completed scaffold entries in each student notebook during circulation. Specific check: does the student use the word ions for ionic conduction and delocalised electrons for metallic conduction? Does the student distinguish between the two? At the graphite resolution, teacher asks a cold-call exit question: 'In one sentence, why does graphite conduct electricity but diamond does not — even though both are made of carbon?' Students write the answer in their notebooks before the DQB phase begins.

	forces, which is why graphite is soft. Students annotate their notebooks with a labelled diagram of graphite layers contrasted with the diamond tetrahedral network.	atom, layers held by van der Waals forces. Teacher says: 'This is the same carbon atom. The bonding arrangement alone changes every property. This is the heart of our driving question.' Teacher connects explicitly: 'Write this in your notebooks as your Phenomenon Connection statement for today.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** DNA as the "transforming principle" Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Covalent and Ionic Bonding (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students (6 minutes) return to the class Driving Question Board. Each student collects two sticky notes. On the first note they write one question from the board that they can now answer with evidence from the investigation. On the second note they write one new question the graphite-diamond resolution has raised for them or one question that remains unanswered. Students post the answered questions in a green column (We Can Now Explain) and the new or remaining questions in a yellow column (Still Wondering). The teacher reads three or four of the new yellow notes aloud to the class and previews how Lessons 10 through 13 will address them.	Teacher directs students to the DQB and says: 'Look at the questions we posted in Lessons 1 and 7. Find one question you can now answer using evidence from our investigation. Write your answer on a green sticky note — include the evidence. Then write one question the graphite explanation has opened up for you — maybe about uses, maybe about other carbon forms, maybe about why anyone would need a substance that is both a non-metal and a conductor.' WAIT TIME: 3 minutes for writing. Teacher facilitates posting and reads three yellow notes aloud, responding: 'That question about why graphite is used in batteries — we will investigate exactly that in Lesson 10 when we look at real uses in Kenyan dry-cell batteries.' Teacher ensures the core driving question card is visible and says: 'We have now built most of the answer to our big question. Diamond and graphite differ entirely because of HOW their carbon atoms are bonded. Four bonds in 3-D gives diamond its hardness and insulation. Three bonds in layers with one free electron gives graphite its softness and conductivity. Structure determines properties. Properties determine uses. That is the story of chemical	Differentiated sticky-note posting (answered vs. still wondering) makes the class thinking visible and celebrates explanatory progress while honouring genuine remaining curiosity. The teacher preview of Lessons 10 to 13 provides narrative continuity.	Teacher scans the green notes for accuracy of evidence-based explanations. A green note that says only 'NaCl conducts because it is ionic' without mentioning free-moving ions is flagged for follow-up. Teacher photographs the DQB at the end of the lesson for the class record and for use in Lesson 13 final revision.

		bonding.'		
Model Building Phase 📹 **VIDEO:** [Creating a Poof and Not a Giant Kaboom] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Covalent and Ionic Bonding (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students (4 minutes) complete a structured exit ticket in their notebooks. The exit ticket has three parts: (1) Complete the comparison table — fill in any remaining cells for the four structure types across the properties: conductivity in solid state, conductivity in solution or molten state, melting point category (high or low), solubility in water, and example substance. (2) Write a Phenomenon Connection sentence: 'Diamond and graphite have different properties because...' using today is explanation. (3) Circle the bond type that explains the anomalous conductivity of graphite: ionic bond, metallic bond, covalent bond with delocalised electrons, hydrogen bond. Students hand in the exit ticket or teacher circulates to mark the circled answer before dismissal.	Teacher says: 'You have four minutes for your exit ticket. Part 3 is the key question for today — choose carefully and be ready to justify your choice.' WAIT TIME: 4 minutes of silent individual work. Teacher circulates and stamps or ticks completed tickets. Teacher closes with: 'Next lesson we move from properties to uses — we will find diamond at the Nairobi gem cutters, graphite in the dry-cell battery in your torch at home, aluminium in the mabati roofing on Kenyan buildings, and SiO_2 in the glass of this very window. Every use is explained by the bonding we have just described. See you in Lesson 10.' Teacher collects exit tickets for formative review before Lesson 10.	The three-part exit ticket assesses knowledge (comparison table), explanatory reasoning (phenomenon connection sentence) and conceptual discrimination (identifying the correct bond type for graphite). The closing teacher narrative bridges to Lesson 10 using Kenyan contexts — mabati roofing, dry-cell batteries, gem cutting — to sustain relevance and motivation.	Exit ticket Part 1 reveals table completeness; Part 2 reveals quality of causal reasoning about the phenomenon; Part 3 reveals conceptual precision about delocalised electrons in covalent structures. Teacher uses Part 3 errors to plan targeted re-teaching at the start of Lesson 10 if needed. Expected correct answer for Part 3: covalent bond with delocalised electrons.

D. TEACHER REFLECTION

Before Lesson 10, review all exit tickets and tally: (1) How many students correctly identified delocalised electrons as the explanation for graphite conductivity? If fewer than 70 percent, open Lesson 10 with a 3-minute revisit of the graphite diagram before moving to uses. (2) How many comparison tables are fully completed? Students with two or more blank cells need a completed reference table. (3) Do any Phenomenon Connection sentences conflate ionic and covalent bonding? These students may need a paired peer-explanation task early in Lesson 10. Also reflect: did the graphite resolution land as a genuine aha moment, or did it feel like teacher-delivered information? If the latter, consider whether a short PhET simulation or the graphite conductivity live demo from Lesson 1 could be revisited at the start of Lesson 10 to re-ground the explanation in observable evidence before moving to uses.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed from their Lesson 8 data that NaCl conducts only when dissolved, sugar does not conduct at all, aluminium conducts in the solid state, SiO_2 does not conduct, and graphite — a giant covalent structure like SiO_2 — conducts electricity, making it an anomaly among the four structure types tested.
What did I learn?	Students learned that the four structure types differ systematically in conductivity, melting point and solubility because of differences in the particles present and the forces or bonds between them: ionic lattices require free-moving ions for conduction;

	metallic lattices have permanently delocalised electrons; simple molecular substances have no ions or free electrons in any state; giant covalent structures generally have no free particles for conduction. Graphite is the critical exception: its three-bond-per-carbon arrangement leaves one delocalised electron per atom free to move between layers, giving it metallic-like conductivity despite being a covalent structure of non-metal atoms.
How does this explain the phenomenon?	The properties of a substance are determined not merely by which atoms are present but by how those atoms are bonded and arranged. Diamond and graphite both consist entirely of carbon atoms, yet their different bonding arrangements — four bonds in a 3-D tetrahedral network versus three bonds in a layered hexagonal structure with delocalised electrons — produce completely opposite properties: hardness versus softness, electrical insulation versus electrical conductivity, transparency versus opacity. This is the structural explanation for the anchoring phenomenon and the central answer to the driving question.

LESSON 10 (80 minutes): Resultant Structures & Uses: From Bond Type to Real-World Application	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To relate bond types and resultant structures to the uses of elements, molecules and compounds in everyday Kenyan life.
Knowledge	Students know that the physical properties of a substance (hardness, conductivity, melting point, solubility) arise directly from its bond type and resultant structure, and can match specific substances (diamond, graphite, NaCl, Al, SiO_2) to their uses on the basis of those properties.
Skills	Students can (i) construct a properties-to-uses argument for at least four substances using evidence gathered in previous lessons; (ii) brainstorm and evaluate Kenyan examples of materials whose uses are explained by their bonding; (iii) present findings clearly to peers in a paired format.
Attitudes	Students appreciate that understanding chemical bonding is not abstract — it directly explains the materials woven into Kenyan daily life, from the pencil in a student's hand to the aluminium sufuria on a jiko, and show care (Love) for peers during group activities.
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	Lesson 10 is the synthesis lesson of the evidence-gathering arc. Students have spent nine lessons building evidence about valence electrons, bond types, Lewis structures, and physical-property investigations. Now they turn that evidence outward: Why does the Nairobi jeweller use diamond to cut glass but graphite to lubricate the cutter's guide rail? Why is NaCl rubbed into nyama choma to preserve it? Why is aluminium chosen for jua-kali cookware? Why is SiO_2 melted to make windows? Every answer circles back to the anchoring paradox — the same element (carbon) bonded differently produces radically different outcomes — and proves that bond type and structure are the master key to material selection.
Safety Notes	No hazardous chemicals are used in this lesson. If physical samples (NaCl crystals, graphite pencil, aluminium foil) are handled, remind students not to taste any laboratory sample even if the substance is food-grade. Wash hands after handling samples. Broken glass or ceramic tiles from previous lessons must be disposed of in the sharps container.

B. LESSON OVERVIEW

In this 80-minute synthesis lesson, students move from evidence to application. They open by revisiting the anchoring phenomenon (diamond vs. graphite) and articulating what nine lessons of evidence have taught them. They then work through a structured Claim–Evidence–Reasoning (CER) activity in pairs, matching five key substances (diamond, graphite, NaCl, aluminium, SiO_2) to real Kenyan uses and justifying each match with bond-type and structural evidence. A 'Kenyan Materials Brainstorm' gallery walk broadens the picture beyond the five anchor substances. Pairs present one substance each to the class, followed by a whole-class DQB update and a cumulative model-revision task. The lesson closes with an exit ticket that asks students to evaluate a novel material — whether copper wire or an NaCl solution could replace graphite in a dry-cell battery — requiring them to apply bonding principles to an unfamiliar scenario.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown three	Display the three objects	Activation of Prior Knowledge via	Circulate and read over shoulders

📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	unlabelled objects on the front bench: a small piece of aluminium foil (from a Kenyan mandazi wrapping), a glass bottle (common Kenyan soda bottle, made of SiO_2-based glass), and a graphite rod from a disassembled size-D dry-cell battery (torch battery, ubiquitous in Kenyan homes). Without being told the bond types, students individually write a prediction in their notebooks responding to the prompt: 'Choose ONE of these three objects. Predict what bond type holds its atoms together, and predict TWO physical properties you would expect that substance to have based on that bond type. Then suggest why that bond type makes it suitable for its use in this object.' Students work in silence for 4 minutes, then share with a shoulder partner for 2 minutes before three cold-calls share with the class.	prominently. READ the prediction prompt aloud slowly, then post it on the board. Say: 'You have everything you need from the last nine lessons — trust your evidence.' Allow 4 minutes of SILENT individual writing (enforce silence; circulate and note misconceptions without correcting yet). At 4 minutes say: 'Share your prediction with your shoulder partner — you have 2 minutes. Convince them with bonding evidence.' After partner share, cold-call EXACTLY 3 students (choose one per object): 'Amina, tell us your prediction for the aluminium foil — what bond, what properties, what use?' Apply 12-second WAIT TIME before accepting an answer. Do NOT confirm or deny predictions — say: 'Hold that thought. Let's see what the evidence says.'	Prediction: By committing a bond-type prediction to writing before instruction, students surface their current mental models. This creates cognitive tension when evidence later confirms or challenges the prediction, deepening encoding of the correct reasoning.	during silent writing. Look for: (i) correct bond-type identification (metallic for Al, giant covalent/giant atomic for SiO_2 glass, giant covalent/layered covalent for graphite); (ii) at least two plausible properties linked to bond type; (iii) a use-justification that mentions structure. Flag students who confuse graphite with simple molecular — they need targeted support in Phase 3. Do NOT intervene yet.
Observe Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students engage in two sequential evidence-gathering activities. ACTIVITY A (10 min) — 'Substance Dossiers': Each pair receives a printed or screen-displayed 'Substance Dossier' card for one of five substances (diamond, graphite, NaCl, aluminium, SiO_2). Each dossier contains: the substance name, its bond type and structural description (from prior lessons), a data table of physical properties (melting point, hardness, conductivity, solubility), and three photographs of Kenyan uses (e.g., for graphite: a pencil, a dry-cell battery cross-section, a lubricant canister from a Nairobi garage). Pairs study their dossier and complete a 3-column table:	Distribute dossier cards to pairs before class begins or project on screen. Say: 'Your pair has one substance to become expert on. In 10 minutes you must be able to explain to the class: what holds it together, what that structure gives it, and where a Kenyan would encounter it.' During Activity A, circulate and ask probing questions (do NOT give answers): 'Why does that melting point make sense for that bond type?' and 'Where have you seen evidence for that claim in a previous lesson?' At the 10-minute mark say: 'Stand up — gallery walk time. Take your notebook and three sticky notes. You have 10 minutes to visit all six posters. On Mystery Material X, I want your best	Jigsaw Evidence Gathering + Gallery Walk: The dossier system creates distributed expertise — each pair develops deep knowledge of one substance and surface knowledge of four others via the gallery walk. This mirrors how scientists specialise and then share findings at conferences, making the knowledge-sharing phase (Phase 3) feel authentic and purposeful.	Collect one sticky note per pair from the Mystery Material X poster at the end of the gallery walk. Check for: (i) correct identification of giant covalent/giant atomic structure; (ii) use of hardness and melting-point evidence; (iii) analogy to diamond (both giant covalent, both very hard). Any pair hypothesising 'ionic' or 'metallic' for SiC needs explicit re-teaching during Phase 3. Also scan 3-column tables during Activity A circulation — watch for the common error of attributing graphite's conductivity to covalent bonds rather than to delocalised electrons.

	Property | Structural Explanation | Kenyan Use. ACTIVITY B (10 min) — 'Kenyan Materials Gallery Walk': Six A3 posters are fixed around the room, each with a large photograph of a Kenyan material in use: (1) jua-kali aluminium sufuria, (2) Lake Magadi soda ash ($NaCl/Na_2CO_3$ context), (3) pencil graphite on exam paper, (4) Nairobi skyscraper glass façade (SiO_2), (5) diamond-tipped tile cutter at a Mombasa hardware shop, (6) a NEW mystery poster — silicon carbide (SiC) grinding wheel used in a Kisumu workshop, labelled only 'Mystery Material X — extremely hard, very high melting point, does not conduct electricity.' Pairs circulate with sticky notes, adding bond-type hypotheses and use-justifications to each poster.	hypothesis about bond type.' During gallery walk, park yourself near the Mystery Material X poster and listen to student reasoning without intervening. Cold-call 2 students at the mystery poster: 'Kariuki, what bond type are you hypothesising for Mystery X and why?' Apply 10-second WAIT TIME.		
Explain Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The Explain phase has three parts. PART A — Pair Presentations (15 min): Each of the five substance-expert pairs presents their completed 3-column table to the class in a strict 2-minute slot. The presenting pair must answer one question from the class and one from the teacher. The class completes a 'listening grid' — a blank version of the 3-column table for all five substances — filling it in as each pair presents. PART B — CER Synthesis (7 min): Individually, students write a Claim–Evidence–Reasoning (CER) paragraph responding to: 'A jua-kali artisan in Nairobi needs to choose between an aluminium rod and a graphite rod to conduct electricity along a handle fitting. Using bond type and structure, justify which material is better suited and explain why the other is	For Part A, use a visible 2-minute timer. After each presentation say: 'One question from the floor — who has evidence that supports or challenges what was just presented?' Cold-call 1 student per presentation (5 total). Apply 12-second WAIT TIME each time. For graphite presentation, if students do not mention delocalised electrons, prompt: 'You said graphite conducts — which part of graphite's structure makes that possible? Think back to Lesson 7.' For Part B, say: 'This is individual silent work — 7 minutes. I am looking for the words metallic bonding, delocalised electrons, giant covalent, layered structure, and a reference back to our diamond-and-graphite phenomenon.' Circulate; stamp or initial completed CER paragraphs. For Part C, draw a quick structural	Claim–Evidence–Reasoning (CER) Writing: CER forces students to distinguish between an assertion (claim), the empirical data that supports it (evidence from previous lessons), and the logical connection between them (reasoning via bonding principles). This is the core of scientific argumentation (NGSS Science & Engineering Practice 7: Engaging in Argument from Evidence) and directly advances the driving question.	Collect CER paragraphs at the end of Part B (or photograph with teacher device). Assess against a 3-point rubric: (1) Claim — correct material identified with a reason; (2) Evidence — at least two structural/property facts cited; (3) Reasoning — logical link made using bonding vocabulary. Target: ≥80% of students score 2–3 points. Students scoring 0–1 are identified for a 5-minute small-group re-teach at the start of the next lesson using the carbon paradox as the re-entry point.

	unsuitable.' Students must use the terms: metallic bonding, delocalised electrons, giant covalent structure, layered structure, and reference the anchoring phenomenon. PART C — Mystery Material X Reveal (3 min): Teacher reveals that Mystery X is silicon carbide (SiC), a giant covalent structure similar to diamond — used in grinding wheels because of extreme hardness and high melting point. Students annotate their gallery-walk sticky notes with the correct explanation.	comparison on the board: diamond (C–C tetrahedral network) vs. SiC (Si–C tetrahedral network, same geometry, similar hardness). Ask: 'What does this tell us about what controls hardness?' Cold-call 2 students, 10-second WAIT TIME each.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. Students re-read the anchoring driving question aloud together: 'Why do diamond and graphite — both made of only carbon atoms — have such completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for?' Each student writes on a sticky note: one thing they can NOW fully explain about the carbon paradox (green note) and one question that today's lesson has raised or that remains unresolved (orange note). Notes are placed on the DQB in the appropriate column. The teacher reads three green notes and three orange notes aloud to the class, narrating how the green notes represent accumulated understanding and the orange notes will guide Lessons 11–13.	Hand out pre-cut green and orange sticky notes. Say: 'We have been asking this driving question since Lesson 1. Today, after 10 lessons of evidence, I want to see how much of that question you can now answer with confidence.' Read the driving question aloud together — ask the class to read it chorally. Allow 4 minutes for individual note-writing. Collect DQB updates. Read three green notes aloud (choose ones that explicitly reference today's substances), affirming: 'This student has connected graphite's layered structure to its use in a dry-cell battery — that is exactly the kind of reasoning a materials scientist uses.' Read three orange notes and say: 'These unresolved questions — about intermolecular forces, about why metals are malleable, about how SiO_2 forms glass — are exactly what Lessons 11 to 13 will address.' Do NOT answer the orange-note questions yet.	Driving Question Board Metacognitive Check-in: The DQB is a living document of the class's collective understanding. Separating 'what I can now explain' from 'what remains puzzling' builds metacognitive awareness — students learn to distinguish between knowing and not-yet-knowing, a foundational scientific habit. The public sharing of both types of notes normalises intellectual humility.	Read all green notes before the next lesson. Look for: (i) specific structural language (tetrahedral network, layered/hexagonal sheets, delocalised electrons, ionic lattice, metallic lattice, giant covalent); (ii) explicit connection to a Kenyan use from today; (iii) reference to the original diamond-graphite paradox. Any class where fewer than 60% of green notes contain structural language signals that Phases 2–3 need reinforcement in Lesson 11's opening.
Model Building	Students open their cumulative model portfolios (started in Lesson	Distribute model portfolios. Say: 'Your model has been growing	Cumulative Model Revision with Use Annotation: Adding 'uses'	Review model portfolios before Lesson 11. Assess the 'Carbon

Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	1). They revise or extend their existing models in two steps. STEP 1 (8 min) — 'Uses Annotation': Students annotate their existing structural diagrams for diamond, graphite, NaCl, aluminium, and SiO_2 with at least one specific Kenyan use per substance, drawing an arrow from the structural feature that explains the use (e.g., an arrow from 'delocalised electrons' in the graphite model pointing to 'dry-cell battery electrode — allows electron flow'). Students who have not completed all five structural diagrams use this time to complete the missing ones with peer support. STEP 2 (7 min) — 'Comparison Panel': Students add a new page to their portfolio titled 'The Carbon Paradox — Explained'. On it they draw a two-column comparison of diamond and graphite, with rows for: bond type, structural description, three physical properties, and three Kenyan uses. This panel must be self-sufficient — a reader with no prior knowledge of the unit should be able to understand the paradox from this page alone.	since Lesson 1. Today you are going to make it useful — annotate every structure with at least one Kenyan use and the structural reason for that use.' Circulate during Step 1 and prompt: 'I can see you have drawn the graphite layers correctly — now draw an arrow and tell me what property those layers give graphite that makes it useful in a pencil.' After 8 minutes say: 'Now turn to a blank page. Title it: The Carbon Paradox — Explained. You have 7 minutes. This is your personal synthesis — make it clear enough for a Form 2 student to understand.' At the 2-minute warning say: 'If you have not started the Kenyan uses column, start there now — this is the bridge between chemistry and real life.' Collect portfolios at the end of the lesson for teacher review; return at the start of Lesson 11.	annotations to structural models forces students to traverse the full causal chain: atomic structure → bond type → resultant structure → physical properties → real-world application. This is the highest-order cognitive move in the unit and directly answers the driving question. The 'Carbon Paradox — Explained' panel acts as a self-assessment artefact revealing the depth of each student's conceptual integration.	Paradox — Explained' panel against three criteria: (A) Structural accuracy — correct bond type and structural description for both diamond and graphite; (B) Property–structure linkage — at least two properties correctly derived from structure for each substance; (C) Kenyan use relevance — at least two specific, correct Kenyan uses per substance with structural justification. Flag any portfolio missing criterion A for targeted re-teaching. Use exemplary panels (anonymised) as anchor models for Lesson 11's opening.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Were students able to move independently from 'bond type' to 'property' to 'use' without prompting, or did they still need structural scaffolding? If most students required prompting during the CER task, the property–structure connection (covered in Lessons 7–9) needs consolidation in Lesson 11's warm-up. (2) Did the Kenyan contexts feel authentic to students, or did they treat the examples as decorative? If students did not spontaneously generate their own Kenyan examples during the gallery walk, consider bringing in more tangible local objects in Lesson 11 (a piece of aluminium sufuria, a soda bottle, a broken pencil). (3) How did students perform on the Mystery Material X (SiC) task? Strong performance here signals genuine structural reasoning — they are not just memorising diamond and graphite but transferring the giant covalent framework to a new substance, which is the target competency. (4) Review the orange sticky notes from the DQB. Do the unresolved questions cluster around metallic bonding, intermolecular forces, or period trends? Use this to prioritise content emphasis in Lessons 11–13. (5) Were the 2-minute pair presentations equitable? Note which student in each pair did most of the talking and engineer roles in Lesson 11 so the quieter partner leads.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined Substance Dossier cards containing physical-property data tables and Kenyan-use photographs for diamond, graphite, NaCl, aluminium, and SiO_2, and conducted a gallery walk around six Kenyan-context material posters including a mystery substance (silicon carbide) used in a Kisumu grinding wheel.
What did I learn?	Students learned that the real-world use of a material is determined by its bond type and resultant structure: diamond's 3-D tetrahedral giant covalent network makes it the hardest natural substance (used in cutting tools); graphite's layered structure with delocalised electrons makes it a conductor and lubricant (used in dry-cell batteries and pencils); NaCl's ionic lattice makes it soluble and non-toxic (used in food preservation); aluminium's metallic lattice makes it malleable and conductive (used in Kenyan cookware and construction); SiO_2's giant covalent network gives it a very high melting point and transparency (used in glass-making).
How does this explain the phenomenon?	Students explained, through Claim–Evidence–Reasoning paragraphs and cumulative model annotations, how the bond type of each substance produces the specific physical properties that make it suitable for its Kenyan use — and extended this reasoning to a novel giant covalent substance (SiC), bringing the anchoring paradox (diamond vs. graphite) to a full structural resolution.

LESSON 11 (80 minutes): Animations, Simulations & Digital Literacy: Seeing Chemical Bonding in Motion	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To use digital animations and simulations to visualise chemical bonding and resultant structures, and to critically evaluate what digital models show accurately and what they simplify.
Knowledge	Learners will know that digital simulations model ionic, covalent, dative covalent, metallic, hydrogen bond, and Van der Waals structures; that all models are simplifications; and that the 3-D tetrahedral network of diamond and the layered hexagonal structure of graphite arise directly from how carbon atoms bond.
Skills	Learners will be able to: watch and interpret bonding animations; compare simulation outputs against their own Lewis diagrams and physical-property data from earlier lessons; identify strengths and limitations of each digital model; and articulate how simulated structures explain the properties of diamond, graphite, NaCl, and aluminium metal.
Attitudes	Learners appreciate the power of digital literacy as a learning tool while maintaining critical thinking — recognising that every model, digital or hand-drawn, is an approximation of reality. Learners show care and respect for peers during structured discussion (Value: Love).
Key Inquiry Question	How does the type of chemical bond in a substance determine its physical properties and uses?
Purpose in Storyline	In Lessons 1–10 learners built Lewis diagrams, investigated physical properties experimentally, and brainstormed structure–property relationships. Lesson 11 now brings those findings into dialogue with curated digital animations and simulations, giving learners a dynamic, 3-D visualisation of the same bonding concepts. By critically evaluating each simulation against their own evidence, learners deepen their understanding of why diamond and graphite — both pure carbon — are so different, and they consolidate digital literacy as a core competency before the project work in Lesson 12.
Safety Notes	No hazardous chemicals are used in this lesson. If learners use shared devices, remind them to handle tablets/laptops with care. Ensure screen brightness is comfortable to avoid eye strain during extended viewing. If the school uses an internet connection, supervise browsing to approved simulation sites only (e.g. PhET Interactive Simulations). Offline ARES platform videos may be used where internet is unavailable.

B. LESSON OVERVIEW

Learners work in groups of four at devices (tablets, laptops, or the ARES offline platform) to watch three curated animations/simulations covering: (1) ionic bonding and the NaCl lattice, (2) covalent and giant covalent structures including diamond and graphite, and (3) metallic bonding and the electron-sea model for aluminium. After each simulation, groups complete a structured 'Simulation Critique Card' comparing what they see on screen against their Lewis diagrams and physical-property tables from earlier lessons. A whole-class structured debrief uses the Carbon Paradox phenomenon as the anchor: students must explain why the diamond simulation shows no free electrons and the graphite simulation shows delocalised electrons moving between layers. The lesson closes with a DQB update and a model-revision step in which learners annotate their cumulative carbon bonding model with new insights from the simulations.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write a 'Digital Prediction Card': Before watching any simulation, they sketch from memory (a) the structure of diamond, (b) the structure of graphite, and (c) the NaCl lattice — labelling bond types, approximate bond angles, and whether free electrons are present. They then write one sentence predicting: 'I think the simulation will show _______ about graphite's conductivity that our earlier experiment could not show, because _______.' Cards are kept face-down until after Phase 2.	Display the anchoring phenomenon images (diamond chip and graphite pencil) on the board. Say: 'Before we open a single simulation, I want you to show me what YOU already know. Sketch the three structures from memory — no notes, no phones. You have four minutes. Go.' [Allow 4 minutes of silent individual work.] After 4 minutes say: 'Now write your one prediction sentence about graphite conductivity. What do you think the simulation will reveal that our bulb-and-wire experiment last week could not?' [Allow 2 minutes.] Cold-call THREE learners to read their prediction sentences aloud. After each, ask the class: 'Hands up — who made a similar prediction?' [Wait 10 seconds.] Do NOT confirm or correct predictions — say: 'Hold that thought. The simulation will answer you shortly.'	Activation of Prior Knowledge through Retrieval Sketching — by drawing structures from memory before seeing the simulation, learners surface what they actually understand versus what they merely recognise, creating a productive cognitive gap that the simulation will fill.	Teacher circulates and scans sketches: observable indicator is whether learners correctly recall tetrahedral geometry for diamond vs. layered hexagons for graphite. Note learners who draw graphite without delocalised electrons — these students need targeted questioning during Phase 3.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four rotate through three simulation stations (10 minutes each), each accompanied by a printed 'Simulation Critique Card' with four columns: (A) What does the simulation SHOW? (B) How does this MATCH our Lewis diagrams / property tables from earlier lessons? (C) What does the simulation SIMPLIFY or leave out? (D) One new thing this simulation helped me understand about the Carbon Paradox. Station 1 — Ionic Bonding (NaCl lattice): Learners watch the ARES-embedded animation of Na^+ and Cl^- ions arranging into a 3-D lattice, observing the transfer of one valence electron from sodium to chlorine, the resulting electrostatic attraction, and the high melting point implication. Station 2 —	Before groups begin, give explicit instructions: 'You have exactly 10 minutes per station. One person operates the device, one person writes on the Critique Card, one person is the timekeeper, and one person is the presenter who will share your findings in the debrief. Rotate roles at each station.' [Project a visible 10-minute countdown timer.] Circulate continuously. At Station 2 (diamond/graphite), crouch to group level and ask: 'Point to where the free electron is in the diamond simulation.' [Wait 10 seconds.] If learners point incorrectly, say: 'Look again — can you find ANY free electron in diamond's structure?' At Station 3 pause a group and ask: 'Your conductivity experiment showed	Structured Observation with a Critique Card Framework — the four-column card forces learners to move beyond passive watching into active comparison and evaluation, directly linking digital evidence to physical evidence gathered in previous lessons. This is a core Digital Literacy competency (KICD).	Collect one Critique Card per group at the end of Phase 2. Observable indicators: (1) Column B entries correctly reference specific Lewis diagrams or melting point data from earlier lessons; (2) Column D explicitly mentions electrons, bond geometry, or layers in relation to the Carbon Paradox. Groups whose Column D entries are vague ('it showed bonding') receive a probing question from the teacher before Phase 3.

	Giant Covalent Structures (Diamond & Graphite): Learners use the PhET 'Molecule Shapes' simulation or ARES equivalent to build and rotate a tetrahedral carbon unit, then watch an animation of graphite's layered structure showing pi-electron delocalisation between layers. Station 3 — Metallic Bonding (Aluminium): Learners watch an electron-sea animation for aluminium, observing the lattice of positive ions surrounded by delocalised electrons, connecting this to conductivity and malleability data from Lesson 9.	aluminium conducts — where in THIS simulation can you SEE why that happens?' [Wait 12 seconds.] Cold-call two groups at Station 1 to read column (C) of their Critique Card aloud before the station rotation ends.		
Explain Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class structured debrief anchored to the Carbon Paradox. Each group's designated presenter shares one key finding from their Critique Cards. The class then works through three 'Simulation vs. Reality' discussion questions projected on the board: (1) 'The graphite simulation showed electrons moving freely between layers. How does this explain our conductivity experiment result from Lesson 8 — where graphite lit the bulb but diamond did not?' (2) 'The NaCl simulation showed ions in a rigid lattice. How does this explain why NaCl dissolved in water but diamond did not, in our solubility investigation?' (3) 'What is ONE thing about the diamond structure that NO simulation can fully show — and why does that limitation matter for a jeweller in Nairobi using a diamond-tipped cutting tool?' Learners write individual 'Explanation Exit Sentences' answering: 'Digital simulations help us understand the Carbon Paradox because ________, but they simplify the reality by	Begin debrief by projecting the two Carbon Paradox images. Say: 'Every simulation you just watched ultimately explains THIS — two substances, both 100% carbon, completely different. Let's use your Critique Cards to prove it.' Call on each group's presenter in turn. After each presentation, ask the class: 'Does anyone want to add to or challenge what Group ________ said?' [Wait 10 seconds each time.] For Discussion Question 1, cold-call a learner who previously drew graphite WITHOUT delocalised electrons in Phase 1 — give them the opportunity to self-correct: 'Amina, in your prediction sketch earlier you didn't show free electrons in graphite. After the simulation, what would you change?' For Discussion Question 3, say: 'This is a harder question — I want you to discuss in your group for 90 seconds before I call on anyone.' [Circulate, listen, then cold-call TWO groups.] Close by saying: 'Write your Explanation Exit Sentence individually — no group	Claim-Evidence-Reasoning (CER) Structured Debrief — learners are required to link each simulation observation (evidence) back to a specific experimental result from a previous lesson (claim) and articulate a mechanistic explanation (reasoning). This directly addresses KICD SLO (b): investigating the relationship between bond types and physical properties.	Collect Explanation Exit Sentences at the door. Observable success indicator: the sentence must name a specific bond type or structural feature (e.g., 'delocalised electrons between graphite layers', 'rigid 3-D tetrahedral network in diamond') AND identify a specific simplification (e.g., 'simulations don't show the actual scale of the lattice' or 'animations show electrons as dots but electrons are probability clouds'). Sentences that only state general observations indicate the learner needs re-teaching in Lesson 12.

	_______.'	discussion. I will read these before Lesson 12.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the GREEN note they write one thing they can now answer about the driving question ('Why do diamond and graphite have such different properties?') thanks to the simulations. On the ORANGE note they write one new question the simulations raised — something the digital model showed that they do not yet fully understand. Learners post both notes on the class DQB under the appropriate column headings: 'We can now explain…' and 'We still wonder…'. The class briefly reads the new orange questions aloud to check whether any can be answered immediately from prior lessons.	Distribute sticky notes and say: 'The simulations should have answered some questions on our DQB — and probably raised new ones. Take three minutes to write both notes.' [Allow 3 minutes of silent writing.] As learners post notes, scan for misconceptions on orange notes. Read THREE orange notes aloud to the class and ask: 'Has anyone written something on their GREEN note that answers THIS question?' [Wait 10 seconds per question.] If a common new question emerges (e.g., 'Why do Van der Waals forces between graphite layers allow them to slide?'), say: 'That is an excellent question — let's mark it with a star. We will return to it in our project work.' Update the DQB legend visibly: 'Lesson 11 — Digital Evidence Added.'	Two-Colour DQB Update (Progress Tracking + New Inquiry) — the dual sticky-note format makes visible both the growing explanatory power of the class AND the productive questions that remain, reinforcing that science is an ongoing process of building and refining models. This directly reflects the KICD Learning to Learn competency.	Teacher photographs the updated DQB. Observable indicator: GREEN notes must reference specific structures or bond types (not vague statements like 'carbon is different'). A class where more than half of GREEN notes are vague signals that the structured debrief in Phase 3 was insufficient and a brief whole-class recap is needed at the start of Lesson 12.
Model Building Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative 'Carbon Bonding Model' — a large poster or notebook page they have been annotating since Lesson 1 of this sub-strand. Using a different coloured pen (reserved for digital evidence), they add three annotations: (1) On the diamond structure, they label the 3-D tetrahedral network and write 'No free electrons → no conductivity — confirmed by simulation AND experiment.' (2) On the graphite structure, they label the delocalised pi electrons between layers and write 'Free electrons → conducts electricity → confirmed by simulation AND experiment.' (3) They add a new 'Model Limitations' box at the bottom of the poster: 'What simulations	Say: 'Open your Carbon Bonding Model — your running poster from this unit. Pick up the [designated colour] pen. You have eight minutes to add three annotations. I will be checking that annotation 1 and 2 each reference BOTH your experimental evidence AND the simulation. That dual reference is what makes your model powerful.' [Circulate and prompt with: 'Where is your experimental evidence from Lesson 8? Write the lesson number next to your annotation so we can trace it.'] With 2 minutes remaining, cold-call FOUR groups to share their 'Model Limitations' box entry. After the fourth group, say: 'Notice — every group identified a DIFFERENT limitation. That tells us that even the best	Cumulative Model Revision with Dual-Source Annotation — by explicitly requiring learners to reference both experimental evidence and simulation evidence in the same annotation, the teacher builds the scientific practice of triangulating across multiple evidence sources (NGSS Science and Engineering Practice 2: Developing and Using Models; Practice 8: Obtaining, Evaluating, and Communicating Information).	Teacher spot-checks five models before learners leave. Observable success indicator: both annotations 1 and 2 contain a cross-reference to at least one prior lesson (e.g., 'from conductivity experiment, Lesson 8') AND to the simulation. The 'Model Limitations' box must name a specific limitation, not a generic one. Models lacking cross-references are flagged — the teacher writes a sticky-note prompt inside the notebook for the learner to complete as homework before Lesson 12.

	cannot show: [learner fills in at least one example, e.g., true atomic scale, quantum nature of electrons, temperature effects on bond vibration].' Groups share one 'Model Limitation' with the class before the lesson ends.	simulation is just one window into reality. The Carbon Paradox is richer than any single model.' Close with: 'In Lesson 12, you will build physical models using locally available materials — maize cobs, bottle tops, string — to do what even the best simulation cannot: put the structure in your hands.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Which simulation station generated the most productive discussion — and which generated the most confusion? What does this tell you about the visual design of the animations versus the conceptual difficulty of the bond type? (2) Review the Explanation Exit Sentences collected at the end of Phase 3: what proportion of learners correctly named a specific structural feature AND a specific simulation limitation? If fewer than 70% did so, consider opening Lesson 12 with a 5-minute whole-class re-examination of the graphite simulation together. (3) Did learners in your class who struggled with Lewis diagram drawing in Lessons 3–5 show renewed understanding when they saw the 3-D simulation? If yes, consider whether your future lessons should integrate digital visualisation earlier in the sequence. (4) Was the 10-minute station rotation sufficient, or did groups need more time at the giant covalent structures station? Note this for future lesson pacing. (5) Did the DQB orange notes reveal any persistent misconceptions — for example, learners confusing delocalised electrons in graphite with free electrons in metals? If so, address this directly in the Lesson 12 project briefing. (6) Reflect on digital equity: did all groups have equal access to functioning devices? If some groups shared one device among six, how did this affect their Critique Card quality? Plan device allocation more carefully for Lesson 12 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched curated animations and simulations of ionic (NaCl lattice), giant covalent (diamond and graphite), and metallic (aluminium) bonding structures, and completed Simulation Critique Cards comparing digital models to their own Lewis diagrams and physical-property data from earlier lessons.
What did I learn?	Learners can explain how digital simulations visualise the 3-D tetrahedral network of diamond (no free electrons, no conductivity) and the layered hexagonal structure of graphite (delocalised pi electrons, conducts electricity), and can articulate at least one specific simplification or limitation of each digital model.
How does this explain the phenomenon?	The Carbon Paradox: diamond and graphite are both 100% carbon, but diamond's 3-D tetrahedral covalent network (no free electrons) makes it hard and non-conducting, while graphite's layered structure with one delocalised electron per carbon atom makes it soft between layers and electrically conducting — a difference that no simulation fully captures but that digital models make far easier to visualise and understand than a static Lewis diagram alone.

LESSON 12 (80 minutes): Project Lesson — 3-D Model Building: Bonding Structures with Locally Available Materials	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To apply cumulative knowledge of chemical bonding by constructing annotated 3-D physical models of assigned bonding structures using locally available materials, and to present and defend model design choices to peers.
Knowledge	Students consolidate understanding of ionic lattice (NaCl), giant covalent networks (diamond, SiO_2), and layered covalent structures (graphite) — including bond type, bond angles, coordination numbers, and how structure determines physical properties.
Skills	Students plan, construct, and annotate a 3-D structural model; apply spatial reasoning to represent bonding geometry; communicate scientific reasoning through a gallery-walk presentation; and evaluate peer models using a structured rubric.
Attitudes	Students demonstrate creativity and imagination in selecting and repurposing local materials; show care (Love value) for peers during collaborative construction; and appreciate environmental conservation by prioritising recycled and natural materials over purchased ones.
Key Inquiry Question	How does the three-dimensional arrangement of bonds in a structure — ionic lattice, tetrahedral network, layered hexagonal sheet, or continuous covalent lattice — explain every physical property we have measured and observed throughout this evidence-gathering sequence?
Purpose in Storyline	This is the creative synthesis lesson of the evidence-gathering sequence. Students have spent Lessons 1–11 observing, measuring, and explaining the properties of diamond, graphite, NaCl, metals, and simple molecules. In Lesson 12 they must externalise their understanding into a tangible, annotated 3-D object — making the invisible bond geometry visible and defensible. The completed models become the centrepiece of the Lesson 13 final gallery and class explanation of the Carbon Paradox driving question.
Safety Notes	Supervise use of craft knives, scissors, and wire cutters — all cutting to be done on a flat board, not in hand. Clay and paint residue — students wash hands before handling food. Ensure no student has allergies to maize, bean, or other organic materials used. Remind students to handle bottle-top wire edges with care to avoid cuts. Keep workspace tidy to prevent trip hazards.

B. LESSON OVERVIEW
Groups of 3–4 students are each assigned one of four structures: (A) NaCl ionic lattice, (B) graphite layered hexagonal structure, (C) diamond tetrahedral network, or (D) SiO_2 giant covalent lattice. Using a kit of locally sourced materials — coloured clay, bottle tops, maize cobs, string, wire, cardboard, and recycled bottle caps — groups plan, build, and annotate their 3-D model over 55 minutes. Each model must be labelled with: bond type, bond angle(s), coordination number, and at least three physical properties explained by the structure. The lesson closes with a rapid 'model carousel' in which groups rotate and leave written feedback on sticky notes. The teacher uses targeted questioning during build time and the carousel to surface and correct persistent misconceptions. All work feeds directly into the Lesson 13 final driving-question resolution and class gallery.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Each group receives their assigned structure card (NaCl /	Teacher distributes structure cards and blueprint sheets. Read aloud	Strategy: Anticipatory Planning / Blueprint Sketching. By forcing	Observable indicator: Blueprint sheets show a labelled 2-D sketch

📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	graphite / diamond / SiO_2) and a 'Model Blueprint Sheet'. Before touching any materials, groups spend 8 minutes sketching a 2-D plan of their intended 3-D model on the blueprint sheet, answering three planning prompts: (1) Which material will represent which atom or ion, and why did you choose it? (2) What bond angle(s) must your model show — how will you achieve this physically? (3) Which ONE physical property of your structure will be most visually obvious from your model, and how? Groups write their predictions and material choices on the blueprint. One group planning to build graphite predicts: 'We will use flat cardboard layers stacked loosely to show the weak Van der Waals forces between layers — you can slide them apart like the real graphite does against the tile.' This connects directly back to the pencil-smearing observation from the anchoring phenomenon.	the three planning prompts to the class. Say: 'Before you touch a single piece of wire or clay, I need your group to agree on a plan. A builder in Nairobi does not start pouring concrete before drawing the foundation. Same rule applies here.' Allow 8 minutes of silent group planning. Circulate and crouch to group level — do NOT answer questions about what the model should look like; redirect with: 'What does your Lewis diagram from Lesson 4 tell you about the bond angle?' and 'What did you measure in the conductivity investigation in Lesson 9 — does your model need to show that?' After 8 minutes, cold-call ONE reporter from three different groups (rotate by asking the student to your left at each table) to share their material choice for ONE atom. Record the three choices on the board under the heading 'Our Material–Atom Mapping Predictions'. Do not evaluate — just record. WAIT TIME: 12 seconds after each cold-call before accepting the answer.	students to commit a plan to paper before building, this strategy activates prior knowledge (Lewis diagrams, bond angles, physical properties from Lessons 1–11) and makes implicit mental models explicit and inspectable. The teacher's board record creates a shared class reference point for later comparison — revealing whether the finished models match original intentions and why deviations occurred.	with atom/ion identities, material choices justified by at least one chemical reason (e.g. 'red clay = Na^+ because it is a cation and we need a different colour from Cl^-'), and at least one bond angle stated numerically. Teacher collects and scans three blueprint sheets (one per assigned structure type) to check for bond-angle errors (e.g. a group assigning 90° to diamond's tetrahedral bonds instead of 109.5°) before the build begins.
Observe Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar	Groups collect their materials kit from the front table and begin construction following their blueprint. Each kit contains: coloured modelling clay (two colours), pre-cut wire lengths, string, cardboard sheets, bottle tops (two colours), dried maize kernels, dried beans, and toothpicks. Groups are free to substitute or improvise with additional recycled materials they brought from home (an instruction given in the previous lesson). During construction, students physically embed the chemistry: the NaCl group alternates two clay	At the 12-minute mark, say: 'Materials are available — build phase begins now. Your blueprint is your contract. If you change something, write WHY on the blueprint in red.' Circulate every 4–5 minutes using a rotation pattern (NaCl → diamond → graphite → SiO_2 → repeat). At each group, ask ONE targeted question from the following bank — do not repeat the same question twice at the same group: (a) 'How many neighbours does your central atom have? Count them out loud for me.' (b) 'If I tried to pull your layers apart, what would I be breaking —	Strategy: Embodied / Manipulative Model Construction (3-D Externalisation). Research in science education shows that physically constructing a spatial model requires students to resolve ambiguities that 2-D diagrams mask — e.g. a student cannot place a tetrahedral bond at 90° in clay without immediately seeing it looks 'wrong' relative to their angle-guide card. The side-by-side placement of graphite and diamond models recreates the anchoring phenomenon at the students' own hands, making the bond-geometry explanation of the	Observable indicators during build: (1) Coordination numbers are physically correct and countable on the finished model (NaCl: 6:6; diamond: 4; graphite: 3 within layer; SiO_2: Si surrounded by 4 O). (2) Graphite model has a visible distinction between intra-layer connections (rigid wire) and inter-layer connections (loose string). (3) At least one annotation flag per model explicitly uses the phrase 'because of the bonding structure' in linking a structural feature to a physical property. Teacher uses a 4-item quick checklist on a clipboard to tick off each indicator

readings	colours in a 3-D grid, counting that each Na^+ is surrounded by 6 Cl^- and vice versa (coordination number 6); the diamond group builds a tetrahedral carbon unit and replicates it, checking every bond angle with a printed 109.5° angle-guide card; the graphite group stacks hexagonal cardboard lattices, connecting carbons within a layer with short rigid wire but connecting layers only with loose string to represent weak Van der Waals forces — and leaving one 'electron' bead free per carbon to show delocalisation; the SiO_2 group connects each central maize-kernel 'Si' to four bean 'O' atoms in a tetrahedral arrangement, bridging oxygens to the next Si. At the 35-minute mark, groups begin annotating their model with sticky label flags: bond type, bond angle, coordination number, and three physical properties linked to structure (e.g. on the graphite model: 'Layers slide easily → graphite is soft and leaves a mark on paper and tiles'). The graphite and diamond models sit side by side on the front desk — a deliberate echo of the anchoring phenomenon display from Lesson 1.	a covalent bond or something weaker? Show me on the model.' (c) 'Your bond angle looks closer to 90° than 109.5° — what tool can you use to fix that?' (d) 'Where are the delocalised electrons in your graphite model? I need to SEE them, not just hear about them.' At the 35-minute mark, pause all groups with a hand signal. Say: 'Hands down, eyes up. You have 8 minutes to add your annotation flags. Every model needs: bond type, bond angle, coordination number, and THREE physical properties explained by structure. If you finish early, add a flag explaining the connection back to our carbon paradox phenomenon.' WAIT TIME: After posing any question, count silently to 15 before accepting a response. Cold-call minimum 2 students per group visit who have not yet spoken.	Carbon Paradox visceral rather than abstract. Annotation flags enforce the structure-to-property reasoning chain that is the core sensemaking goal of the entire sub-strand.	as groups complete them.
Explain Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook)	Completed models are arranged in a carousel around the room. One 'ambassador' from each group stays with their model to answer questions; the remaining members rotate clockwise to visit all other models. Visitors carry a 'Peer Review Card' with four sentence starters: (1) 'One thing this model explains clearly about the phenomenon is…' (2) 'A bond-angle or structural detail I would question is…' (3) 'A physical	At the 55-minute mark, say: 'Construction phase ends — pencils and wire down. Ambassadors, stand by your model. Everyone else, pick up your Peer Review Card. You have exactly 4 minutes at each model — I will time-keep. Be a scientist, not a cheerleader: write what is chemically correct AND what could be improved.' Use a phone timer and call rotations clearly. During the carousel, position yourself at	Strategy: Structured Academic Controversy / Peer Critique Gallery Walk. The four sentence-starters on the Peer Review Card scaffold productive scientific disagreement — students are explicitly prompted to question bond angles, challenge property labels, and connect to the driving question, rather than offering vague praise. The ambassador role requires deep fluency: explaining to peers demands higher-order	Observable indicators: (1) Peer Review Cards contain at least one specific structural critique (not just 'looks good') — teacher collects 6 cards (one per visiting student) to check quality. (2) During the final cold-call, students use at least two of the following terms correctly: tetrahedral, delocalised electrons, Van der Waals forces, coordination number, giant covalent network, ionic lattice. (3) The 'model revision note' on each

Source: CK-12 🔍 Search ARES for similar readings	property label I want to challenge is…' (4) 'This model connects to our driving question because…' After visiting all three other models (4 minutes per model), students return to their own model, read the sticky-note feedback left by peers, and the group holds a 3-minute huddle to decide: what would they change, and why? Each group writes one 'model revision note' on their blueprint sheet. The carbon paradox is made explicit when the class stands in front of the diamond and graphite models together: the teacher asks the full class — pointing at both models simultaneously — 'Can you now answer, just from looking at these two models you have built with your own hands, why one writes on paper and the other cuts through ceramic?'	the diamond and graphite models together — listen for peer comments and note any persistent misconception (e.g. students who believe graphite conducts because 'it has metal in it' rather than because of delocalised electrons). At the 70-minute mark, bring the class to stand facing the diamond and graphite models. Ask the driving question aloud: 'Both 100% carbon — same atoms. Your models are right here. Tell me: why is one the hardest substance on Earth and the other writes on paper?' Cold-call 3 students (one from the diamond group, one from the graphite group, one from NaCl or SiO_2 group as a 'fresh eyes' perspective). WAIT TIME: 15 seconds after the question before the first cold-call. After the third student, summarise with: 'The answer was in the bonds all along — four tetrahedral bonds locking every atom in place versus three bonds leaving one electron free and layers barely held together. That is the Carbon Paradox resolved.'	consolidation than passive reception. The final whole-class convergence on the diamond-and-graphite pair recreates the anchoring phenomenon moment from Lesson 1, but now students supply the explanation themselves — closing the explanatory arc of 11 prior lessons.	group's blueprint sheet names a specific structural or annotation error — not a cosmetic improvement — demonstrating scientific self-evaluation.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12	Each student individually writes one new sticky note for the class Driving Question Board using the stem: 'I can now explain ______ about the carbon paradox because I built ______.' Examples that emerge: 'I can now explain why diamond does not conduct electricity because I built the tetrahedral model and saw there are NO free electrons — every electron is locked in a covalent bond.' and 'I can now explain why graphite is slippery because I built the layered model and saw that only loose string connects the layers — those are the weak Van	Distribute sticky notes. Say: 'You have 3 minutes — one note per person, no exceptions. Use the stem on the board. If you built the NaCl or SiO_2 model, connect your note to something about diamond or graphite — you are allowed to use someone else's model as evidence.' After students post notes, briefly scan the board aloud, reading two strong examples. Point out one unanswered question from Lesson 1 that now has an arrow connecting to a new answer and say: 'Look at what happened here — a question posted in Lesson 1	Strategy: Driving Question Board — Cumulative Knowledge Tracking. The DQB functions as a class-level working memory and metacognitive mirror. The sentence stem 'I can now explain ______ because I built ______' forces students to link the concrete physical act of model construction to abstract chemical explanation — operationalising the NGSS practice of Developing and Using Models. Arrow-linking to earlier questions makes knowledge growth visible and motivating, especially for students who posted 'I don't know why…' notes in	Observable indicator: Every student's sticky note names a specific structural feature (not just 'bonding') and links it to a specific observable property. Teacher photographs the DQB after class to audit note quality and identify any student whose note is too vague (e.g. 'I understand bonding now') — these students receive a targeted question at the start of Lesson 13.

🔍 Search ARES for similar readings	der Waals forces that let layers slide.' Students physically walk to the DQB and place their note under the heading 'STRUCTURE EXPLAINS PROPERTIES'. They also review notes placed in earlier lessons (marked in a different colour) and draw an arrow connecting any earlier unanswered question to a new note that now answers it.	has just been answered by a model you built today with wire and clay. That is what science looks like.' Remind students: 'In Lesson 13 — our final lesson — you will use your model AND your DQB notes to write the full class answer to our driving question. Make sure your model and annotation flags are stored safely.'	earlier lessons.	
Model Building Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 3 minutes, each group does one last check of their completed model against the 'Model Completeness Checklist' posted on the board: ✓ Two atom/ion types visually distinguishable; ✓ Correct coordination number countable; ✓ Bond angle labelled (numerically); ✓ Bond type named on a flag; ✓ Three physical properties explained by structure; ✓ One explicit connection to the Carbon Paradox driving question. Any missing item is added immediately. Groups then place their model on the designated display shelf/table at the front of the room, where it will remain for Lesson 13's final gallery and driving-question resolution. Students complete a 60-second exit reflection on the back of their blueprint sheet: 'The one structural feature of my assigned model that most powerfully explains a physical property is ______ because ______.'	Read the Model Completeness Checklist aloud item by item, pausing after each for groups to self-check. Say: 'If your model is missing any item, you have 90 seconds — fix it now.' After 90 seconds: 'Models on the display shelf, gently. These will be the centrepiece of our final lesson.' Collect blueprint sheets (both the original plan and the revision note) — these are used by the teacher to write formative feedback returned at the start of Lesson 13. Close with: 'Look at what is on that shelf. Clay, wire, bottle tops, maize, cardboard — and inside those shapes is every answer to the question we asked in Lesson 1 about the diamond and the pencil. Tomorrow we bring it all together.' Assign no written homework; instead, ask students to 'bring one question you still want answered about any of the four structures — write it down tonight and bring it on paper to Lesson 13.'	Strategy: Self-Assessment Against a Public Checklist (Metacognitive Closure). A publicly posted, item-by-item checklist externalises the success criteria, allowing students to self-regulate completeness without teacher prompting — building the 'learning to learn' core competency. The 60-second exit reflection on the blueprint sheet is a low-stakes written consolidation that surfaces residual gaps before the final lesson. The teacher's instruction to 'bring one remaining question' primes metacognitive awareness and gives Lesson 13 its opening warm-up material.	Observable indicators: (1) All models on the display shelf have a minimum of 5 of 6 checklist items visibly present. (2) Exit reflections on blueprint sheets name a specific structural feature (e.g. 'tetrahedral bonding with bond angle 109.5°' not just 'strong bonds') and a specific property (e.g. 'hardness' or 'electrical insulation'). Teacher sorts exit reflections into three piles — fully correct, partially correct, misconception present — to plan targeted Lesson 13 opening corrections.

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) MODEL ACCURACY — Were the bond angles and coordination numbers on the finished models chemically correct? Where did spatial reasoning break down (e.g. did the diamond group struggle to extend the tetrahedral unit beyond the central atom)? What scaffolding would help — pre-bent wire angle-guides? Printed 3-D template nets? (2) MATERIAL CHOICES — Did locally sourced materials help or hinder accurate geometry? Did any group's material

choice introduce a misconception (e.g. using identical-sized clay balls for Na^+ and Cl^-, obscuring the ion-size difference)? (3) CAROUSEL QUALITY — Were Peer Review Card critiques scientifically substantive, or did students default to aesthetic feedback? If feedback was superficial, consider adding a mandatory 'one chemical term must appear in every critique' rule for future iterations. (4) DQB PROGRESSION — How many students were able to draw an arrow from an unanswered Lesson 1 question to a new answer today? This ratio is a proxy for the cumulative coherence of the storyline — a low ratio suggests earlier lessons did not sufficiently foreground the driving question. (5) PCI INTEGRATION — Did the use of recycled and local materials generate genuine discussion about environmental conservation, or did it pass without comment? Purposefully naming this connection ('by using a maize cob instead of a bought plastic ball, you are doing exactly what environmental conservation looks like in chemistry class') increases PCI salience. (6) DIFFERENTIATION NOTES — Which groups needed the most teacher support during build time? Were high-achieving students sufficiently challenged by the SiO_2 structure (the most geometrically complex)? Consider offering an extension: 'Add a second SiO_2 unit and show how the bridging oxygens connect to the next Si — this is what the entire quartz crystal looks like at the nanoscale.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students physically constructed 3-D models of NaCl ionic lattice, graphite layered structure, diamond tetrahedral network, and SiO_2 giant covalent lattice using clay, wire, bottle tops, maize, string, and cardboard. During the carousel gallery walk, students observed each other's completed models and recorded peer critiques on Peer Review Cards. The anchoring phenomenon was recreated when the diamond and graphite models were placed side by side on the display shelf — echoing the original Lesson 1 display of the real diamond chip and pencil graphite on the teacher's desk.
What did I learn?	Students learned that the three-dimensional geometry of chemical bonds — 109.5° tetrahedral in diamond and SiO_2, 120° trigonal planar within graphite layers, and the alternating octahedral arrangement in the NaCl lattice — is the direct structural cause of every physical property the class has investigated across Lessons 1–11. Specifically: all four covalent bonds in diamond are used for bonding (no free electrons → insulator; rigid 3-D network → hardest natural substance); graphite's third bond delocalises one electron per carbon (→ conductor) while layers are held only by Van der Waals forces (→ soft, lubricating); NaCl's ionic lattice gives high melting point but allows conduction when molten; SiO_2's continuous covalent network gives extreme hardness and a very high melting point. The lesson also embedded the PCIs of environmental conservation (recycled local materials) and creative thinking (model design).
How does this explain the phenomenon?	The Carbon Paradox driving question can now be answered structurally: diamond and graphite are both pure carbon, but in diamond every carbon forms four covalent bonds in a continuous 3-D tetrahedral network — every electron is bonded, the structure is rigid in all three dimensions, and there are no free electrons. In graphite, every carbon forms three covalent bonds in flat hexagonal layers, leaving one electron per atom delocalised across the layer; the layers themselves are held together only by weak Van der Waals forces. This single difference in bonding geometry explains hardness (diamond: bonds resist deformation in all directions; graphite: layers slide), electrical conductivity (diamond: no free electrons; graphite: delocalised electrons carry charge), opacity and lustre, and every other property contrast observed in Lessons 1–11. The 3-D models built today make this invisible nano-scale geometry tangible and directly observable.

LESSON 13 (40 minutes): Model Presentations, Driving Question Board Final Revision and Unit Consolidation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will synthesise all evidence gathered across the unit to construct a complete, evidence-based explanation of the anchoring phenomenon — the Carbon Paradox — and demonstrate how bond type, three-dimensional structure and physical properties are interconnected across all four bonding categories.
Knowledge	Learners will be able to: (1) explain comprehensively how the type of chemical bond and the resultant three-dimensional structure of a substance determines its physical properties and practical uses; (2) distinguish clearly between giant ionic, simple molecular, giant covalent and giant metallic structures using specific examples from Kenyan and global contexts; (3) articulate why diamond and graphite — both pure carbon — have entirely different properties due solely to differences in bonding arrangement.
Skills	Learners will demonstrate: (1) scientific communication skills by presenting 3-D models clearly and accurately to peers; (2) critical evaluation skills by questioning and refining peer explanations during gallery-walk feedback; (3) synthesis skills by connecting evidence from all unit lessons into a unified answer to the driving question; (4) written analytical skills through a structured exit ticket linking bond type to structure to property to use.
Attitudes	Learners will: (1) appreciate that scientific understanding is built incrementally through observation, investigation, modelling and revision; (2) value peer contributions and collaborative knowledge-building; (3) develop confidence in communicating scientific ideas using correct terminology; (4) recognise the relevance of chemistry to everyday Kenyan life — from pencils and jewellers cutting tools in Nairobi to cookware and construction materials nationwide.
Key Inquiry Question	This lesson directly answers both key inquiry questions: (1) How do valence electrons determine the type of bond formed between atoms? — answered through full group presentations showing electron-sharing, transfer and delocalisation in each structure; (2) How does the type of chemical bond in a substance determine its physical properties and uses? — answered through the completed comparison table, the DQB final revision and the exit ticket.
Purpose in Storyline	This is the culminating lesson of the unit storyline. Students have journeyed from pure curiosity (Lesson 1 phenomenon launch) through valence electrons, four bond types, intermolecular forces, investigations, digital literacy and model building. In this lesson all threads converge: 3-D models are presented, the Driving Question Board receives its final evidence-based answers, and the anchoring phenomenon — the Carbon Paradox — is fully explained by students themselves. The teacher role shifts entirely to facilitator as students demonstrate mastery.
Safety Notes	No hazardous chemicals or equipment are used in this lesson. 3-D models from Lesson 12 should be handled carefully to avoid breakage. If wire components are present in models, caution students to avoid sharp ends. Ensure adequate space for gallery-walk movement between model stations without crowding.

B. LESSON OVERVIEW
The lesson opens with a brief predict-phase callback to the anchoring phenomenon, asking students to write their current best explanation of the Carbon Paradox before presentations begin. Groups then take turns presenting their 3-D models (NaCl ionic lattice, graphite layered structure, diamond tetrahedral network, SiO_2 continuous network) while peers conduct a structured gallery-walk with feedback cards. A whole-class sense-making discussion follows, culminating in a completed unit comparison table on the board. The Driving Question Board receives its final revision — every question answered with evidence from the unit. The lesson closes with a teacher-led return to the original diamond-and-graphite demonstration and a structured exit ticket assessing the full bond-type to structure to property to use chain. The 40 minutes

are divided as follows: Predict callback (5 min), Model presentations and gallery walk (15 min), Whole-class consolidation and DQB final revision (12 min), Return to phenomenon and exit ticket (8 min).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students enter the room to find the original anchoring phenomenon items — the graphite pencil stick and the diamond chip cutting-tool display card — placed at the front alongside the class Driving Question Board. Before any presentation begins, students individually write a paragraph-length explanation in their exercise books answering the question: 'Using everything you now know about chemical bonding, explain in your own words why diamond is the hardest naturally occurring substance and a perfect insulator, while graphite is soft enough to write with and conducts electricity — even though both are made only of carbon atoms.' Students are given 4 minutes of silent individual writing time. This activates prior knowledge, surfaces any remaining misconceptions before peer presentations, and gives every learner a personal benchmark they can compare against their understanding by the end of the lesson.	Teacher points to the diamond and graphite samples and says: 'Before your groups present, I want you to go back to where we started. Look at these two substances. You have now spent twelve lessons investigating why they are so different. Without looking at your notes, write your best explanation right now. You have four minutes — silence please, this is your thinking time.' Teacher circulates silently, reading over shoulders and noting any students still writing incomplete explanations — these students receive targeted questioning during the consolidation phase. At the 4-minute mark the teacher says: 'Keep that paragraph. You will compare it to your explanation at the end of the lesson and see how your thinking has grown.' WAIT TIME: Teacher allows a full 4 minutes of uninterrupted silent writing before any further instruction.	Individual written prediction activates schema from all previous lessons and creates a personal cognitive anchor against which new information during presentations will be compared. By committing ideas to paper before hearing peers, every learner is primed to listen critically rather than passively.	Teacher circulates and reads student paragraphs. Key diagnostic questions: Does the student mention tetrahedral bonding and all four covalent bonds in diamond? Does the student mention delocalised electrons and Van der Waals layers in graphite? Are the words 'structure' and 'bonding' used in relation to properties? Students with incomplete answers are mentally flagged for targeted questioning later.
Observe Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos	Groups take turns presenting their 3-D models at the front of the room for approximately 2.5 minutes each (four groups, 10 minutes total), followed by a 5-minute structured gallery walk. Each presenting group must: (1) name the substance and bond type; (2)	Before presentations begin, teacher says: 'Each group has two and a half minutes. I will time you. Your job is not to read from notes — look at your model and talk about it as if you are a scientist explaining your structure to a curious person who has never	Structured peer observation cards ensure all students are active listeners, not passive audiences. The gallery-walk sticky-note protocol builds scientific communication and critical evaluation skills while generating authentic peer-to-peer questions	Teacher collects observation cards at the end of the gallery walk and scans them during the consolidation phase. Peer sticky-note questions reveal misconceptions that need addressing in whole-class discussion. Teacher notes: Are

📖 **HTML:** [Chemical Bonding (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	point to specific features of the model and explain what they represent (bond angles, bond type, free electrons, layers, lattice); (3) state two physical properties caused directly by the bonding arrangement shown; (4) name one real Kenyan use of the substance linked to those properties. During each presentation, non-presenting students complete a structured peer-observation card with four columns: Substance, Bond type shown, Property explained, Kenyan use given. During the 5-minute gallery walk, all models are arranged on desks around the room and students circulate, leaving one written praise comment and one probing question on a sticky note attached to each model they did not build.	heard of chemical bonding. Three things I will listen for: the bond type named correctly, a property linked directly to the structure you are showing, and a Kenyan context.' During presentations, teacher uses strategic questioning — only if a group stalls or makes an error — using prompts such as: 'Can you point to where the delocalised electrons would be in your model?' or 'You said this substance has a high melting point — what in your model explains that?' Teacher does NOT correct errors publicly during the presentation; instead notes them for the consolidation phase. During the gallery walk teacher circulates and reads sticky notes, selecting two or three probing questions from students to read aloud during the consolidation. Teacher says at the start of the gallery walk: 'You have five minutes. Visit every model that is not yours. Write one thing the group explained well and one genuine question you still have about that structure. Be specific — not just good job but what exactly was good and why.' WAIT TIME: After each group presentation, teacher waits 10 seconds before any comment, allowing the class to process and write on their observation cards.	that drive the consolidation discussion. Seeing all four structural models simultaneously allows students to compare and contrast across bond types — a key consolidation strategy.	students correctly identifying bond types from the models? Are property-to-structure links accurate? Are Kenyan contexts relevant and correctly connected?
Explain Phase 📹 **VIDEO:** [Van der Waals Forces - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical	The class assembles for a 12-minute whole-class consolidation. The teacher draws the completed unit comparison table on the board with five columns (Structure type, Example, Bond type, Key physical properties, Kenyan use) and four rows (giant ionic, simple molecular, giant covalent, giant metallic). Students contribute answers from their observation	Teacher begins consolidation by saying: 'I am going to draw our final table. You are going to fill it in. I will not tell you the answers — you have the evidence from twelve lessons and four models. Let us start with the giant ionic row. Someone give me an example substance from our Kenyan context.' After the table is complete, teacher says: 'Now look	The collaborative table-building strategy externalises the unit's conceptual framework so all students can see the pattern simultaneously. Returning to the DQB closes the narrative loop begun in Lesson 1 — every student-generated question is now answered with student-generated evidence. The colour-coded paragraph update gives students a	Teacher uses cold-calling during table completion to ensure broad participation. Quality of DQB answers reveals depth of understanding. Teacher listens for: correct use of the terms delocalised electrons, tetrahedral network, hexagonal layers, Van der Waals forces, giant ionic lattice. Any persistent misconceptions identified here are

[Bonding (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	cards and presentations, completing the table collaboratively. The teacher then reads two or three genuine student questions from the gallery-walk sticky notes and the class discusses them. The teacher returns explicitly to the anchoring phenomenon: the class jointly constructs a full model answer to the driving question on the board, using evidence from the table. The Driving Question Board is then revisited: teacher reads each original student question posted in Lesson 1, and students volunteer answers — pointing to the comparison table, their models, or their exercise-book findings as evidence. Every question on the DQB is answered and the answer is recorded on a new sticky note placed next to the original question. Students update their individual exercise-book paragraph from the predict phase, adding or correcting ideas in a different colour pen.	at the Driving Question Board. In Lesson 1 you asked questions you could not yet answer. Let us answer every single one of them right now, using evidence.' Teacher points to one DQB question at a time and says: 'Who can answer this question and tell me exactly which lesson or which investigation gave you that evidence?' For the carbon paradox specifically, teacher holds up the diamond and graphite samples again and says: 'Final question — the biggest one. Same element, same protons, same electrons. Completely different properties. Someone give me the complete explanation — bond type, bond angle, what happens to the electrons, and what property results.' WAIT TIME: Teacher waits 15 seconds in silence after posing the carbon paradox final question, allowing multiple students to formulate complete answers before calling on anyone. Teacher then takes two different student responses to model the idea that multiple evidence-based explanations can be valid.	visible record of their own conceptual growth across the unit.	directly addressed before the exit ticket.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Van der Waals Forces - Overview](#) Source: CK-12 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Chemical Bonding (Flexbook)](#)	This phase represents the final DQB revision rather than initial creation. Students conduct a structured 4-minute revisit of the full board in silence, reading every original question and every newly added answer. Each student then writes in their exercise book: (1) the question from Lesson 1 that they find most satisfying to now be able to answer — and why; (2) any question about chemical bonding that the unit has not fully answered for them — a genuine remaining curiosity. The remaining-curiosity questions are read aloud by the	Teacher says: 'Take four minutes and read the entire Driving Question Board — every question, every answer. Then in your book write two things: the question whose answer you are most proud of understanding, and any question that chemistry has not fully answered for you yet. Your remaining curiosity is not a failure — it is how science moves forward.' After collecting remaining-curiosity responses (students read them voluntarily), teacher says: 'These questions show that you are thinking like	The most-satisfying-answer reflection metacognitively reinforces learning by asking students to identify their own growth. The remaining-curiosity prompt normalises scientific uncertainty and models authentic scientific thinking — questions lead to more questions. Connecting remaining curiosities to Kenyan universities and real research makes chemistry feel locally alive and professionally relevant.	Most-satisfying-answer choices reveal which concepts have been most deeply understood by the class majority. Remaining curiosities reveal the boundaries of current understanding and inform teacher planning for KCSE revision and future units. Teacher notes whether students can articulate why a question is now answerable — this is evidence of genuine conceptual change, not just memorisation.

Source: CK-12 🔍 Search ARES for similar readings	teacher (anonymously) and briefly acknowledged — the teacher validates them as genuine scientific questions, connecting one or two to KCSE Chemistry topics or real scientific research. For example, a student curious about how graphene (a single layer of graphite) conducts electricity better than graphite itself is told that this is an active area of nanotechnology research and is studied at universities in Nairobi and globally.	chemists. One of you asked about graphene — a single layer of graphite. Scientists at universities across Africa and the world are investigating this right now. Your curiosity is the beginning of research.' Teacher explicitly connects one remaining curiosity to Kenya: 'The question about why aluminium does not rust connects to research on protective coatings being studied at the University of Nairobi School of Engineering — chemistry is happening right here.' WAIT TIME: Teacher allows the full 4 minutes of silent DQB reading before any student speaks.		
Model Building Phase 📹 **VIDEO:** Van der Waals Forces - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Bonding (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	This phase serves as the lesson closure and individual summative formative assessment. Despite the phase label, in this final consolidation lesson the Model Building Phase is the exit ticket phase — students construct a written conceptual model (not a physical model) of the unit's central idea. Each student receives a structured exit ticket card with four sections: (1) BOND TYPE — name all four bond types studied and give one Kenyan example for each; (2) STRUCTURE TO PROPERTY — choose any one substance from the unit and draw a labelled sketch of its bonding structure, then list two physical properties it has because of that structure; (3) PROPERTY TO USE — explain how one specific property of your chosen substance makes it useful in a Kenyan context; (4) THE CARBON PARADOX — in three sentences, explain why diamond and graphite have different properties even though both are	Teacher distributes exit ticket cards and says: 'This is your individual thinking — no group discussion, no notes. Six minutes. Show me what you know.' During the 6 minutes teacher circulates silently. At the 5-minute mark teacher says: 'One more minute — make sure you have compared your opening paragraph to your exit ticket answer. Write that one sentence about how your thinking grew.' When collecting cards teacher says: 'This is not a test for a grade — it is evidence for me of what we need to revisit and evidence for you of how far your thinking has come since Lesson 1. Well done for the work you have put into this unit.' Teacher ends the lesson by holding up the diamond and graphite one final time and saying: 'You walked in thirteen lessons ago not knowing why these two are different. You now know the answer at the level of individual atoms and electrons. That is what chemistry does — it explains the invisible world that	The structured four-section exit ticket assesses the full conceptual chain: bond type identification, structure-to-property reasoning, property-to-use application, and synthesis of the anchoring phenomenon explanation. The before-and-after paragraph comparison is a metacognitive closure strategy that makes conceptual growth visible to each learner personally. The teacher closing statement anchors the entire unit's purpose in the original phenomenon, completing the narrative arc.	Exit tickets provide individual written evidence of mastery across four dimensions: recall (bond types and examples), application (structure-to-property sketch), transfer (property-to-use in Kenyan context), and synthesis (Carbon Paradox explanation). The quality of the Carbon Paradox section specifically reveals whether students have achieved the central SLO of the unit. Teacher analyses exit tickets after the lesson to identify any students requiring individual follow-up before KCSE revision begins. The metacognitive before-and-after sentence reveals whether students are aware of their own learning growth — a key 21st-century competency in the CBC framework.

	made only of carbon atoms. Students complete the exit ticket individually in 6 minutes, then compare their Lesson 13 opening paragraph (written in the predict phase) with their exit ticket answer for the Carbon Paradox section — writing one sentence about how their explanation improved. Exit tickets are submitted to the teacher before students leave.	makes the visible world work.' WAIT TIME: Teacher waits a full 6 minutes without prompting or assisting during individual exit ticket completion.		

D. TEACHER REFLECTION

After the lesson, the teacher should consider the following reflective questions: (1) Did every group present their model with accurate use of bonding terminology, or were there persistent errors in specific bond types that need reinforcing before KCSE revision? (2) Were all DQB questions answered with genuine student-generated evidence, or did some questions remain that need revisiting? (3) Did the exit tickets reveal any systematic misconceptions — for example, confusion between simple molecular and giant covalent structures, or between metallic bonding and ionic bonding in terms of conductivity? (4) Did students make authentic connections to Kenyan contexts, or were their examples generic? If generic, how can the next unit build stronger local-context awareness from the start? (5) Was the 40-minute period sufficient for four group presentations plus consolidation and exit ticket? If time was tight, consider extending group presentations to a separate lesson in future iterations. (6) Did the before-and-after paragraph comparison generate genuine metacognitive reflection, or was it treated as a routine task? How can metacognitive strategies be embedded more regularly across future units? (7) Were the remaining-curiosity questions connected meaningfully to Kenyan research and KCSE pathways, affirming students as future scientists?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In this final lesson, we observed four 3-D models of chemical bonding structures — the NaCl ionic lattice, the graphite layered hexagonal structure, the diamond tetrahedral network and the SiO_2 continuous covalent network — built by student groups from locally available materials. We also observed the original anchoring phenomenon: the diamond chip and the graphite pencil stick, now viewed with thirteen lessons of chemical bonding knowledge. We observed all student-generated questions on the Driving Question Board being answered with evidence gathered across the unit.
What did I learn?	We learned that the type of chemical bond formed between atoms — determined entirely by the number and arrangement of valence electrons — produces a specific three-dimensional structure that directly causes every physical property a substance has. Giant ionic structures (NaCl) have high melting points and conduct electricity only when molten or dissolved. Simple molecular structures (iodine, water) have low melting points because only weak Van der Waals or hydrogen bonds hold molecules together. Giant covalent structures (diamond, SiO_2) are extremely hard with very high melting points because every electron is used in strong covalent bonds throughout a continuous network. Giant metallic structures (aluminium, copper) conduct electricity and heat because delocalised electrons move freely. We also learned that two substances made of the same atoms — diamond and graphite — can have completely opposite properties because the bonding arrangement of those atoms is entirely different.
How does this explain the phenomenon?	We explained the Carbon Paradox fully: in diamond, each carbon atom forms four strong covalent bonds in a tetrahedral arrangement (bond angle 109.5 degrees) creating a rigid three-dimensional network with no free electrons — making diamond the hardest naturally occurring substance and a perfect electrical insulator. In graphite, each carbon atom forms only three covalent bonds in a flat hexagonal arrangement (bond angle 120 degrees) and the fourth electron from each carbon is delocalised between

	the layers — making graphite a good electrical conductor. The layers themselves are held together only by weak Van der Waals forces, so they slide over one another easily, making graphite soft enough to write with and useful as a lubricant. The bond type determines the structure, the structure determines the properties, and the properties determine the uses — from Nairobi jewellers cutting tools to pencils used in every Kenyan classroom.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.